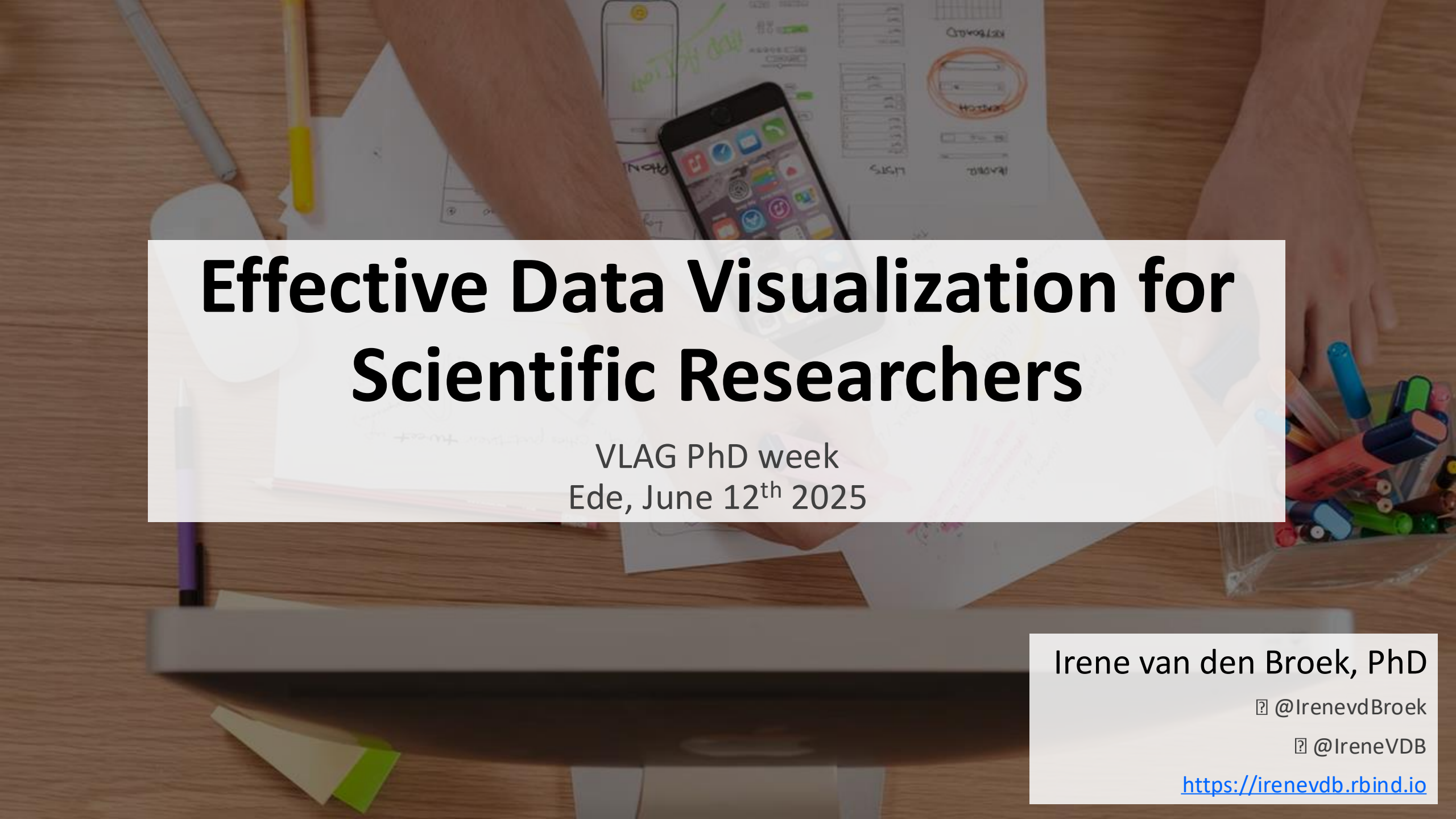


The background image shows a top-down view of a wooden desk. In the center, a black smartphone displays a home screen with various app icons. To the left of the phone, a hand is visible, holding a yellow highlighter. Several sheets of paper with handwritten notes and diagrams are scattered on the desk. One paper has a diagram of a mobile app interface with labels like 'login', 'logout', 'search', and 'list'. Another paper has a diagram of a network or data flow. To the right of the phone, a hand is visible, holding a blue marker. A clear plastic container filled with various colored markers is on the right side of the desk. The overall scene suggests a workspace for research or design.

Effective Data Visualization for Scientific Researchers

VLAG PhD week
Ede, June 12th 2025

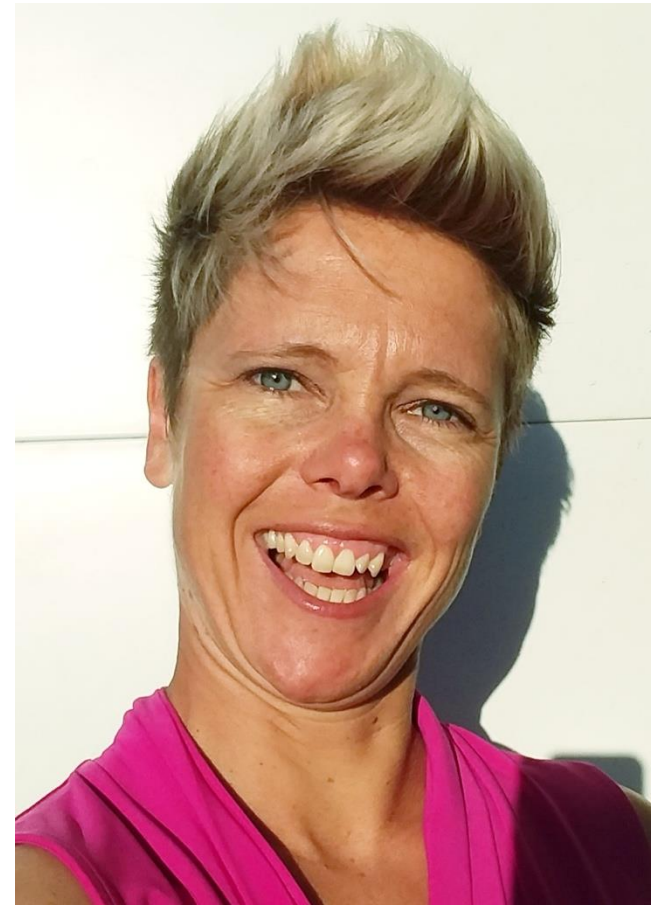
Irene van den Broek, PhD

📧 @IrenevdBroek

📧 @IreneVDB

<https://irenevdb.rbind.io>

Hi there, I'm Irene!



Irene van den Broek

irenevandenbroek@gmail.com

I have a PhD in biomarker research

Liquid chromatography coupled to
tandem mass spectrometry for
the quantitative bioanalysis of
bioactive and potential
biomarker peptides



Irene van den Broek 2010



2005-2010
Utrecht University

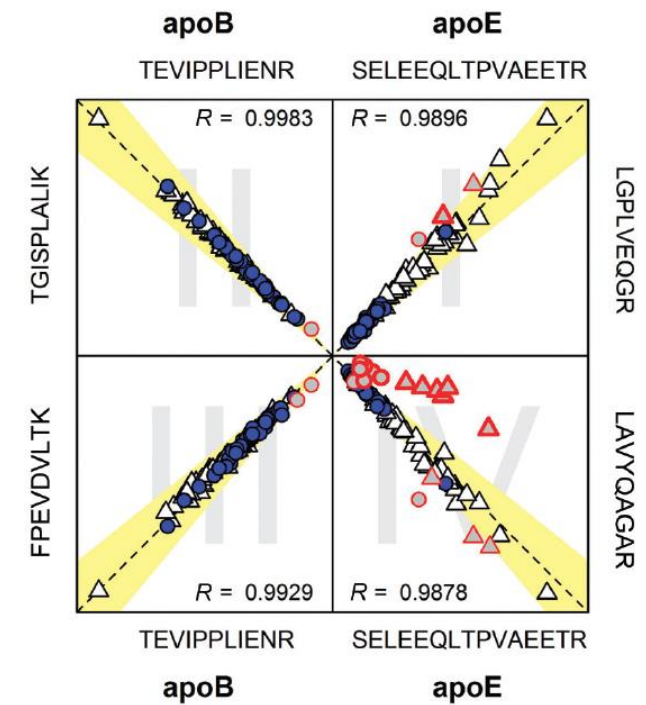
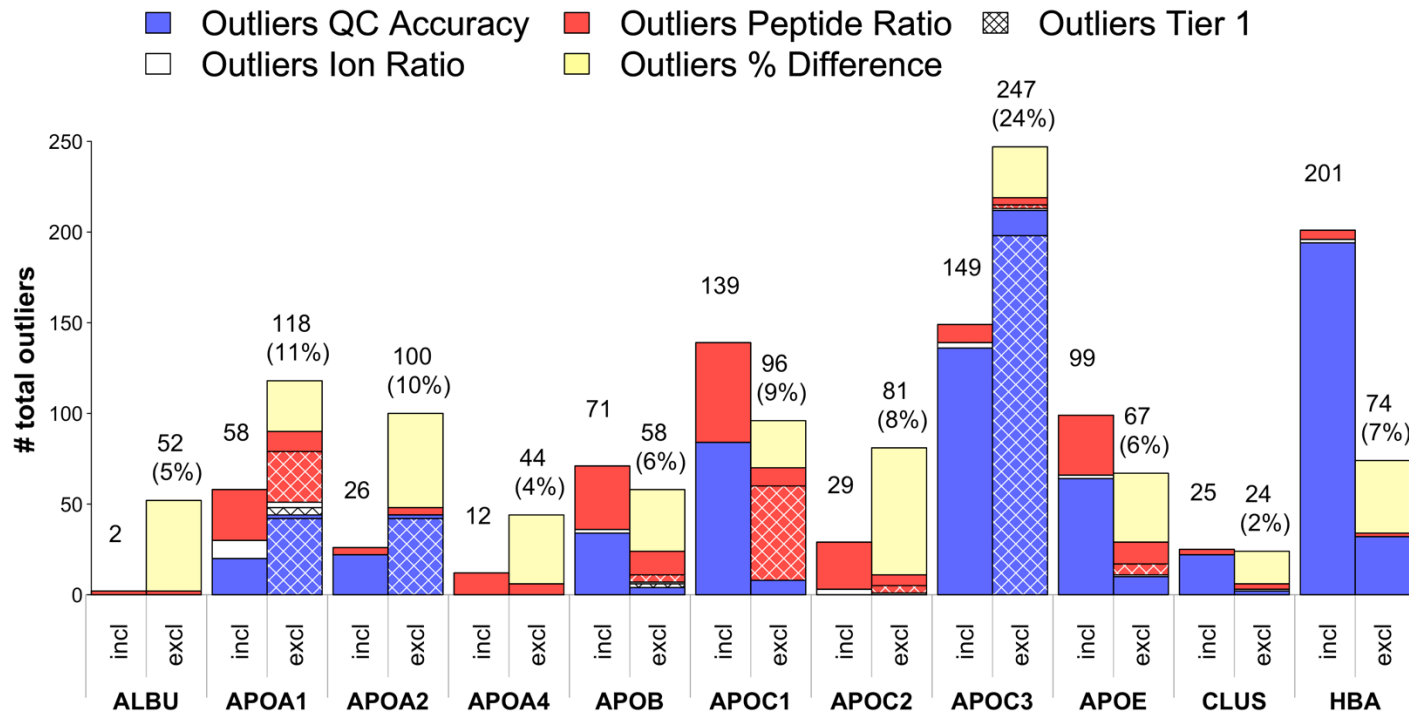


2013-2015
Leiden University
Medical Center

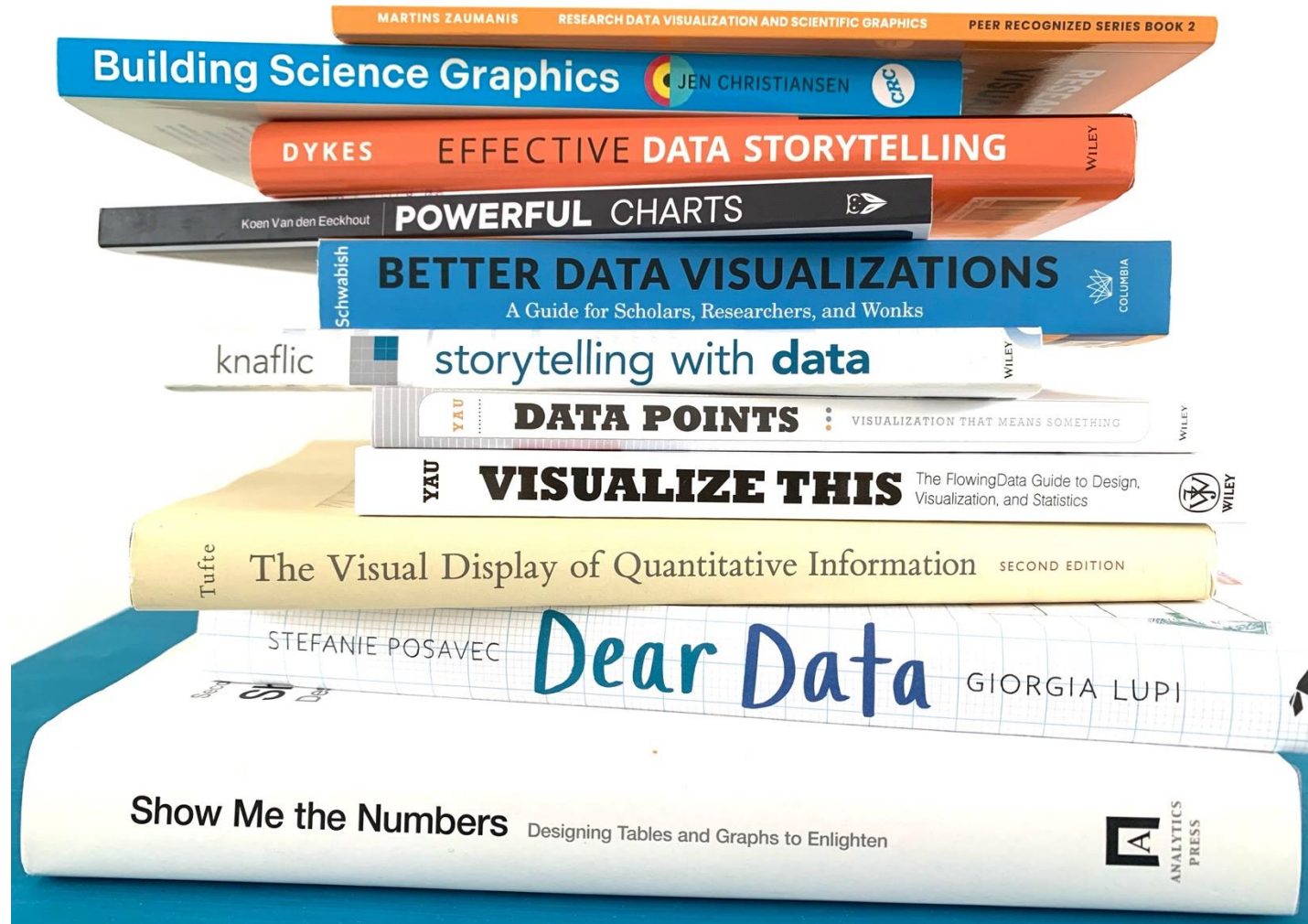


2015-2019
Cedars-Sinai
Medical Center

During my PhD and post-docs, I loved creating charts, preferably as complicated as possible...

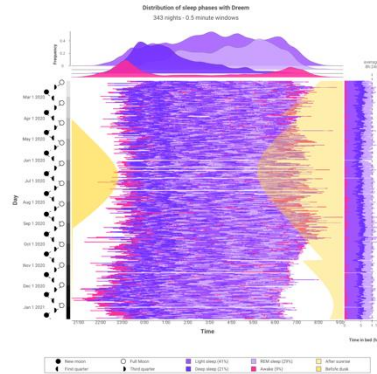


When I learned **Data Visualization** is a field on its own, I was hooked!



I started visualizing my Health and Habits.

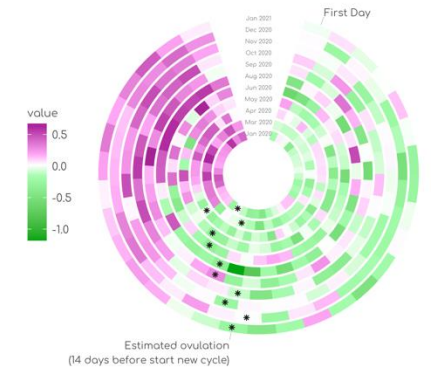
Sleep



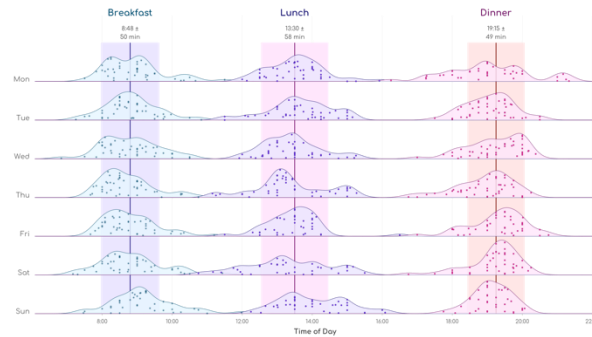
Activity



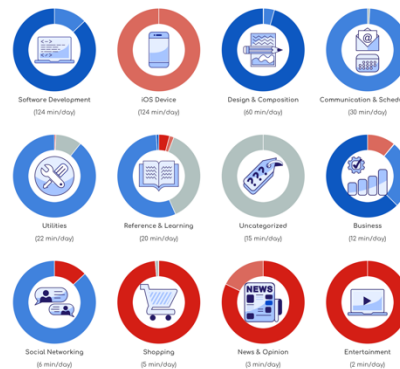
Body



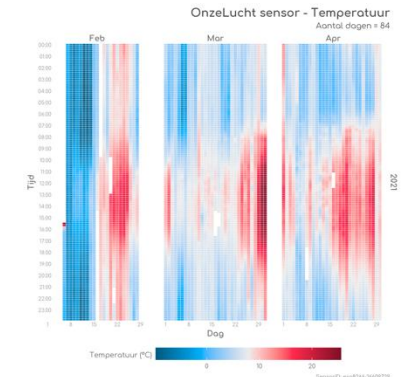
Food



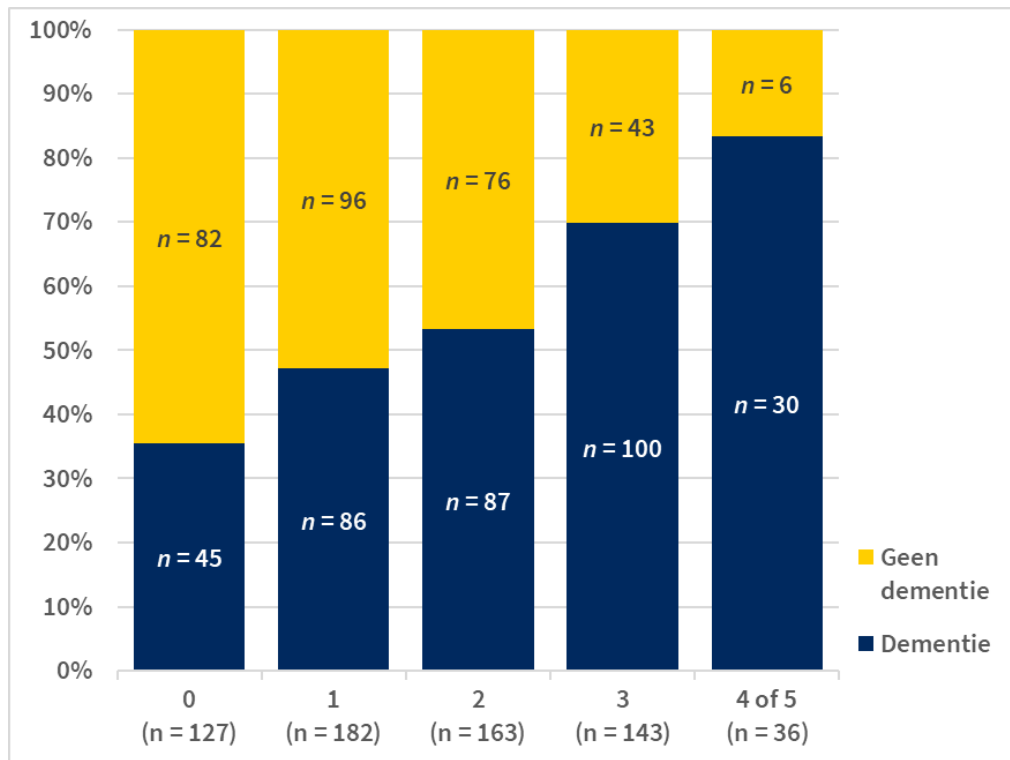
Habits



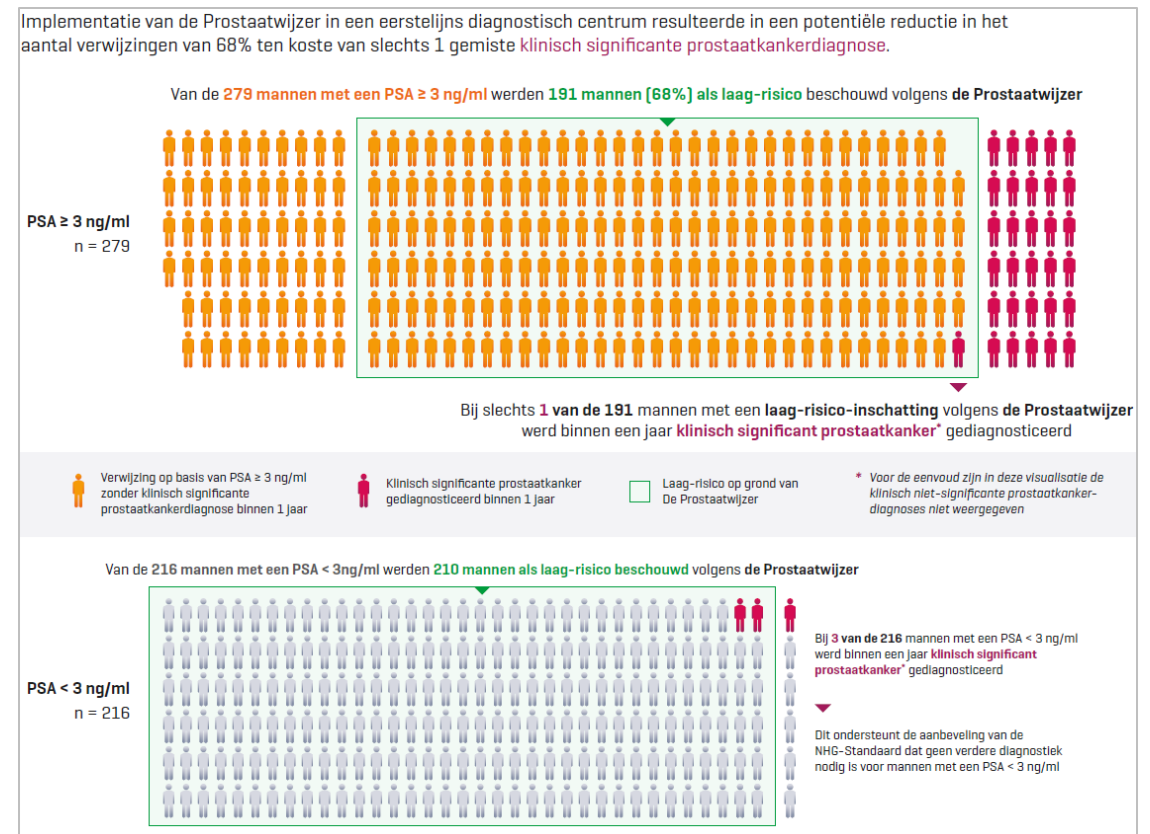
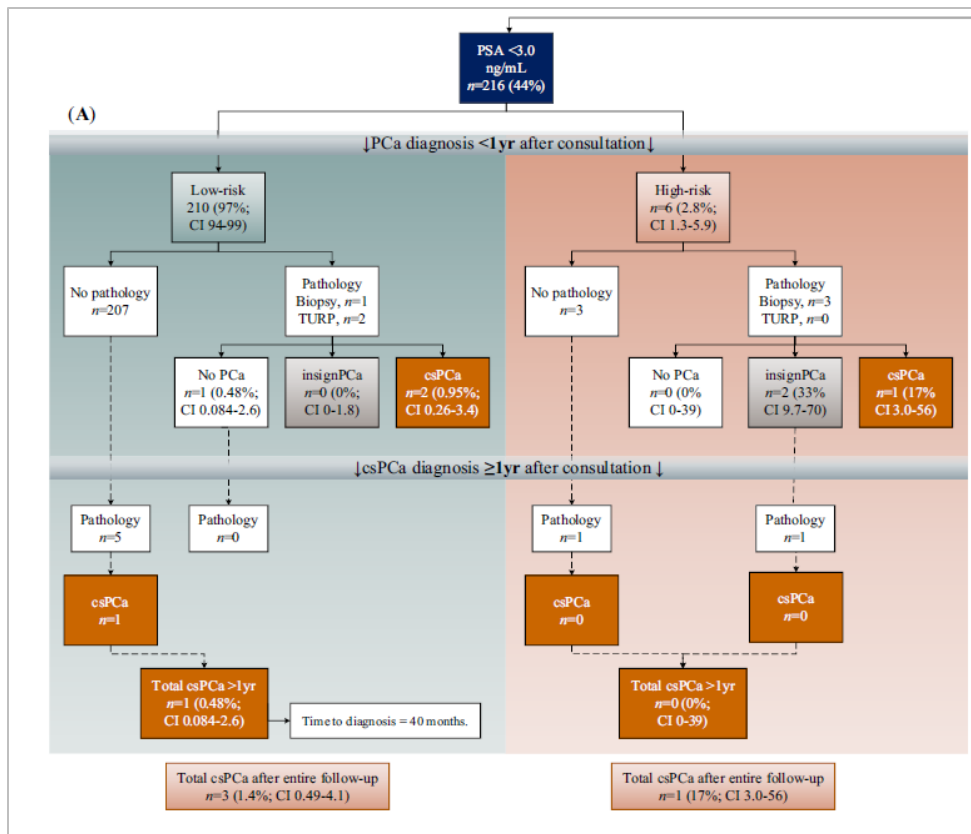
Environment



I help clarify scientific communication...



I help clarify scientific communication...



I help clarify scientific communication...

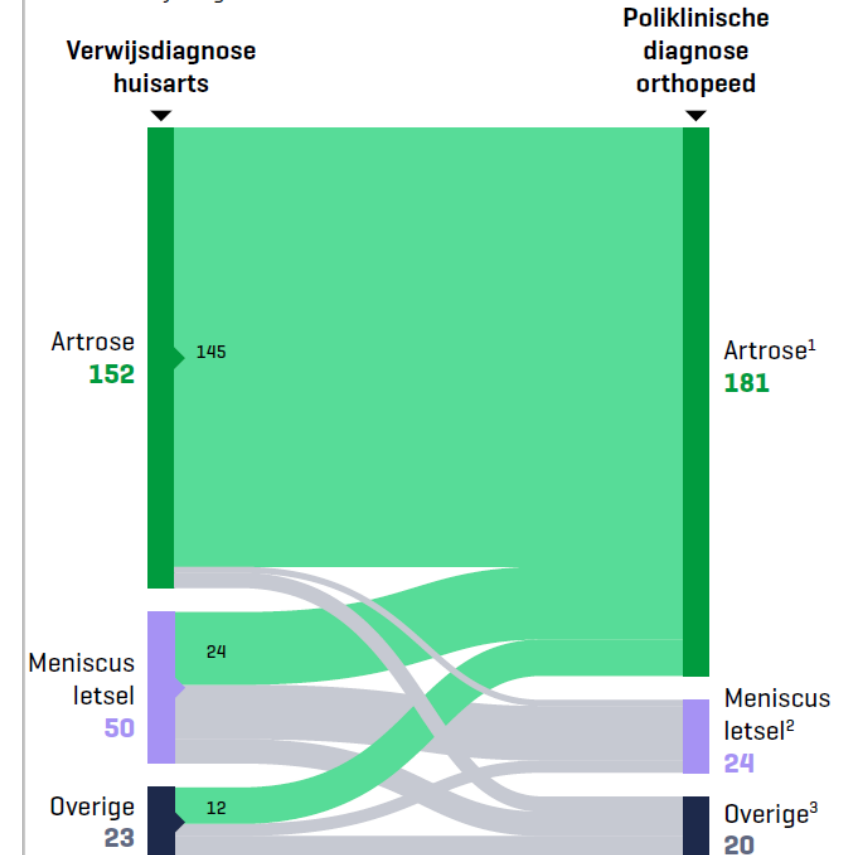
Tabel 1 Diagnoses per leeftijdscategorie

Diagnose		45-54	55-64	65-74	75-84	>85	Total
		N (%)	N (%)	N (%)	N (%)	N (%)	N (%)
Verwijsdiagnose Huisarts (n=276)	artrose	6 (12,5)	45 (51,7)	55 (64,7)	39 (83,0)	7 (87,5)	152 (55,1)
	Meniscus leasie	27 (56,2)	24 (27,5)	20 (23,5)	6 (12,8)	0 (0,0)	77 (27,9)
	Andere	15	18	10	3	1	47 (17,0)
	diagnose**	(31,2)	(20,6)	(11,8)	(4,2)	(12,5)	
Poliklinische diagnose orthopeed N = 329	artrose	21 (32,8)	73 (69,5)	78 (82,1)	47 (90,6)	10 (83,3)	229 (70,0)
	Meniscus leasie	23 (35,9)	15 (14,2)	7 (7,4)	2 (3,8)	0 (0,0)	47 (14,3)
	Andere	20	17	10	4	2	53 (16,0)
	diagnose****	(31,3)	(16,1)	(10,5)	(7,5)	(16,7)	



De huisarts stelt minder vaak de diagnose **artrose** dan de orthopeed

Als huisarts en orthopeed een andere diagnose stelden, was de verwijsdiagnose meestal meniscusletsel



1 De diagnose **artrose** bevat *alle* diagnoses met artrose

2 De diagnose **meniscusletsel** bevat *alle* diagnoses met meniscusletsel, behalve de diagnoses inclusief artrose

3 De diagnose **overige** bevat *alle* diagnoses zonder de diagnose artrose of meniscusletsel

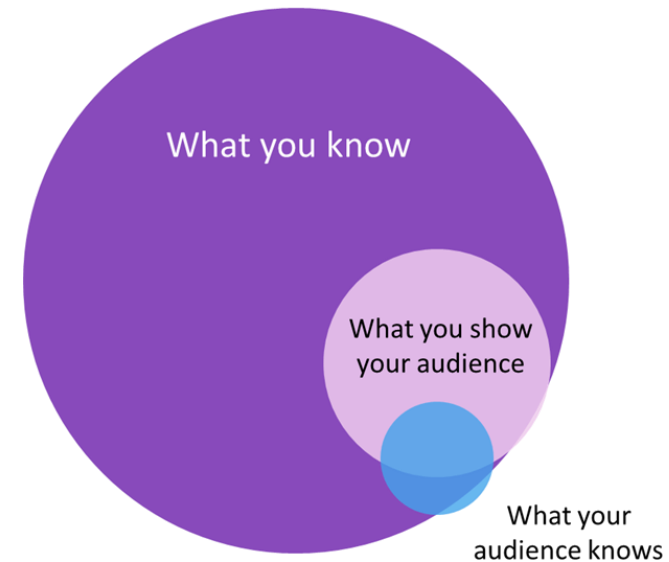
... and love to share with others what I wish I'd learned during my PhD!

Resist the urge to show everything!



Don't underestimate how smart you are!

The curse of knowledge

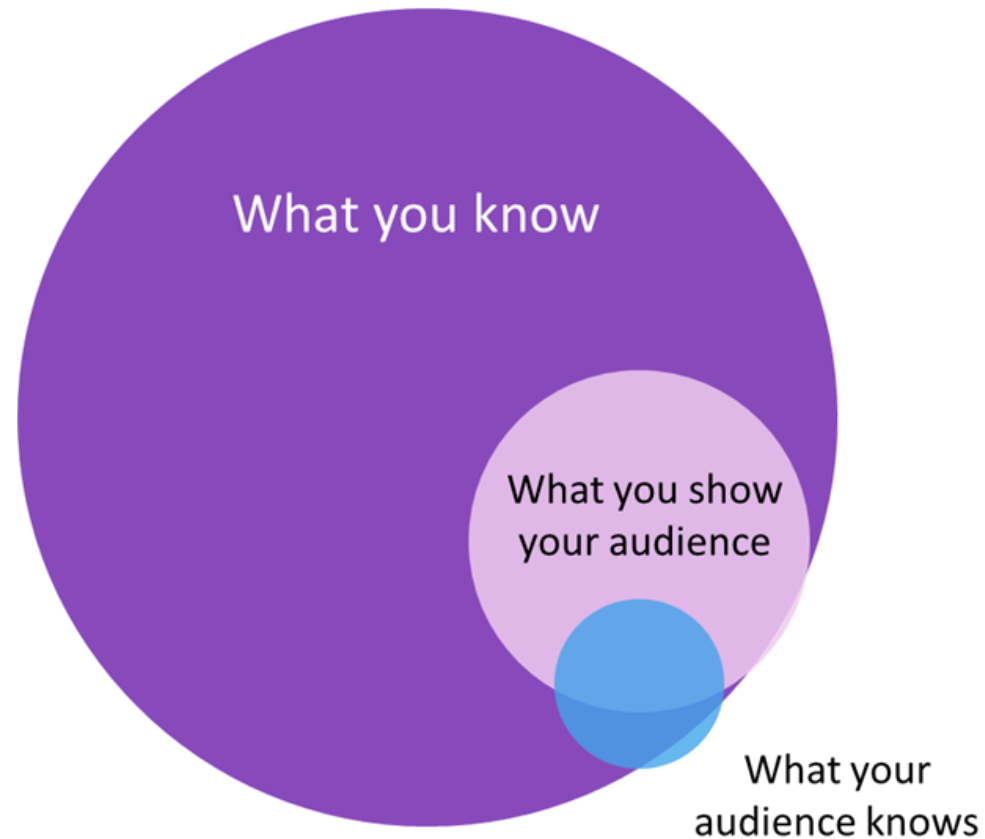


Resist the urge to show everything!



Don't underestimate how smart you are!

The curse of knowledge



Schedule for today:

Explore: Why? What? How?

Set the context

Lunch 12:30-13:30

Enhance: Visual Design

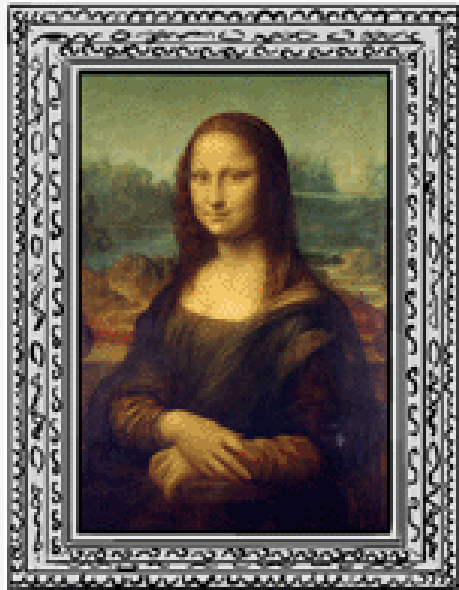
Simplify & Declutter

Engage: The Narrative

Highlight your message

Why visual communication?

Why use visuals in academic research?



A PICTURE IS WORTH A
THOUSAND WORDS.

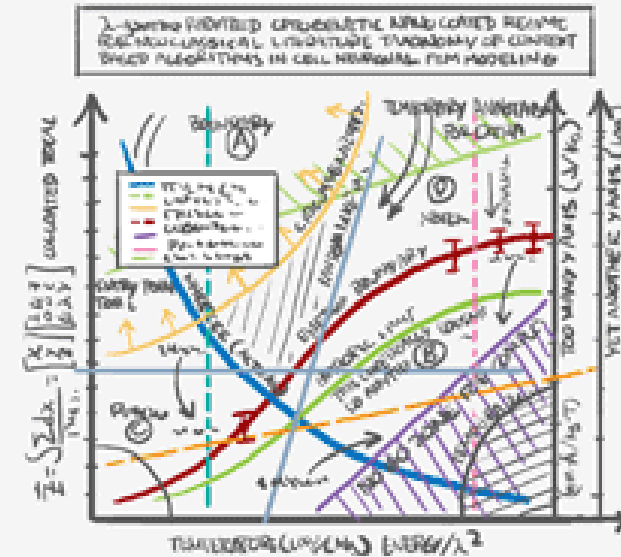


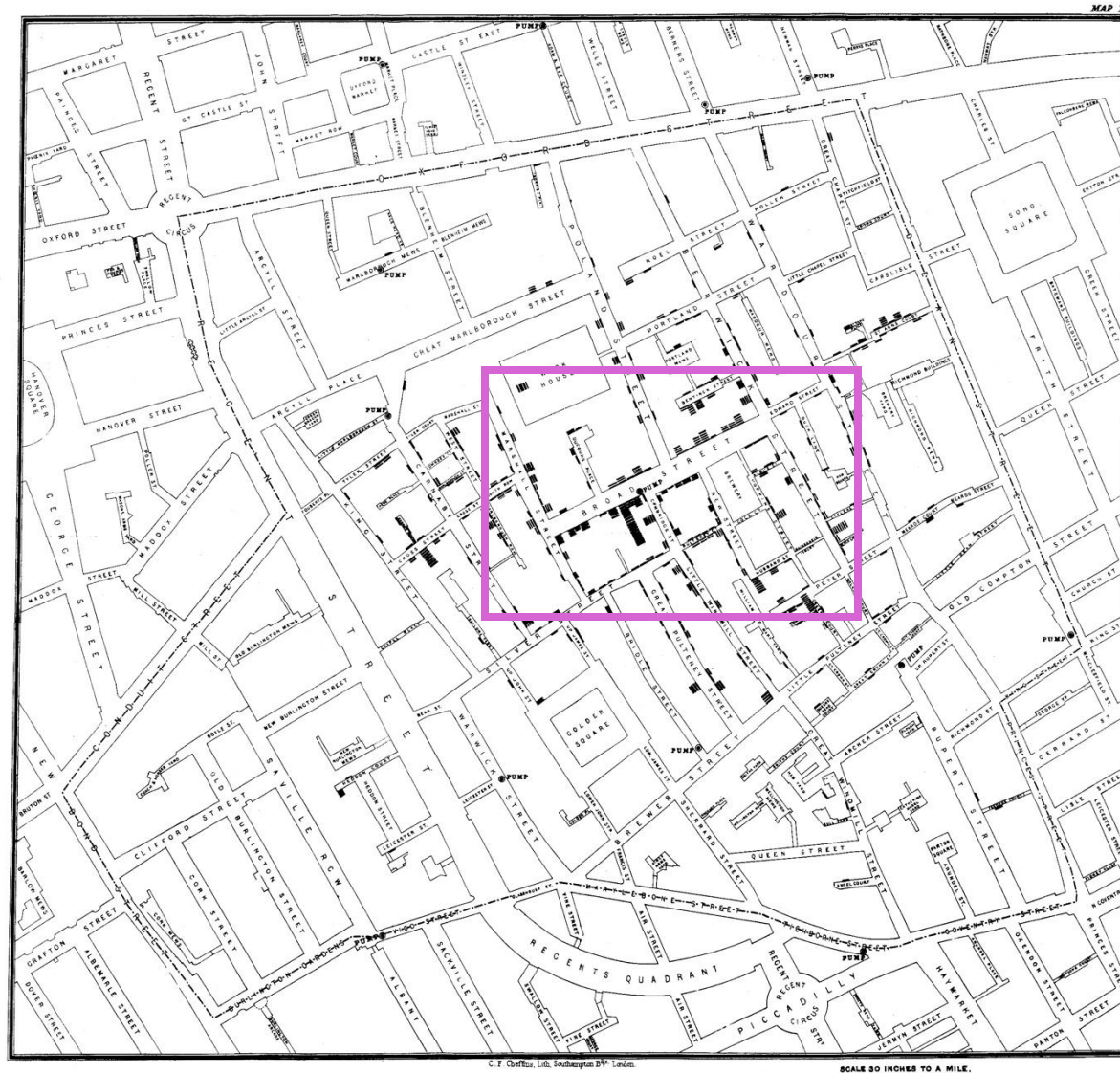
FIGURE 1.

A PICTURE WITH A
THOUSAND WORDS IS
USUALLY WORTH A PH.D.

WOW!

JORGE CHAM © 2017

1854: The cholera map from John Snow



1854: The cholera map from John Snow



What can you see in these numbers?

Anscombe's quartet:

4 datasets

11 x-values

11 y-values

Same total

Same mean

Same standard deviation

	I		II		III		IV	
	x	y	x	y	x	y	x	y
	10	8,04	10	9,14	10	7,46	8	6,58
	8	6,95	8	8,14	8	6,77	8	5,76
	13	7,58	13	8,74	13	12,74	8	7,71
	9	8,81	9	8,77	9	7,11	8	8,84
	11	8,33	11	9,26	11	7,81	8	8,47
	14	9,96	14	8,1	14	8,84	8	7,04
	6	7,24	6	6,13	6	6,08	8	5,25
	4	4,26	4	3,1	4	5,39	19	12,5
	12	10,84	12	9,13	12	8,15	8	5,56
	7	4,82	7	7,26	7	6,42	8	7,91
	5	5,68	5	4,74	5	5,73	8	6,89
SUM	99,00	82,51	99,00	82,51	99,00	82,50	99,00	82,51
AVG	9,00	7,50	9,00	7,50	9,00	7,50	9,00	7,50
STDEV	3,32	2,03	3,32	2,03	3,32	2,03	3,32	2,03

A picture is worth a thousand words!

Anscombe's quartet:

4 datasets

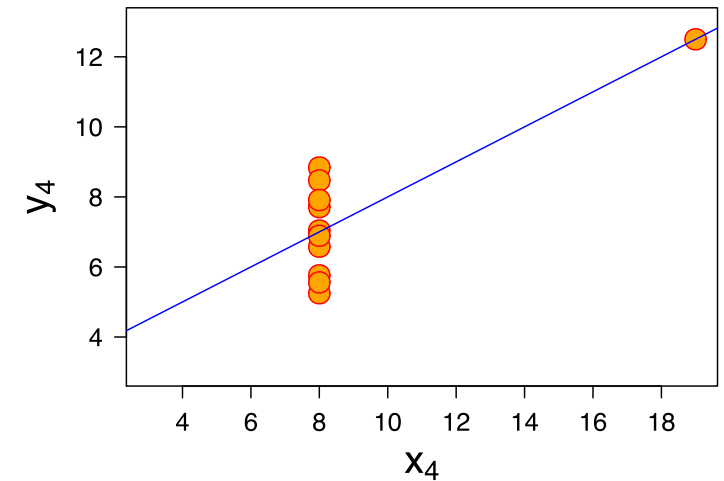
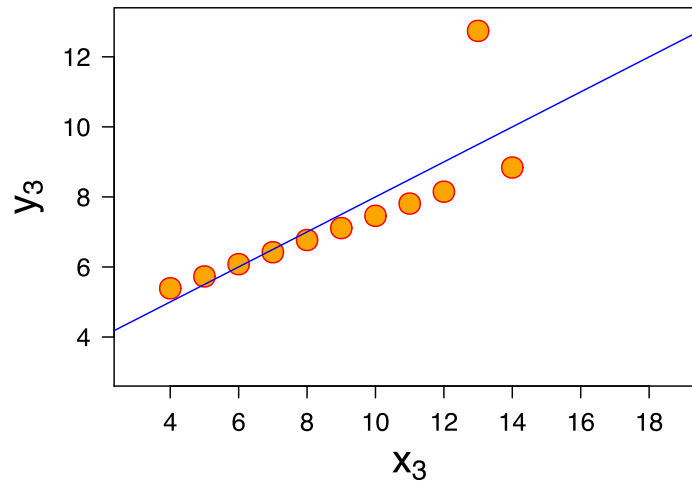
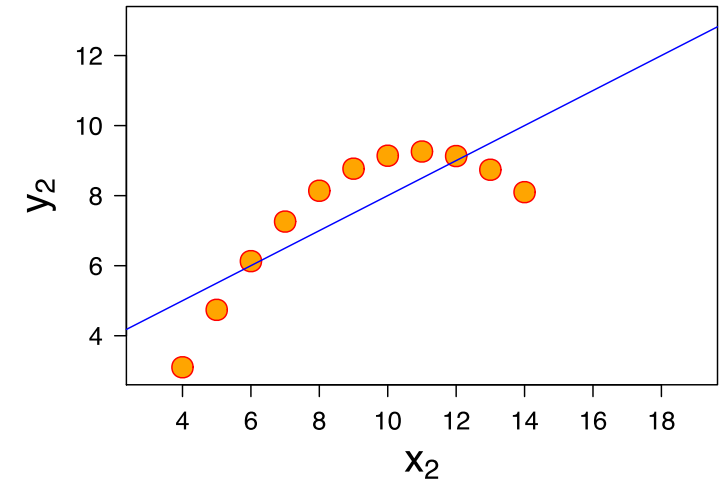
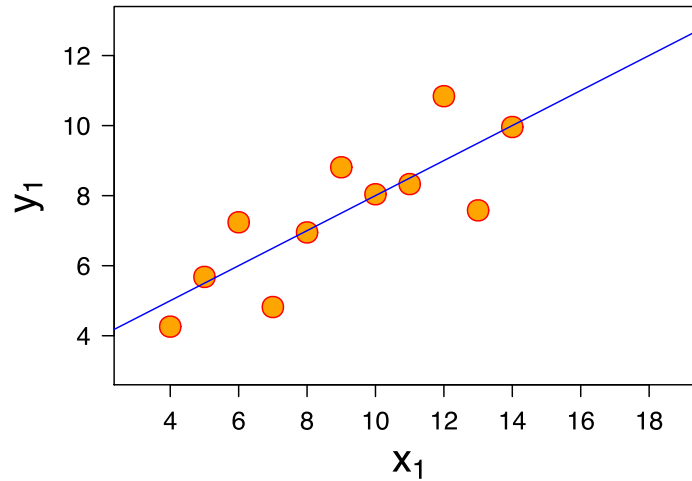
11 x-values

11 y-values

Same total

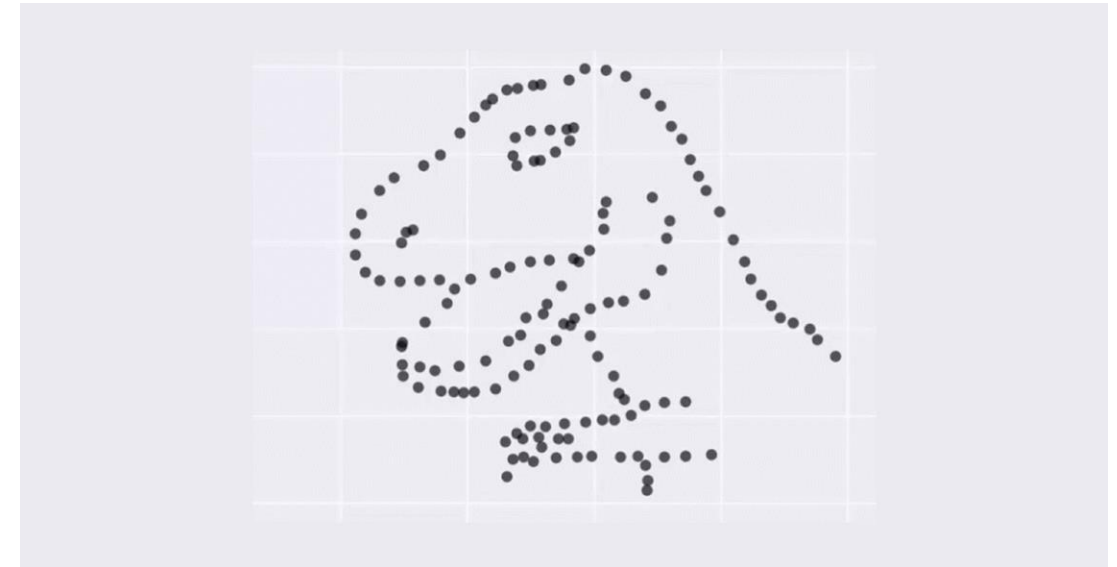
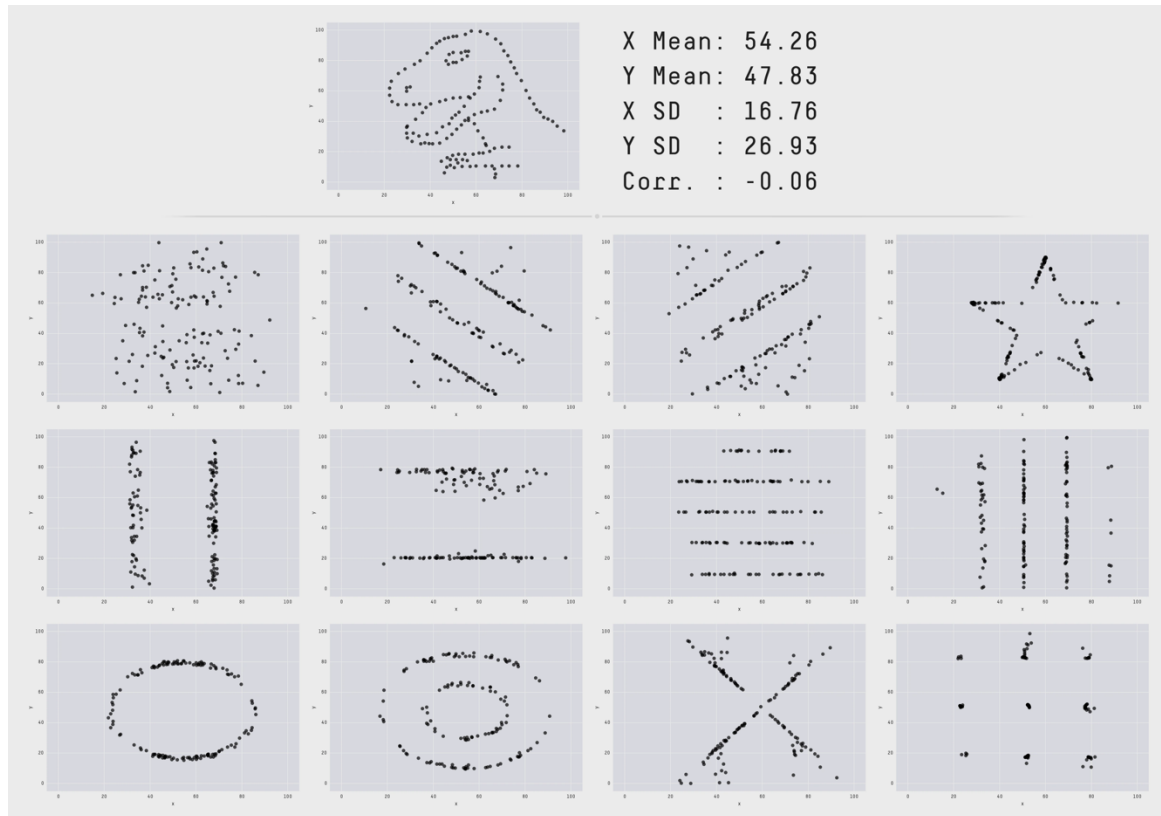
Same mean

Same standard deviation

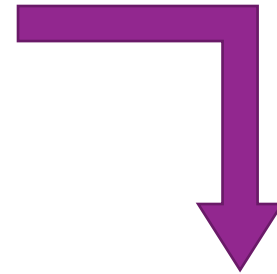
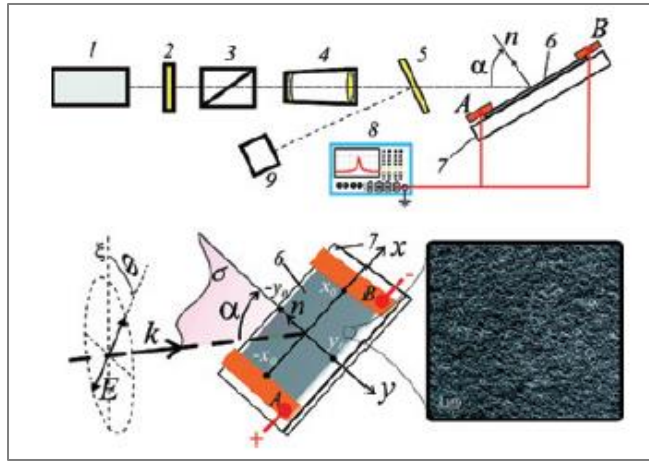


The Datasaurus Dozen 😊:

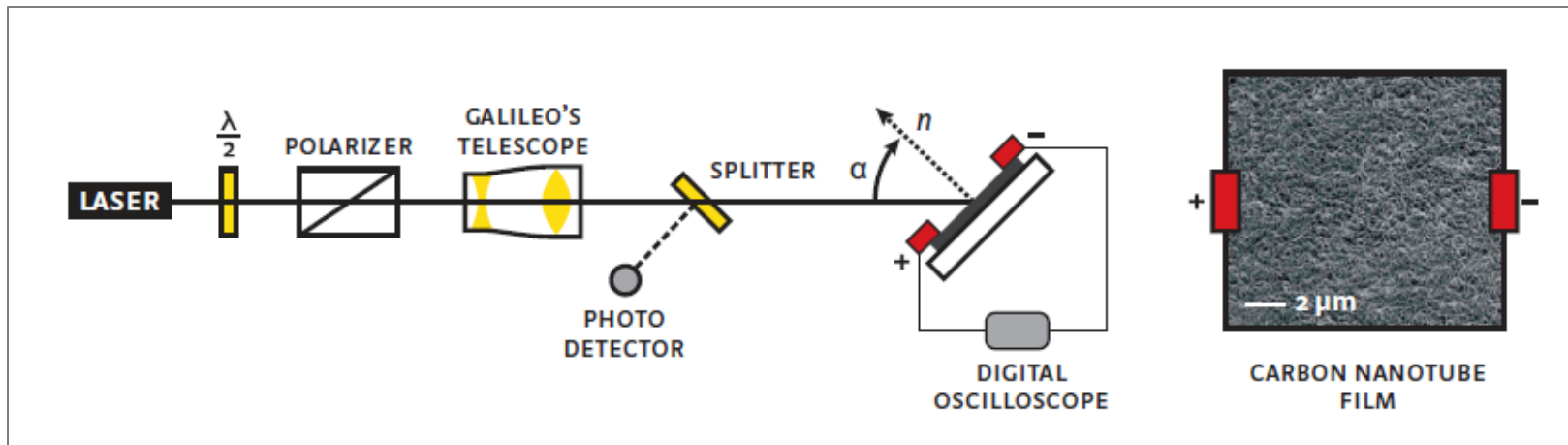
Never trust summary statistics alone!



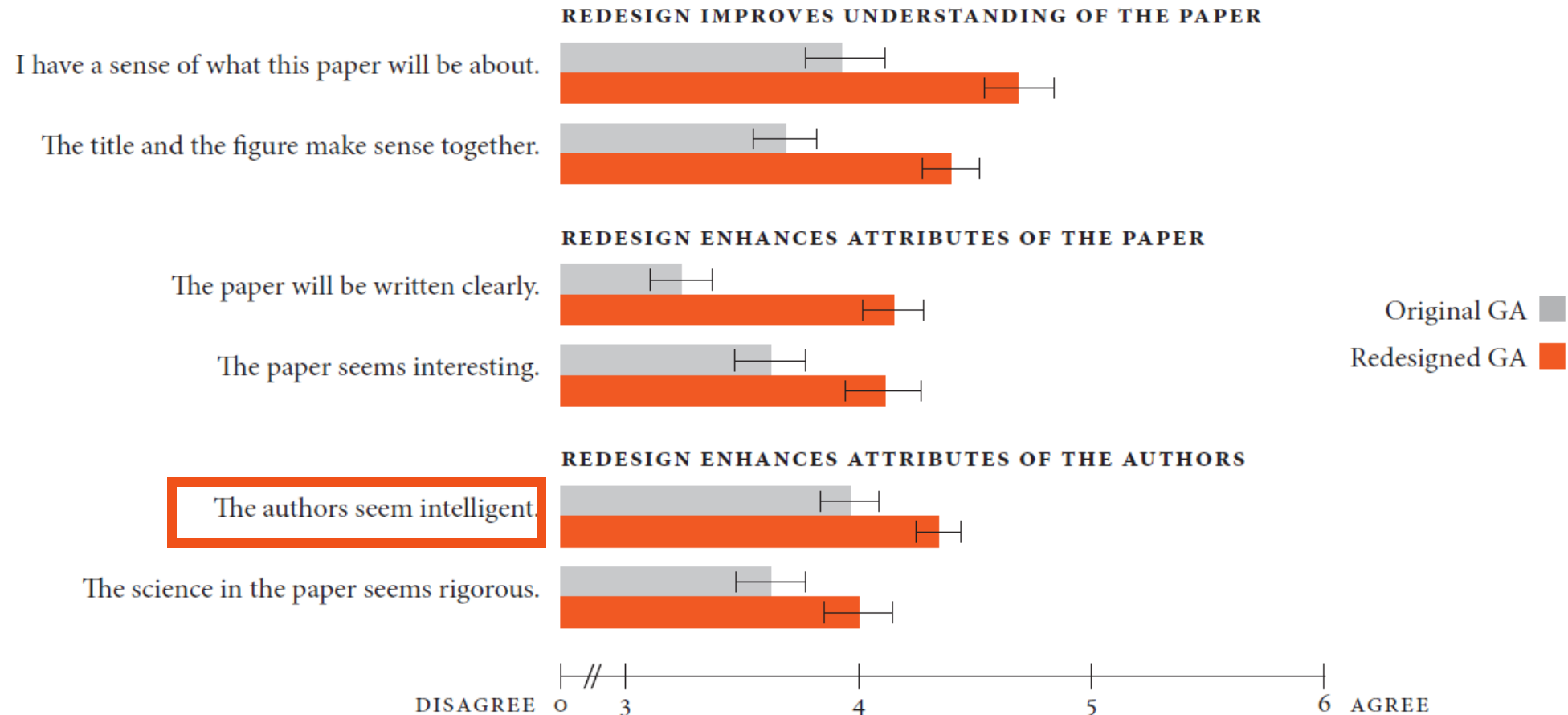
A good visual makes you (look) smarter!



10 graphical abstracts were redesigned based on visual design principles



A good visual makes you (look) smarter!



Why use visuals in academic research?



Provide **insight**

-
- ❑ Publications
 - ❑ Oral presentations
 - ❑ Poster presentations



Grab **attention**

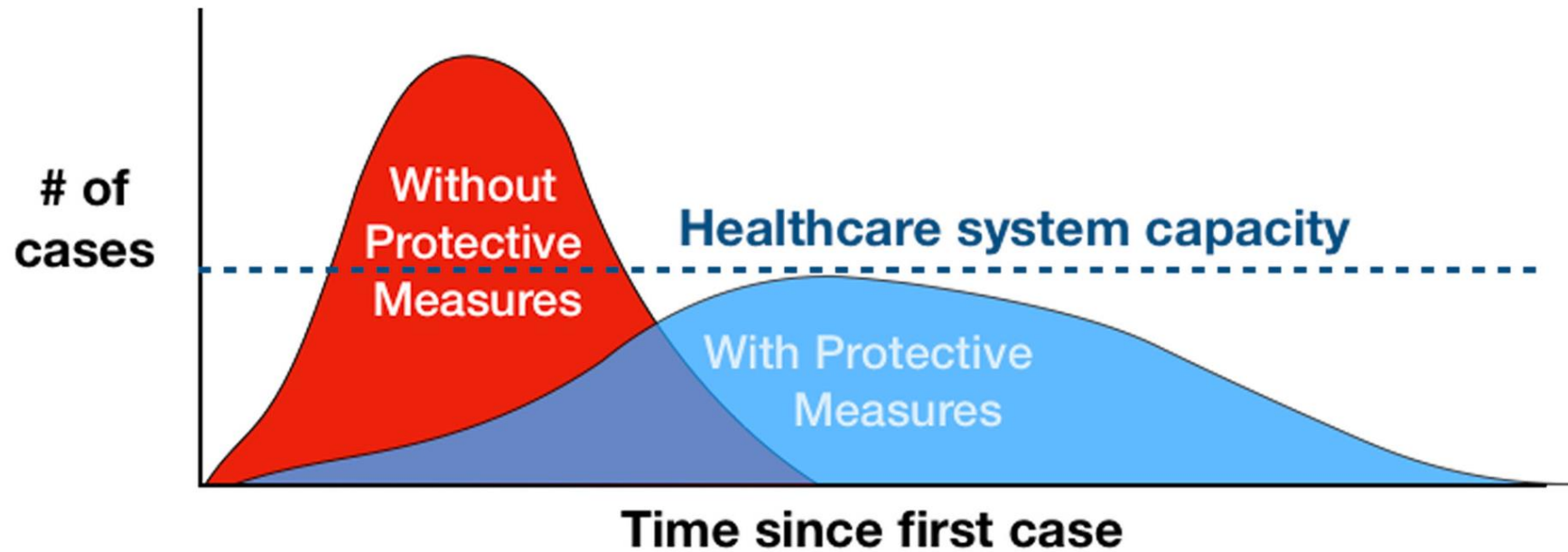
-
- ❑ Graphical abstracts
 - ❑ Social media
 - ❑ General media



Show **competence**

-
- ❑ Grant applications
 - ❑ Portfolio / CV
 - ❑ Pitch

Provide insight...



Adapted from CDC / The Economist

...grab attention...

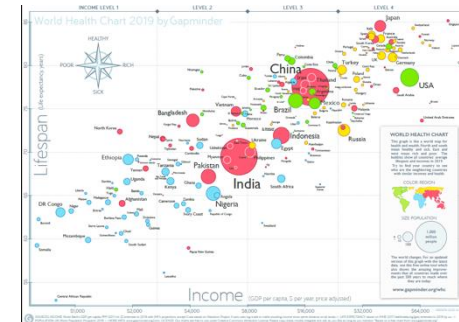
Talking about reaching a broad public!



Prof. Ed Hawkins
#showyourstripes



...and show your competence...



'The world cannot be understood without numbers. But the world cannot be understood with numbers alone.'
- Hans Rosling

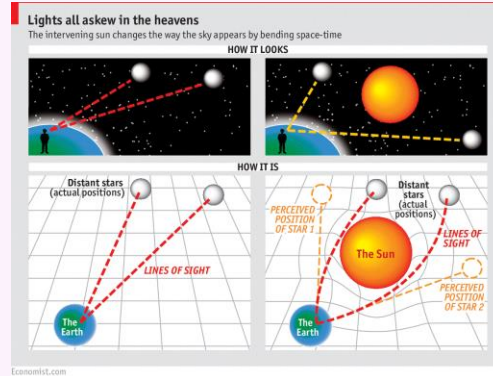
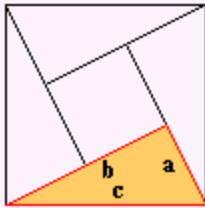
HOW visual communication?

Show the *right* data and show the data *right*.

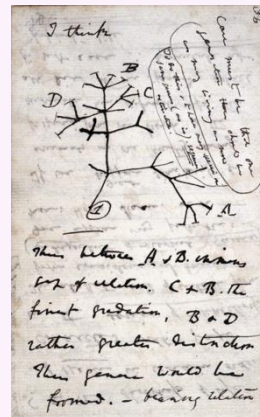
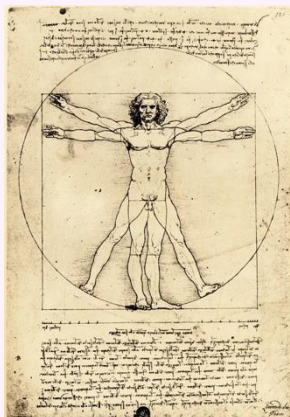
Four types of visualizations

Idea
illustration

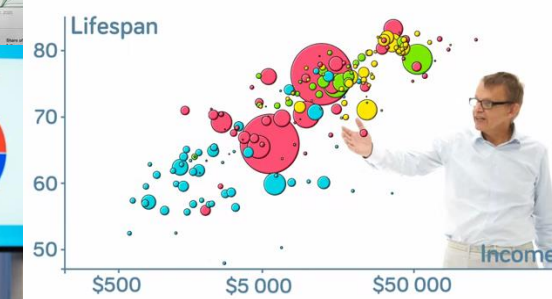
Idea
generation



CONCEPTUAL



EXPLANATORY



'Everyday'
DataViz

DATA-DRIVEN

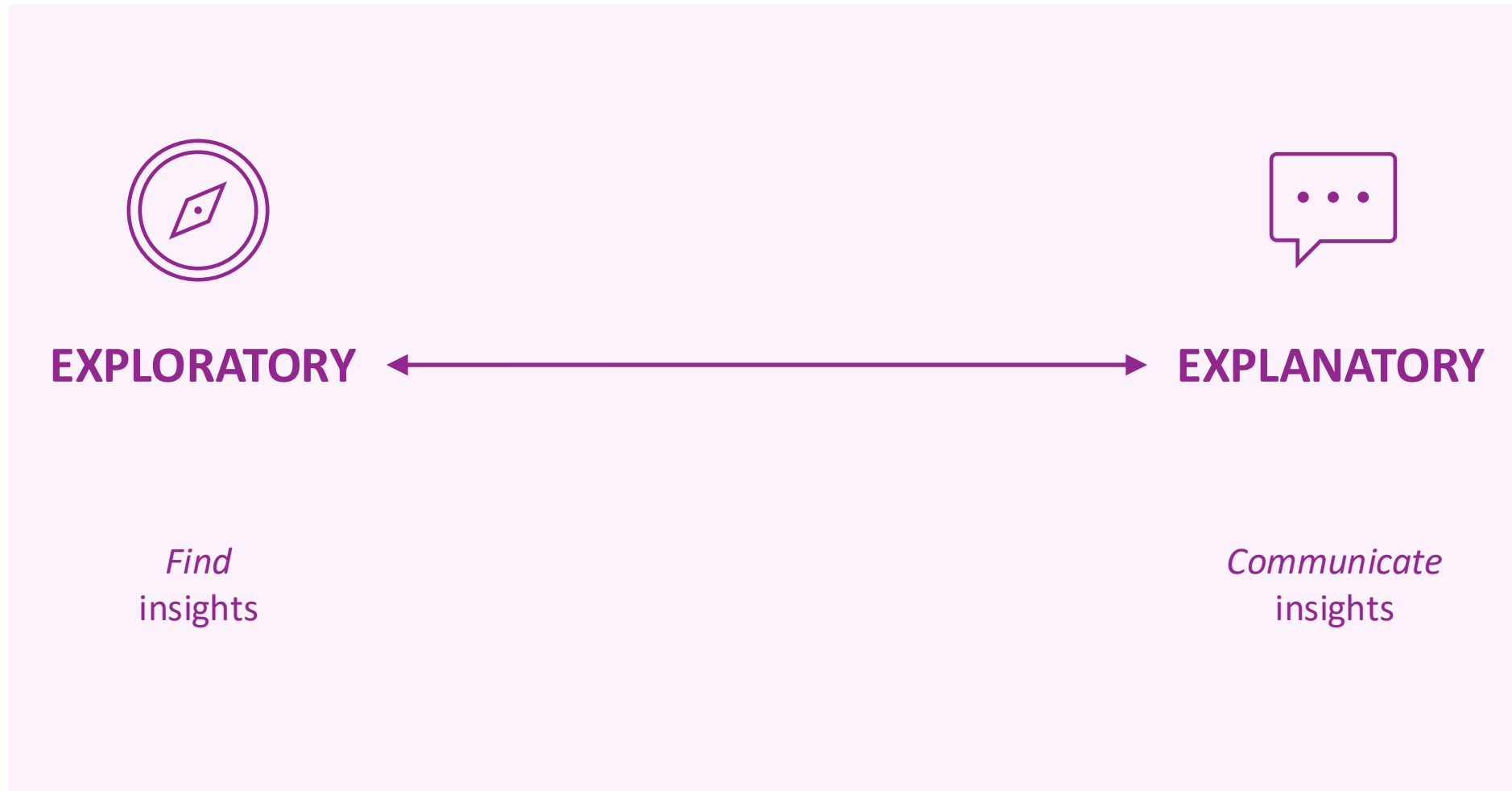


Visual
discovery

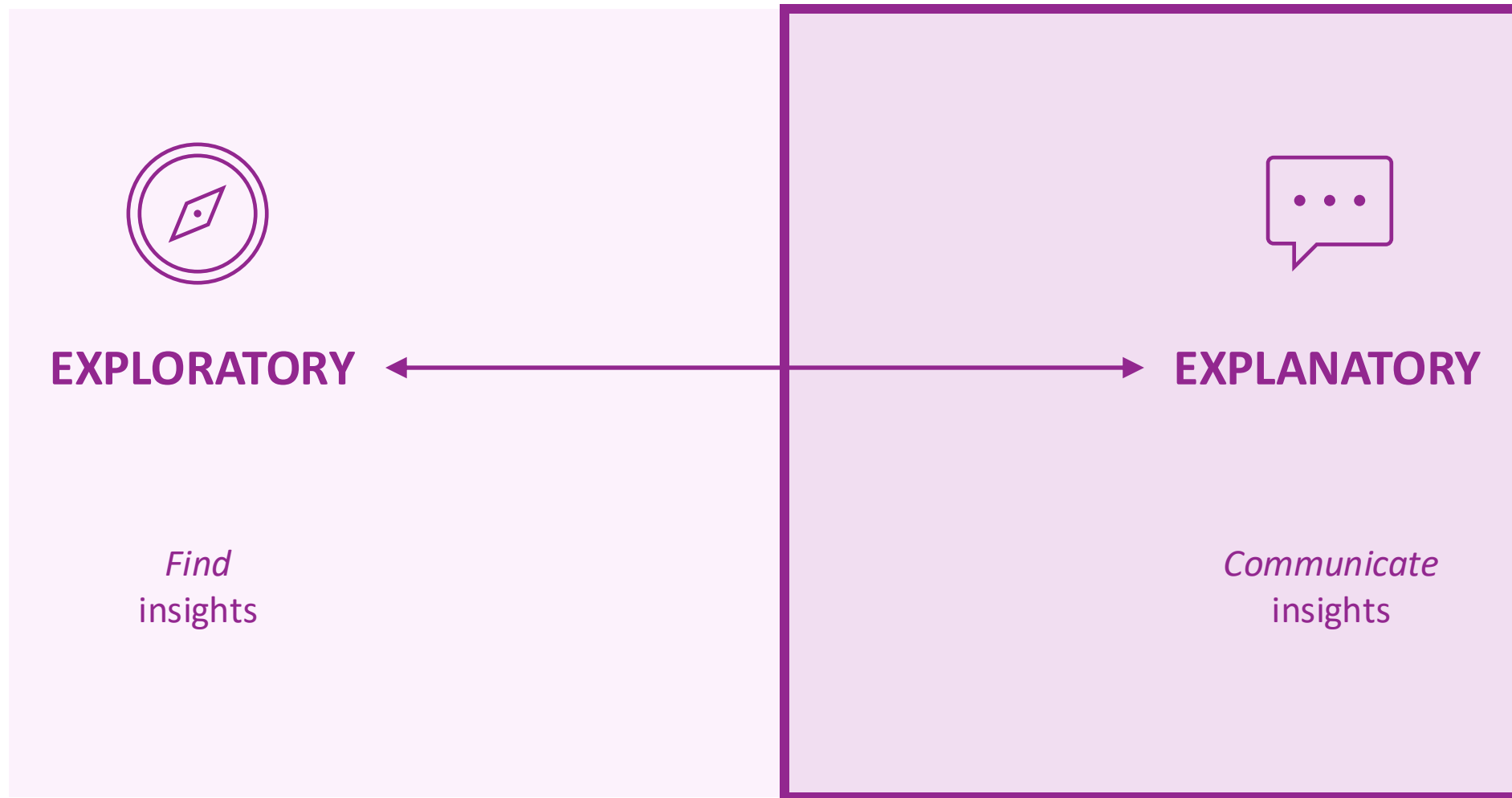
EXPLORATORY

sebastiz.github.io/

Exploratory vs. explanatory data visualization



We will focus on **explanatory** data visualization



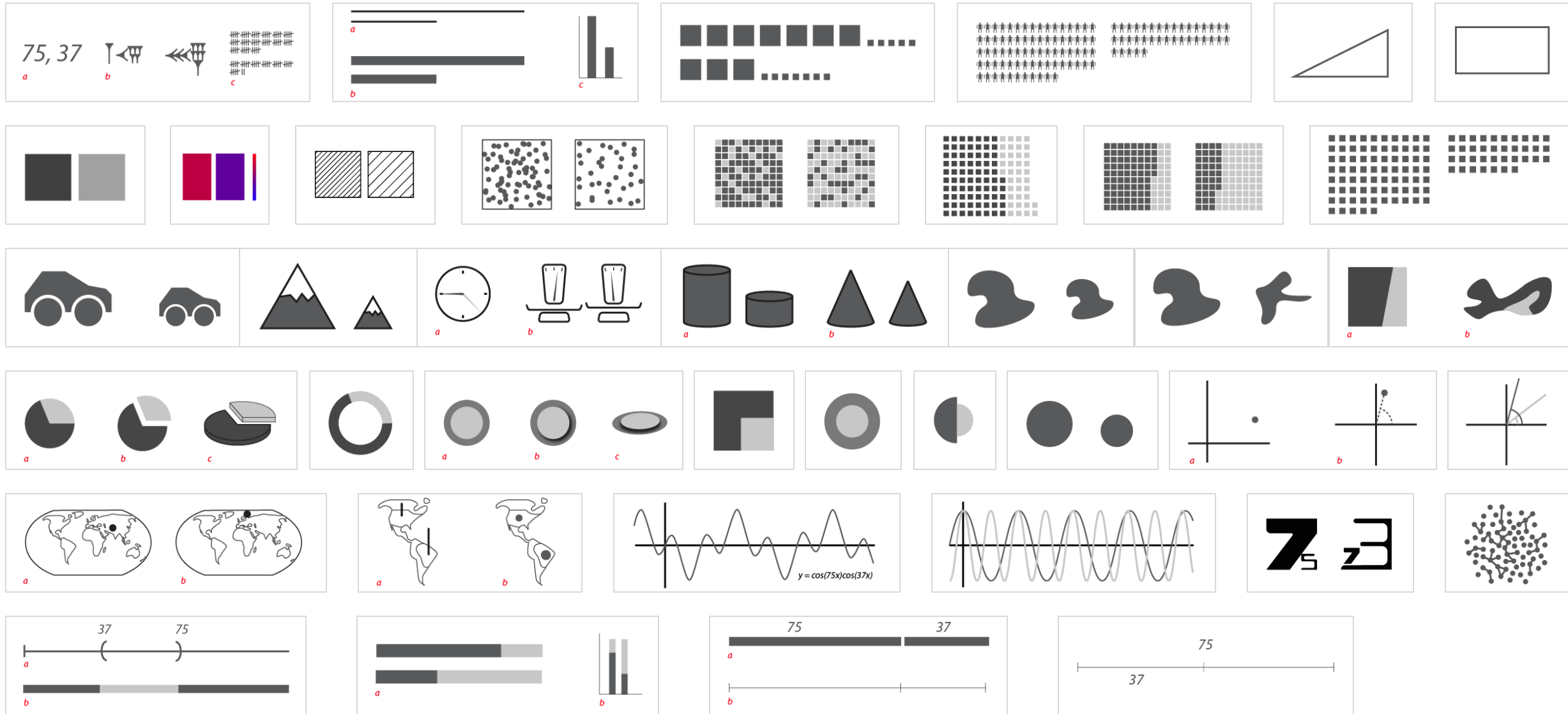
Exercise: In how many ways can you visualize two numbers?

75 and 37



5 min

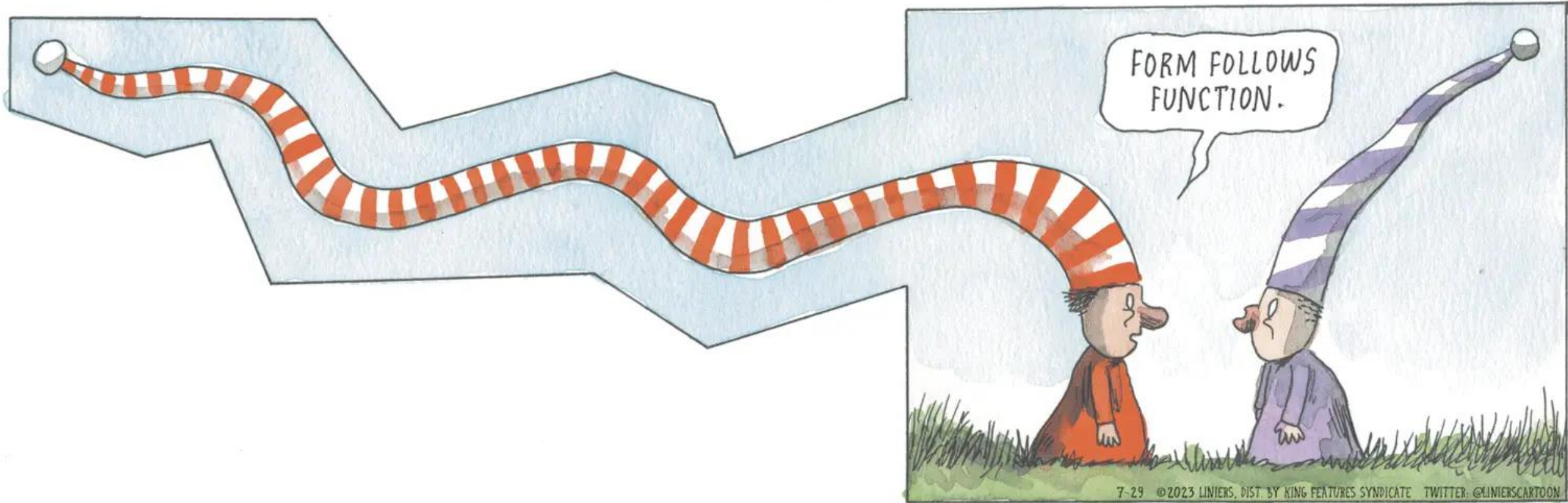
45 ways to communicate two quantities...



So many possibilities. How to choose?



Form follows function!

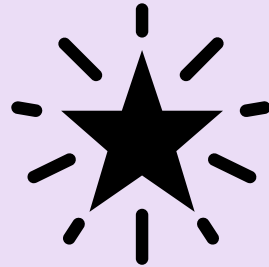


7-29 ©2023 LINIERS, DIST. BY KING FEATURES SYNDICATE TWITTER: @LINIERSCAROON

<https://comicskingdom.com/macanudo/2023-07-29>

Begin with the end in mind

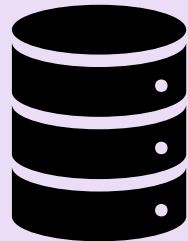
Why?



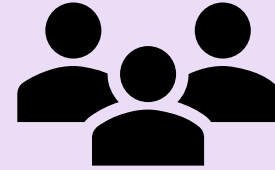
Outcome



Data



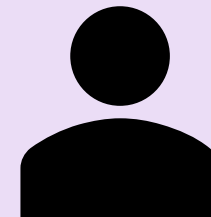
What?



Audience



You



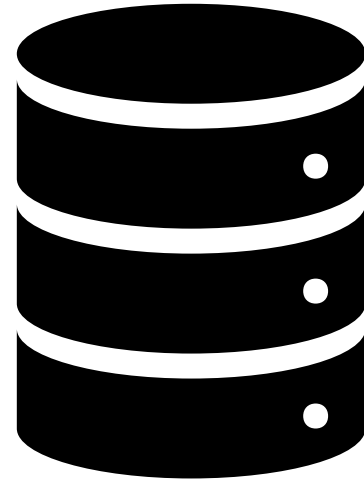
Who?



Where?

What?

Data



What do you want to show?



Comparison



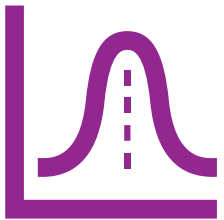
Trend



Part of a whole



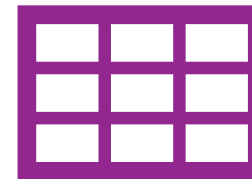
Relationship



Distribution



Location



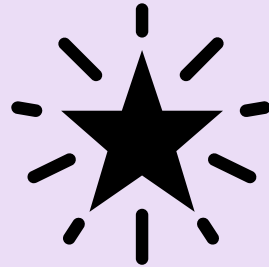
Exact values

What defines the chart type

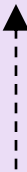


Begin with the end in mind

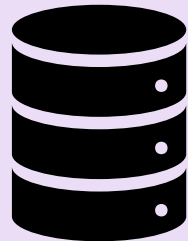
Why?



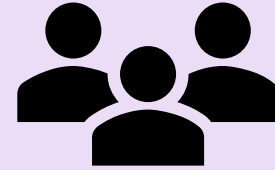
Outcome



Data



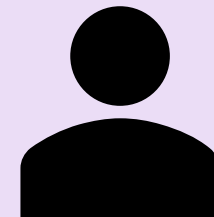
✓
What?



Audience



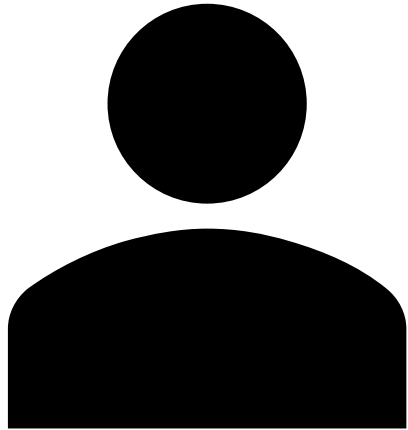
You



Who?

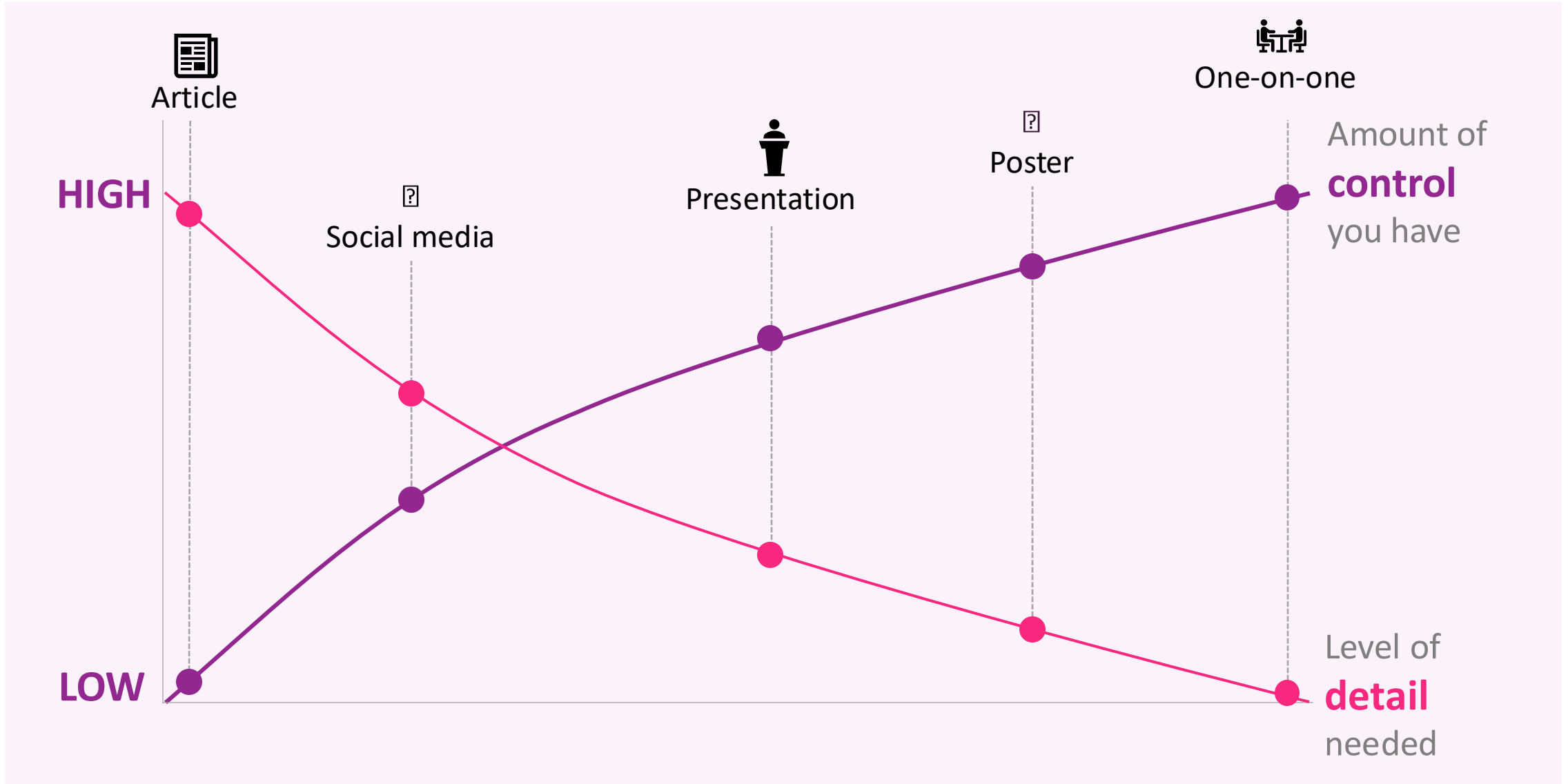
Where?

You

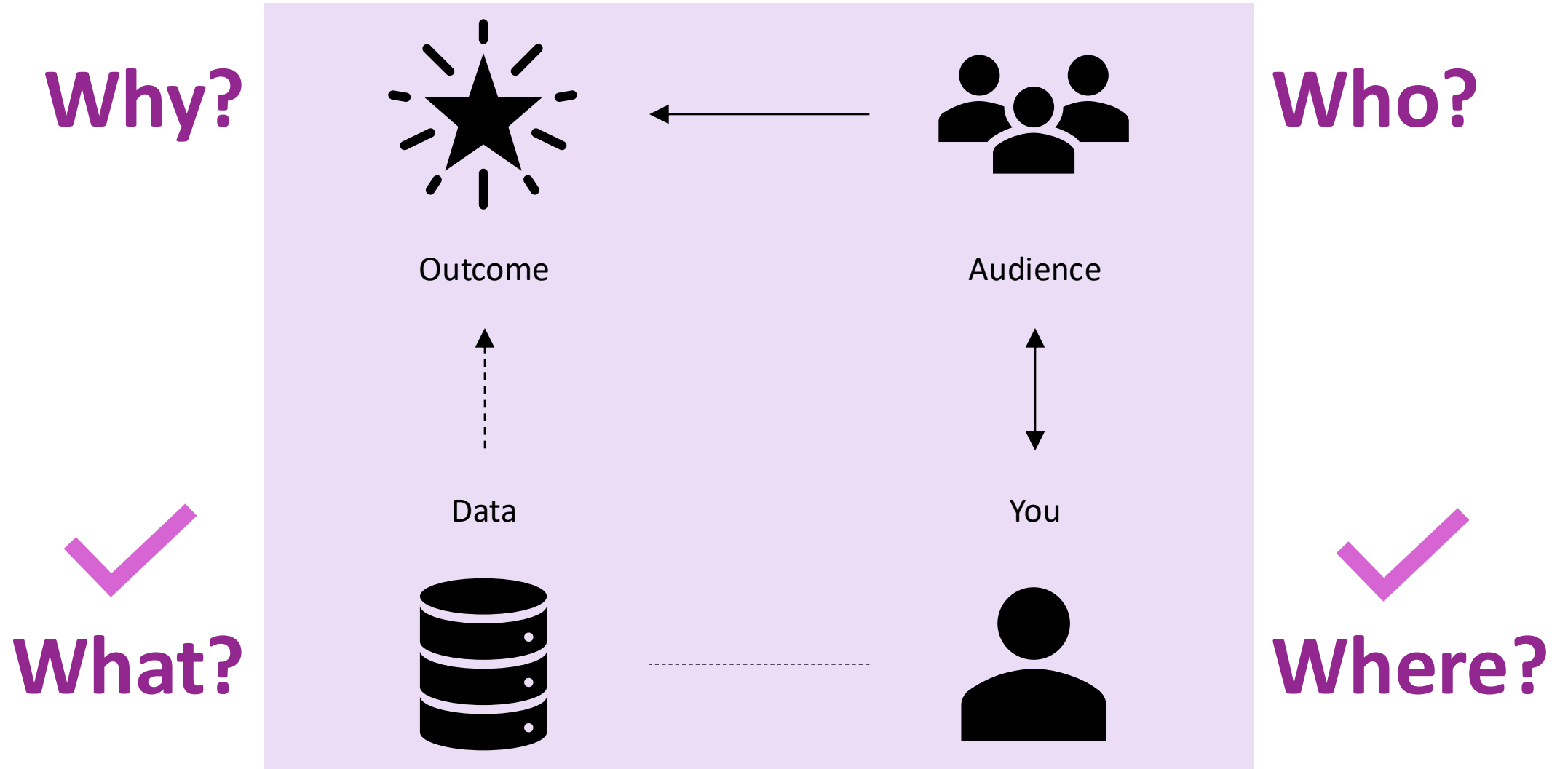


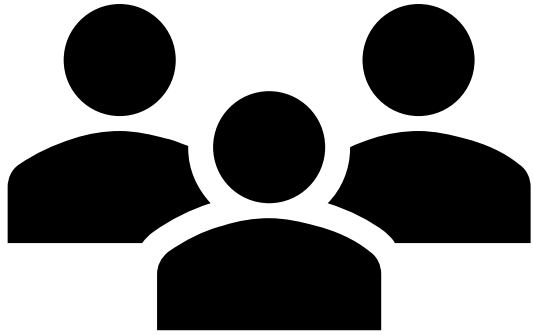
Where?

Where you present defines *how* you present



Begin with the end in mind

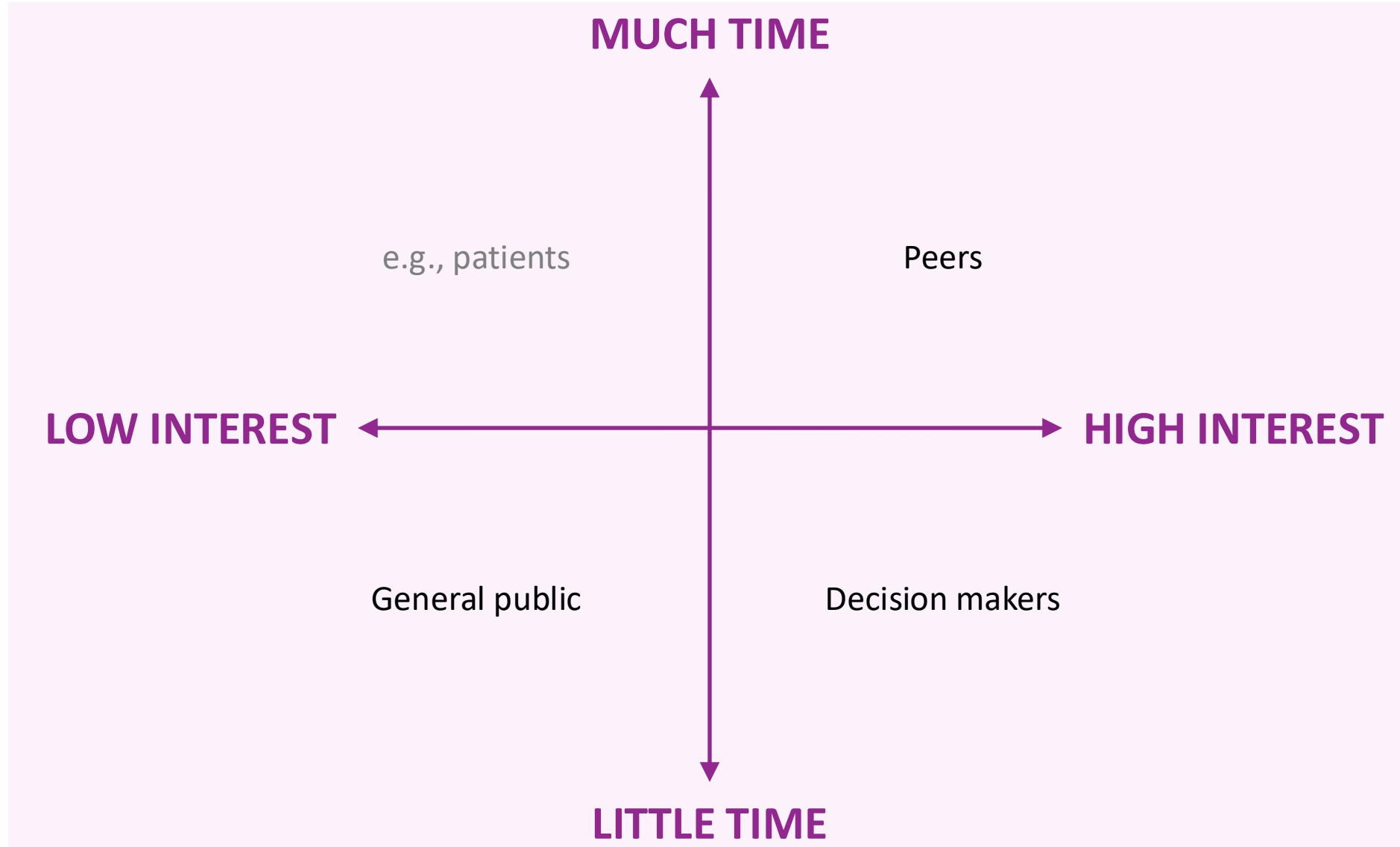




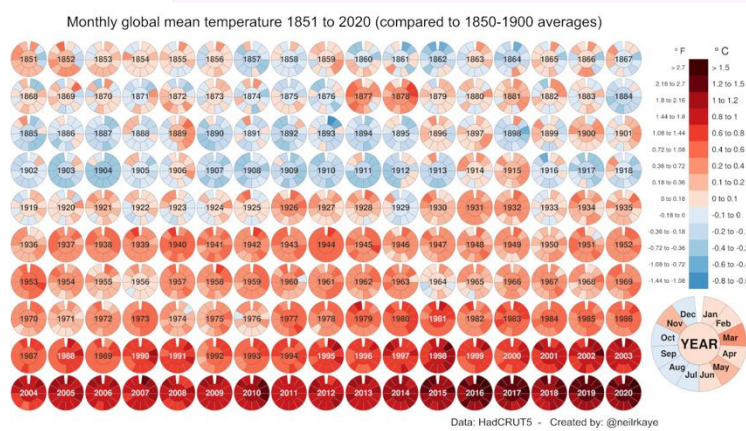
Audience

Who?

Who is your audience?

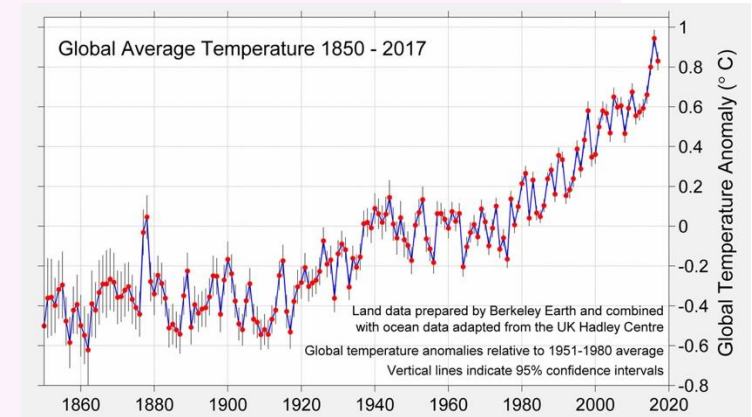


Who defines the level of detail and accuracy



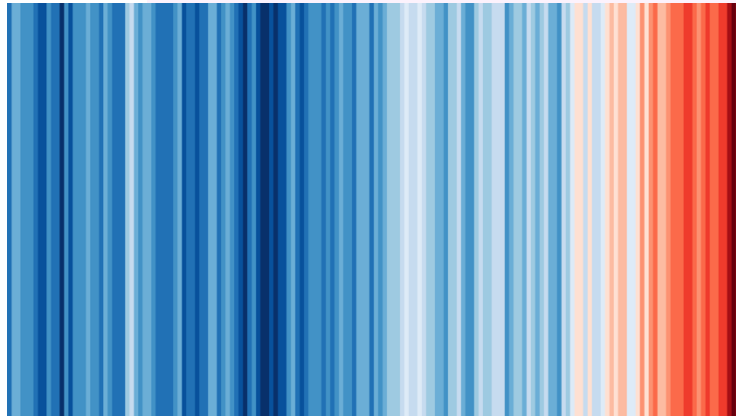
MUCH TIME

Detailed



LOW INTEREST

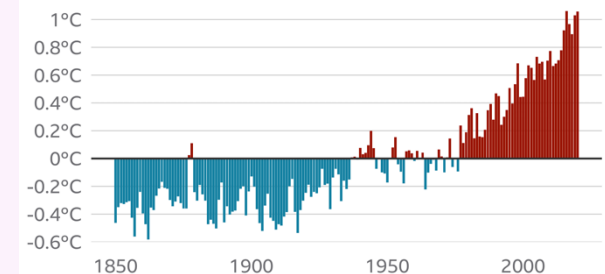
Engaging



HIGH INTEREST

Accurate

The world is getting warmer
Annual mean land and ocean temperature above or below average, 1850 to 2020



Note: Average calculated from 1951 to 1980 data

Source: University of California Berkeley

BBC

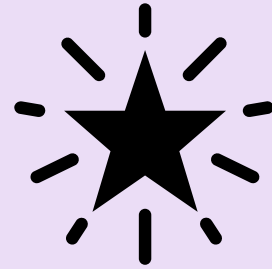
Focused

LITTLE TIME

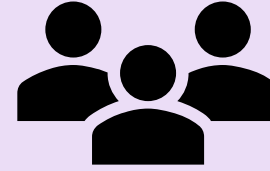
Begin with the end in mind

Why?

Who?



Outcome



Audience



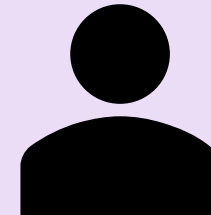
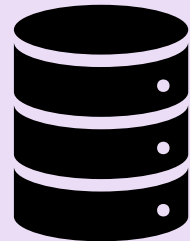
Data



You



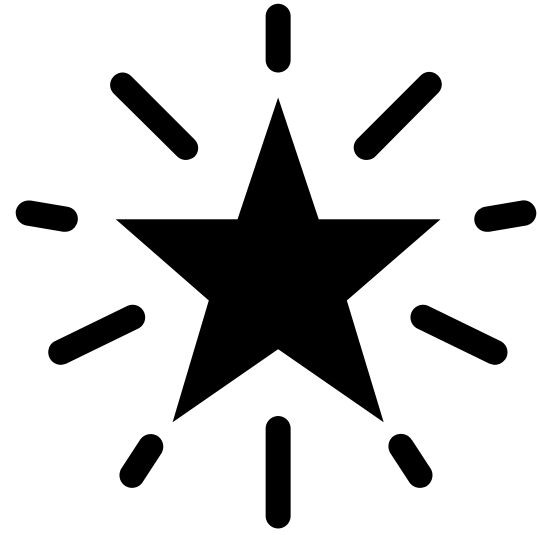
What?



Where?

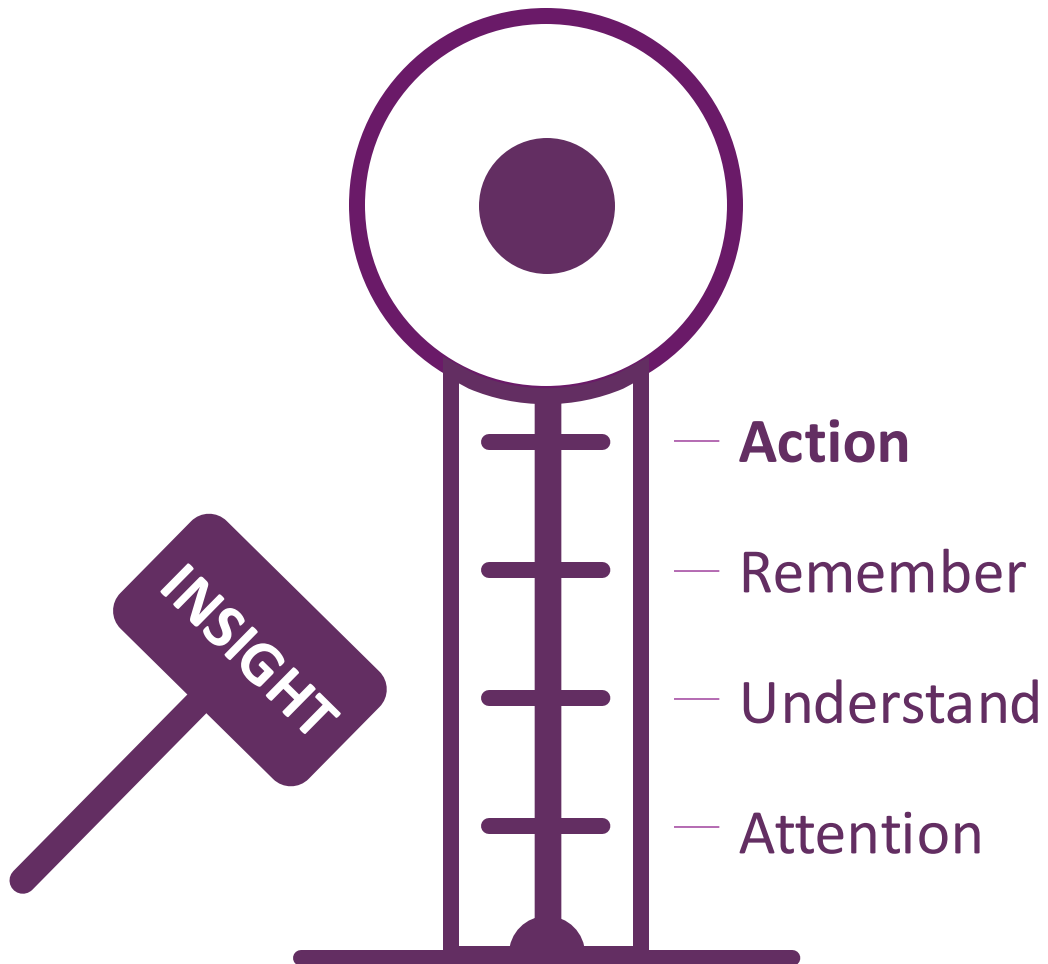


Why?

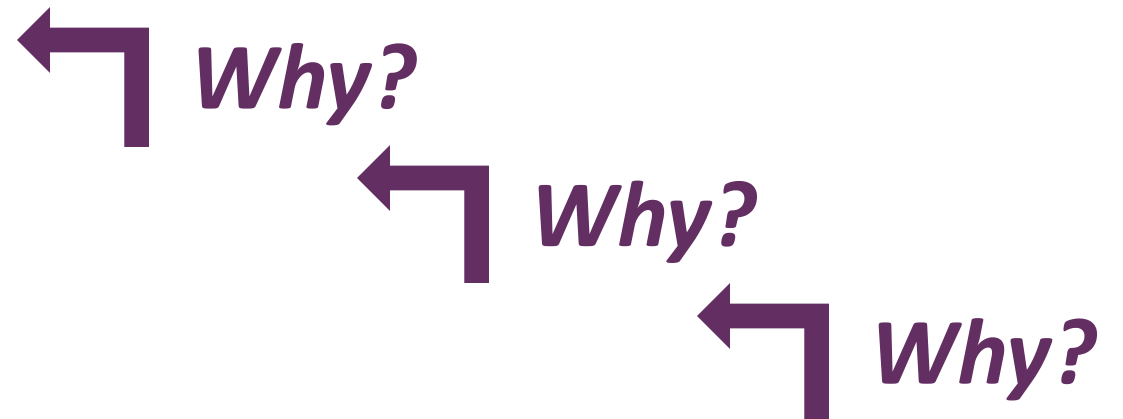


Outcome

Why should your audience care?



What is your **purpose**?
What **action** do you want your audience to take?



What do you need your audience to know or do or never forget?

accept | agree | begin | believe | change | collaborate | commence | create | defend | desire | differentiate | do | emphasize | empower | encourage | engage | establish | examine | facilitate | familiarize | form | implement | include | influence | invest | invigorate | know | learn | like | persuade | plan | promote | pursue | recommend | receive | remember | report | respond | secure | simplify | start | try | understand | validate

- Define your **BIG IDEA**:
 - articulate **your unique point of view**
 - convey **what's at stake**

Exercise: Set the context!



In groups: Share your visual and discuss:

What do you want to show?

Where will you present this visual

Who is your audience?

Why should your audience care?

- *What is the amount of control that you have (and level of detail you need)?*
- *What are they like? Why are they here? How can you solve their problem?*
- **What is your call-to-action?**
- **What is your BIG IDEA?**

What is visual communication?

A look at your work

Exercise: Set the context!



What do you need your audience to know or do or never forget?

accept | agree | begin | believe | change | collaborate | commence | create | defend | desire | differentiate | do | emphasize | empower | encourage | engage | establish | examine | facilitate | familiarize | form | implement | include | influence | invest | invigorate | know | learn | like | persuade | plan | promote | pursue | recommend | receive | remember | report | respond | secure | simplify | start | try | understand | validate

➤ Define your **BIG IDEA**:

- articulate **your** unique point of view
- convey **what's at stake**

Inspired by 990 ideas, Erik Hollnagel's 990, 9/10/10

In groups: Share your visual and discuss:

What do you want to show?

Where will you present this visual

Who is your audience?

Why should your audience care?

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- *What are they like? Why are they here? How can you solve their problem?*
- **What is your call-to-action?**
- **What is your BIG IDEA?**

Part 2: **Visual Design**

“It’s not what you look at that matters, it’s what you see” – Henry David Thoreau

Let's warm-up after lunch 😊

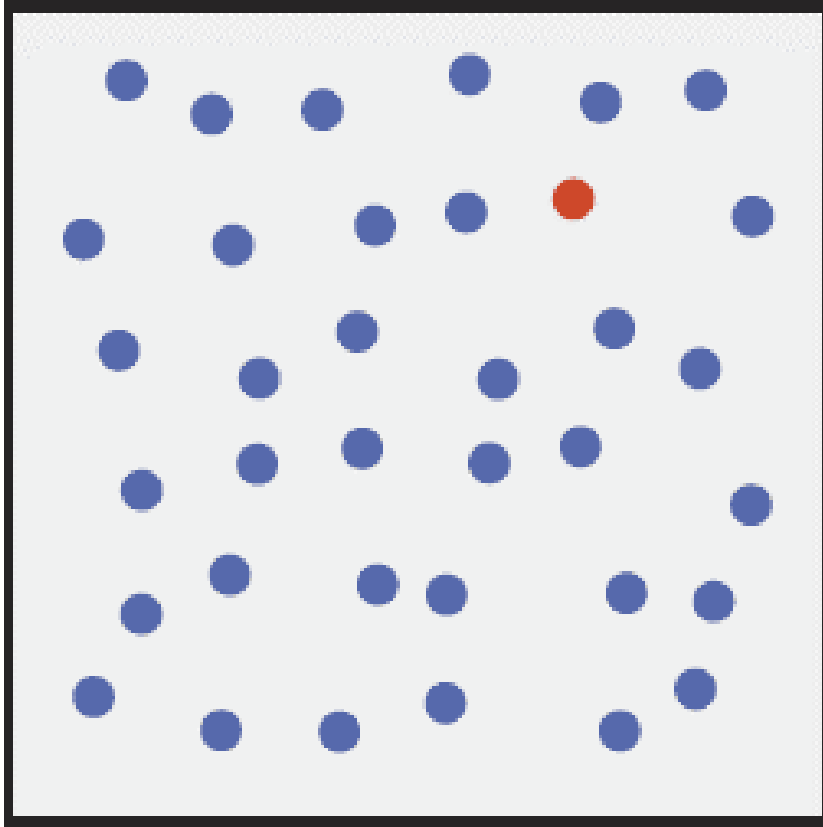
LEFT?

YES?

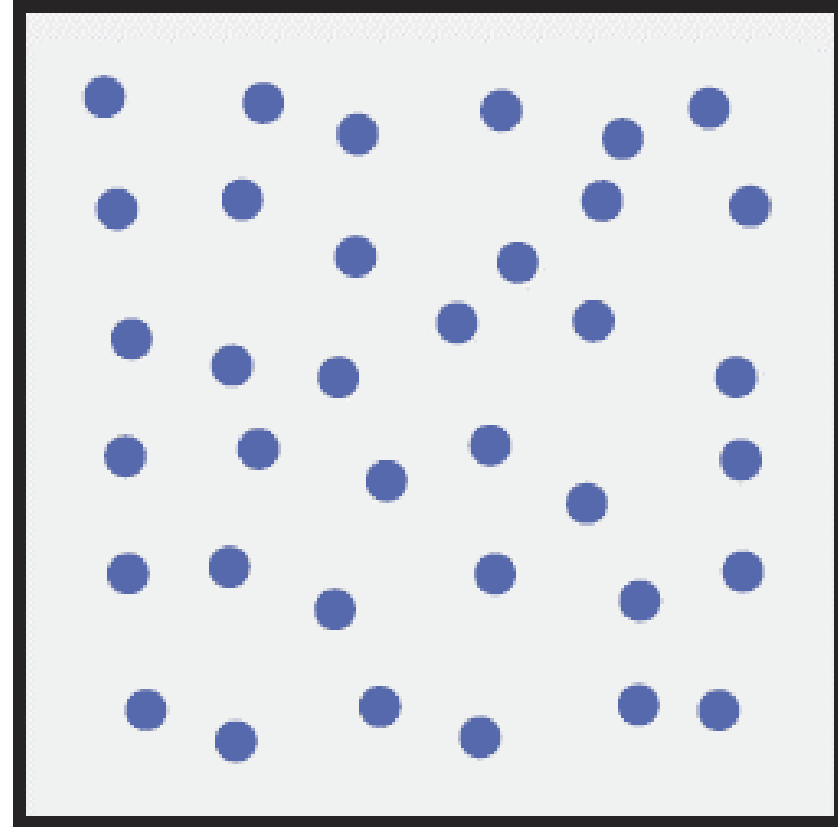


Which panel has a red dot?

A



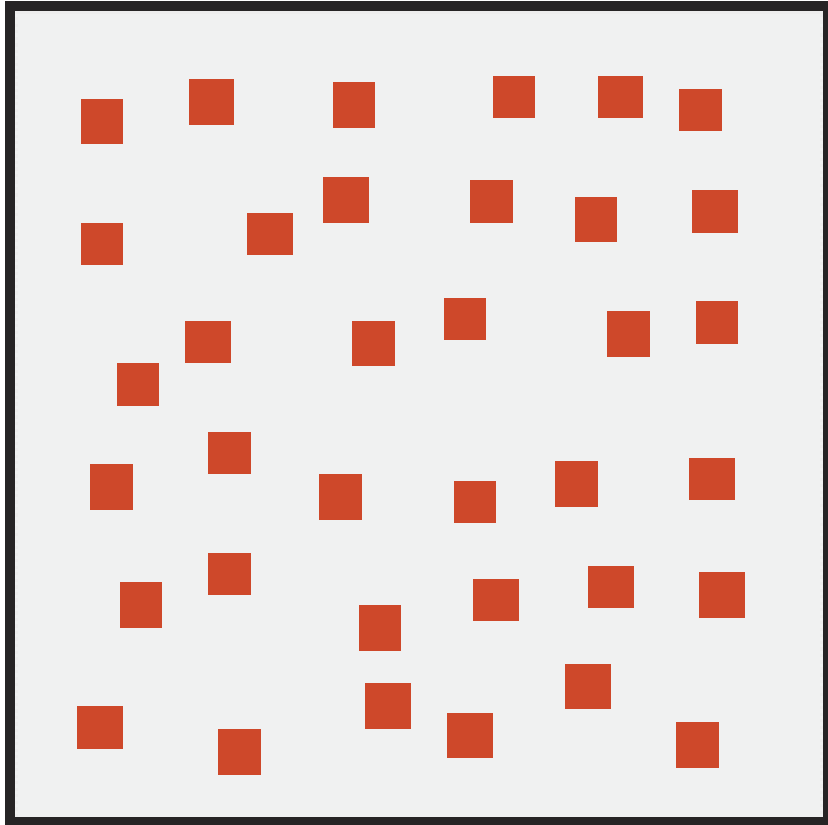
B



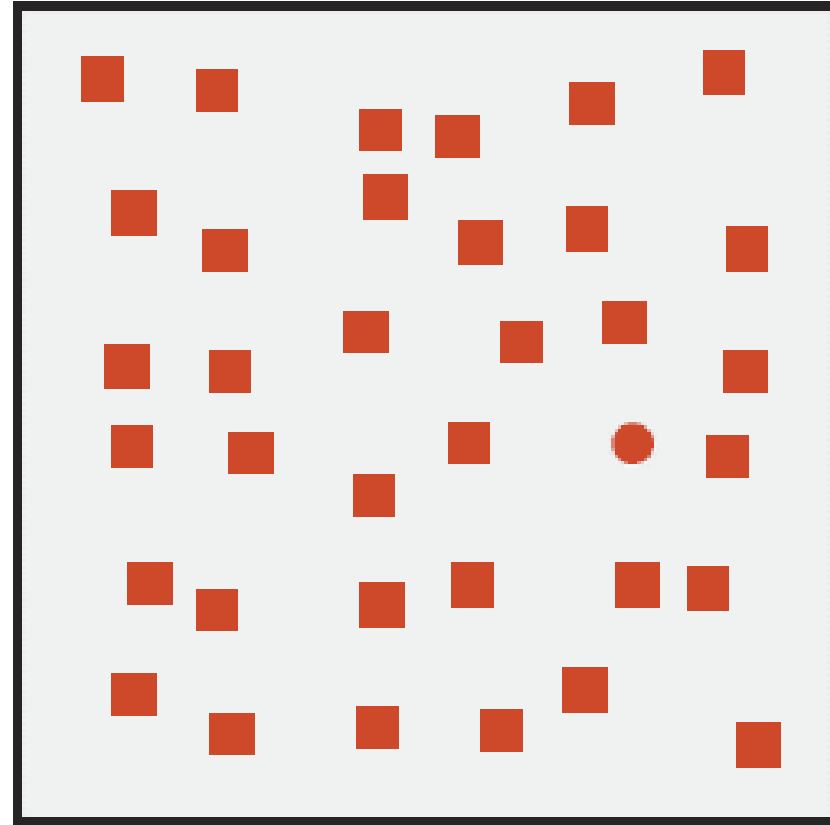
LEFT? = **UP!**

Which panel has a red dot?

A



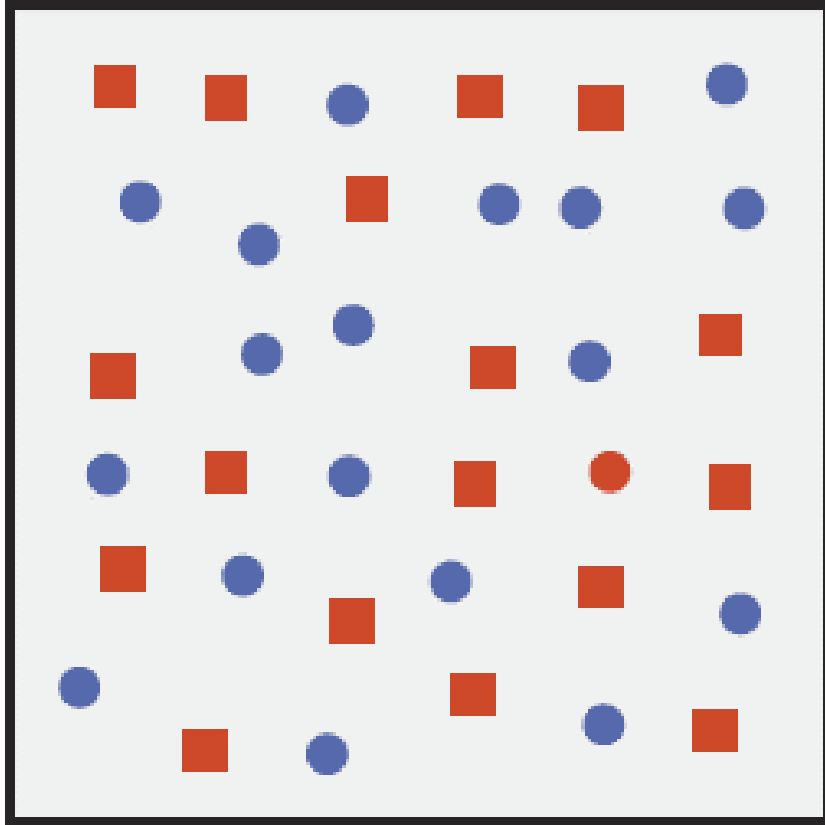
B



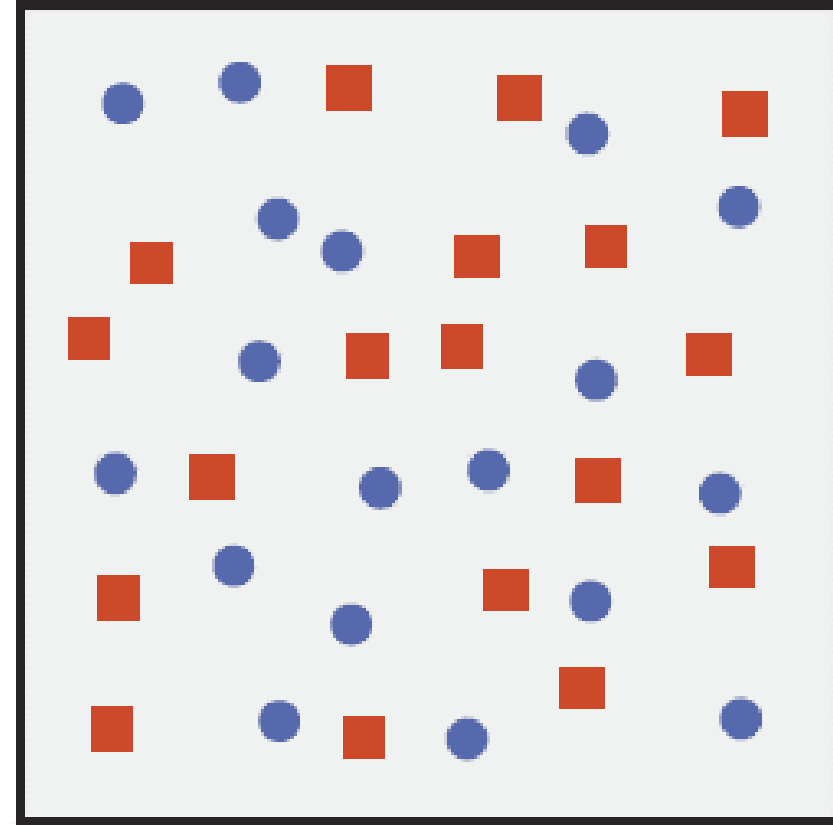
LEFT? = UP!

Which panel has a red dot?

A

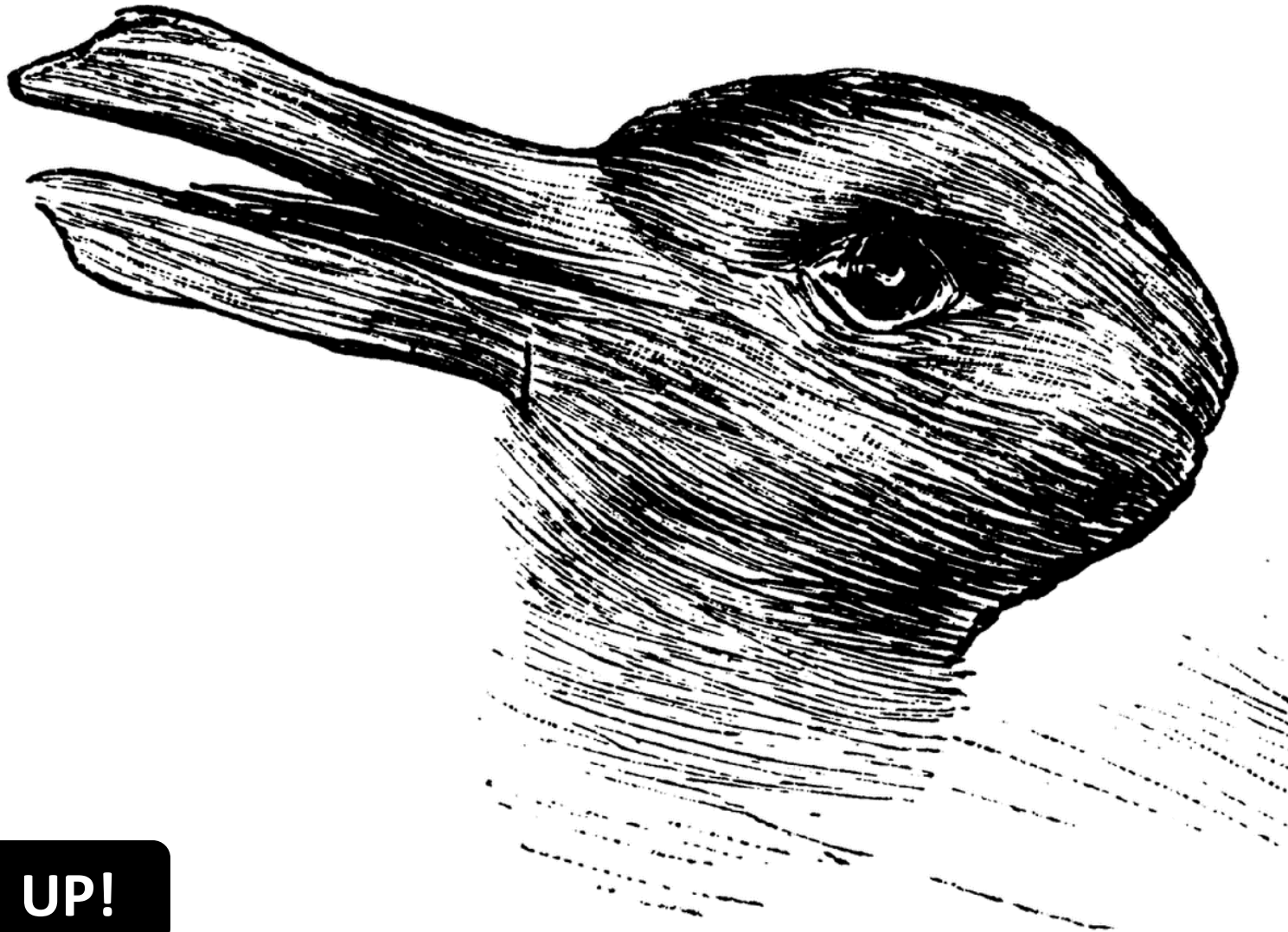


B



LEFT? = UP!

Do you see a Rabbit?

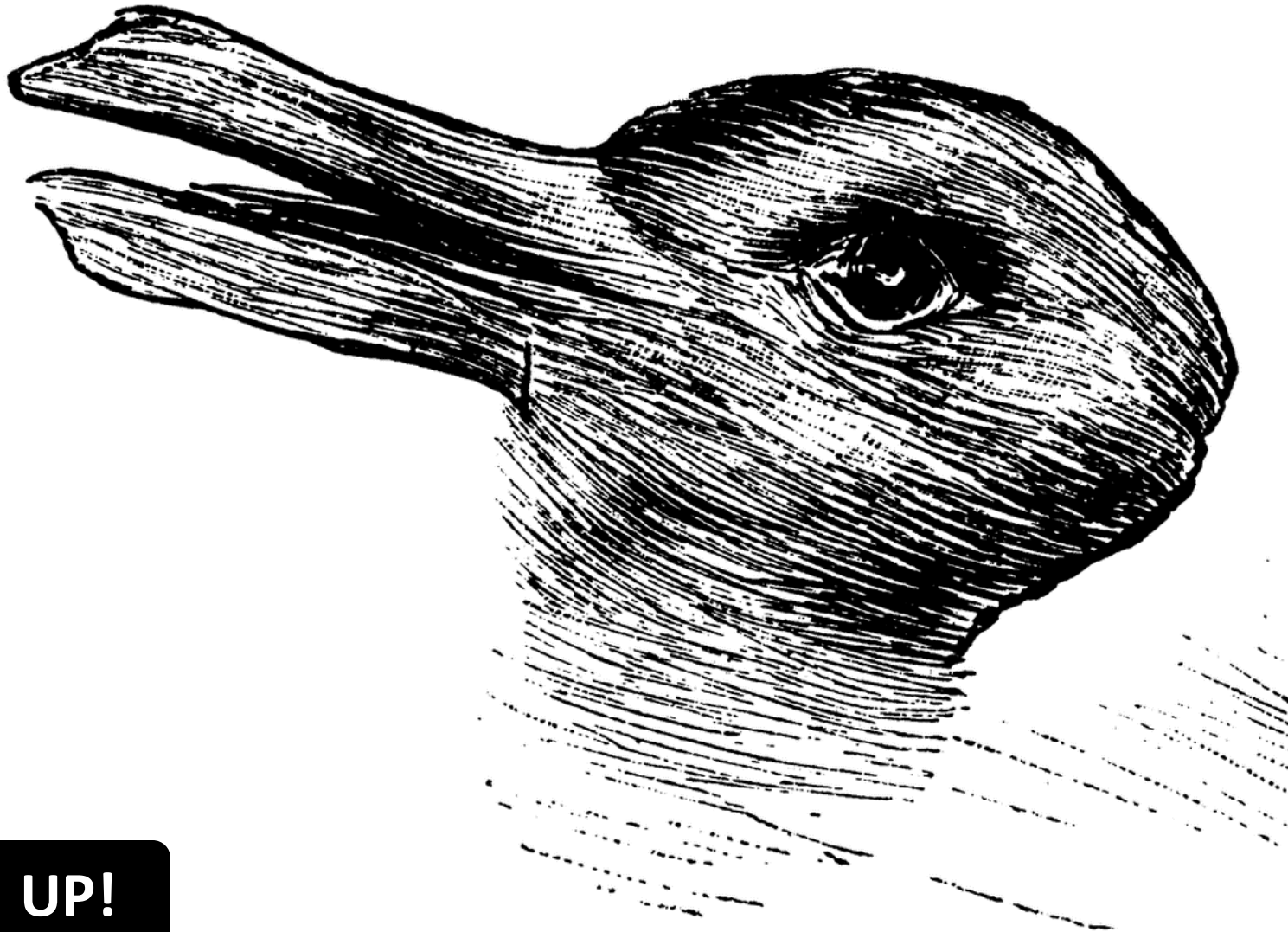


YES?

=

UP!

Do you see a Duck?

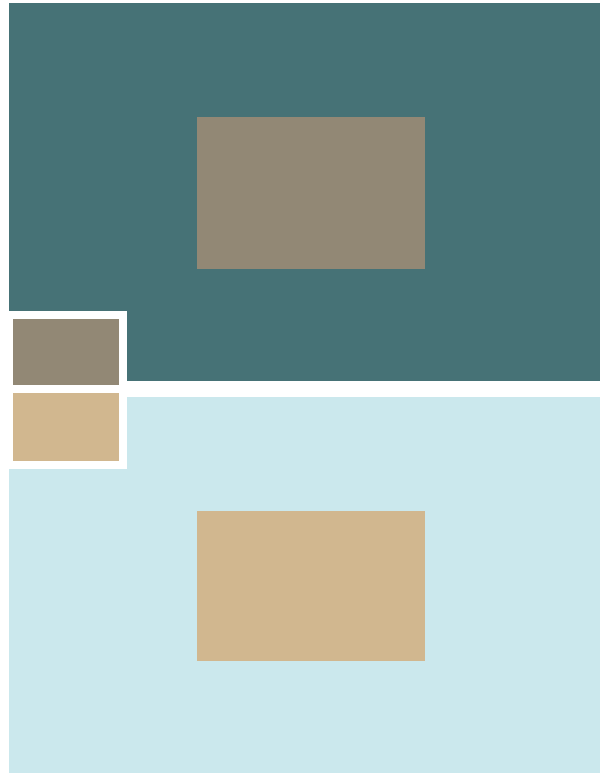


YES?

=

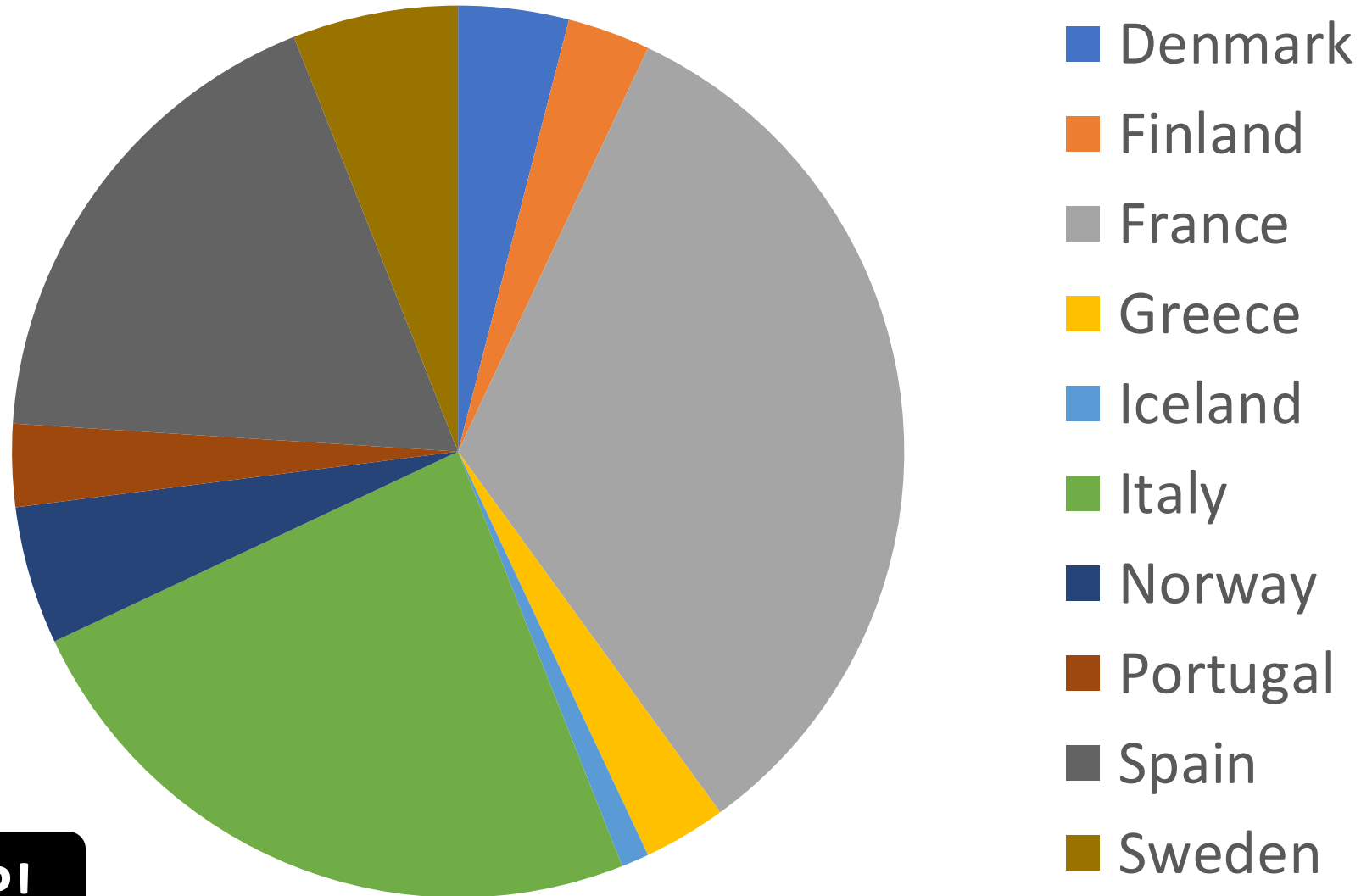
UP!

The inner squares have the same color



YES? = UP!

Norway has more widgets than Finland

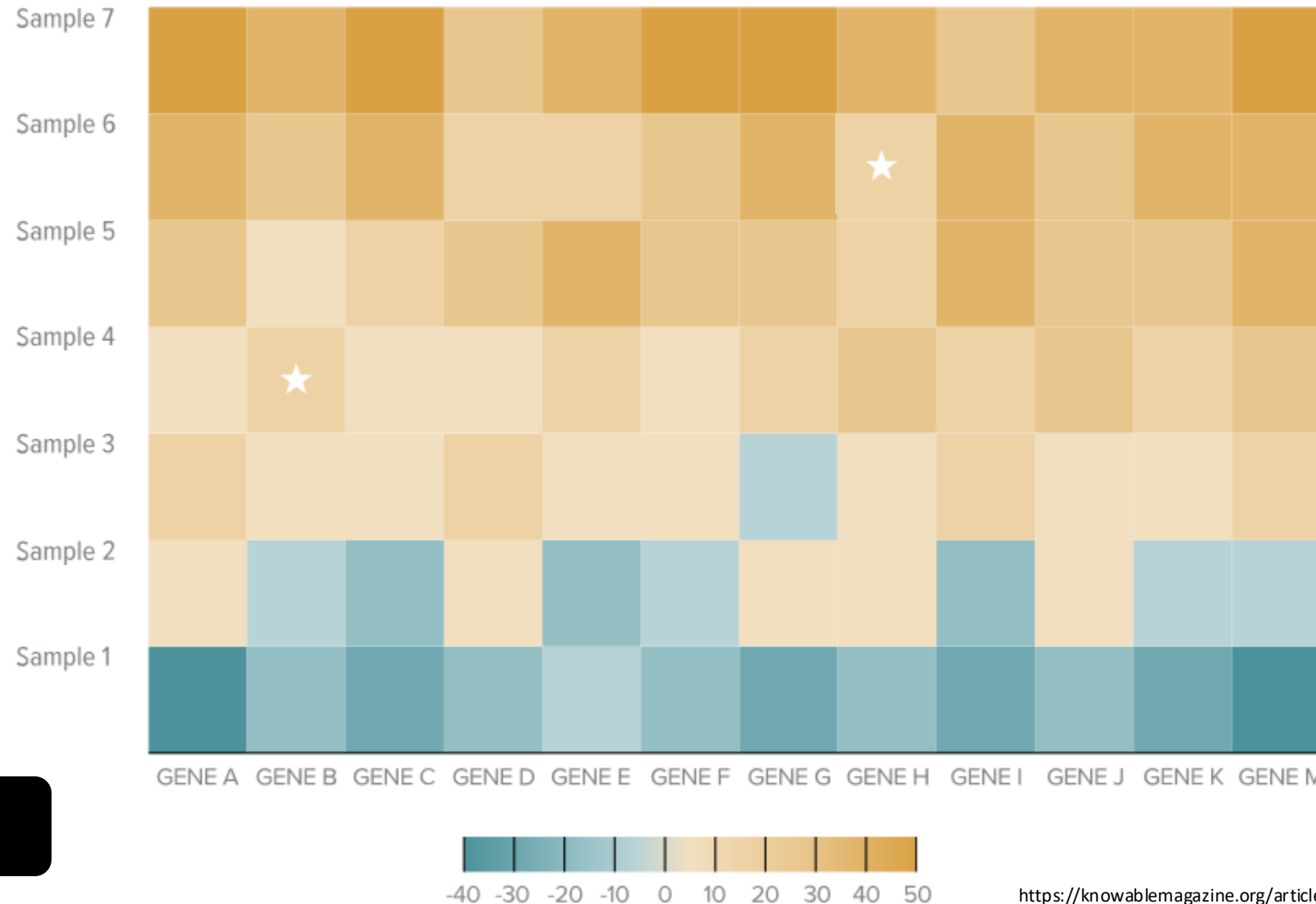


YES?

=

UP!

The areas with a star have the same color

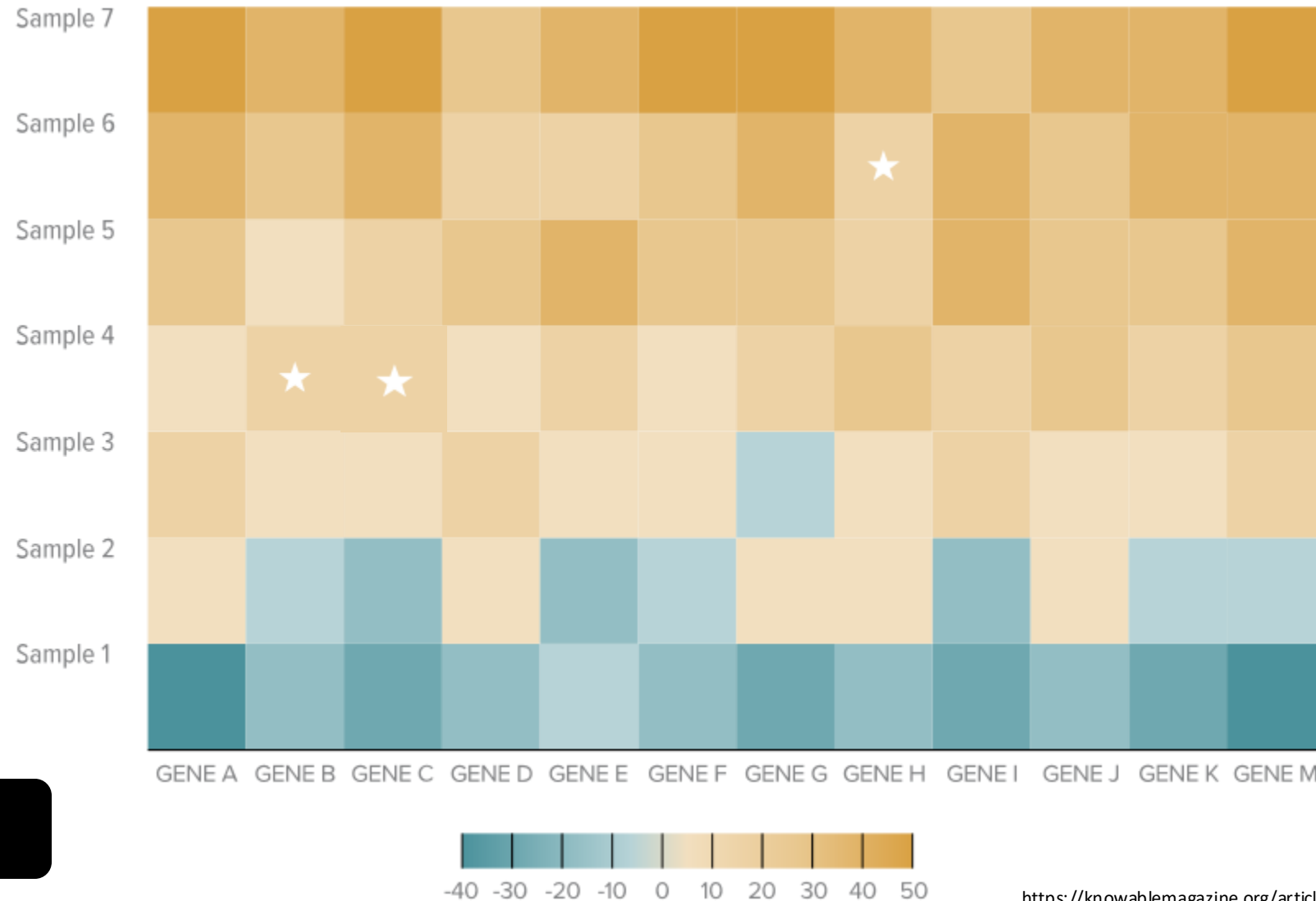


YES?

=

UP!

The areas with a star have the same color



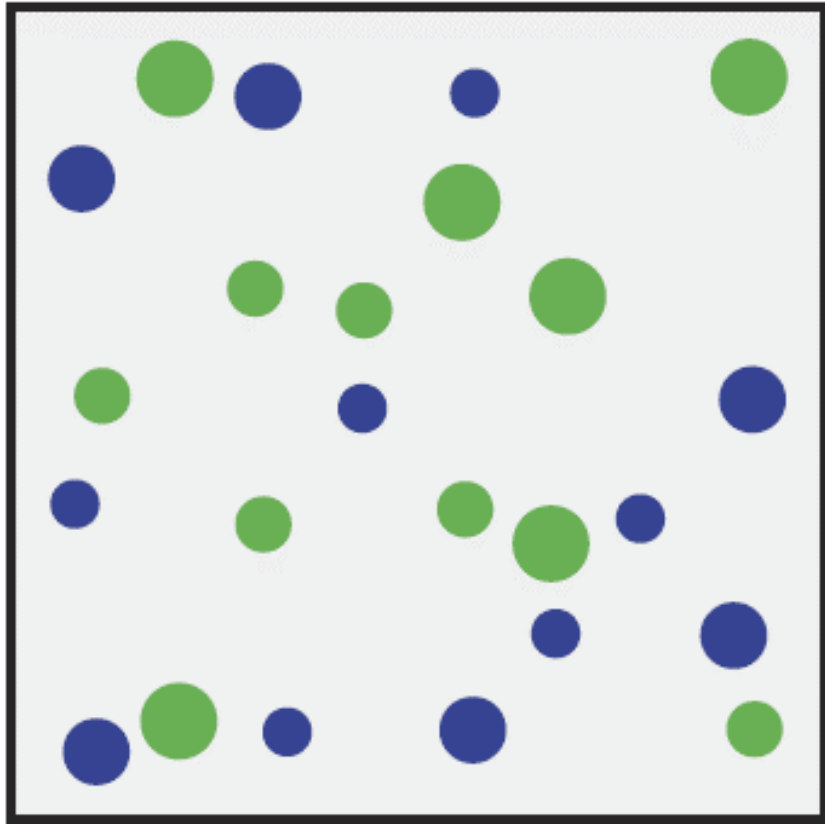
YES?

=

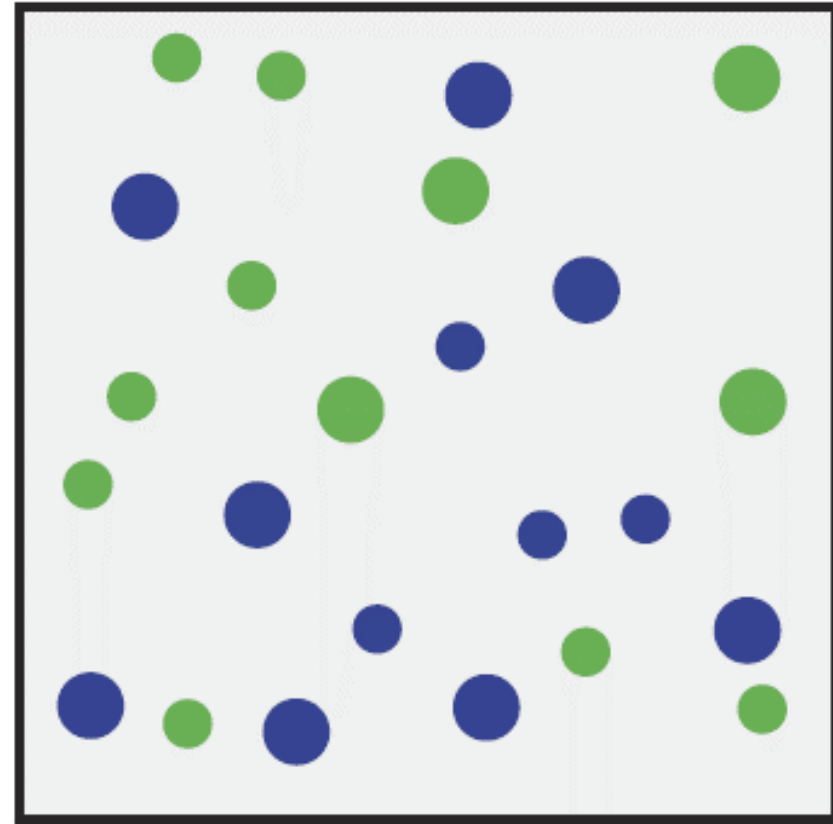
UP!

Which panel has the **biggest green** dots?

A

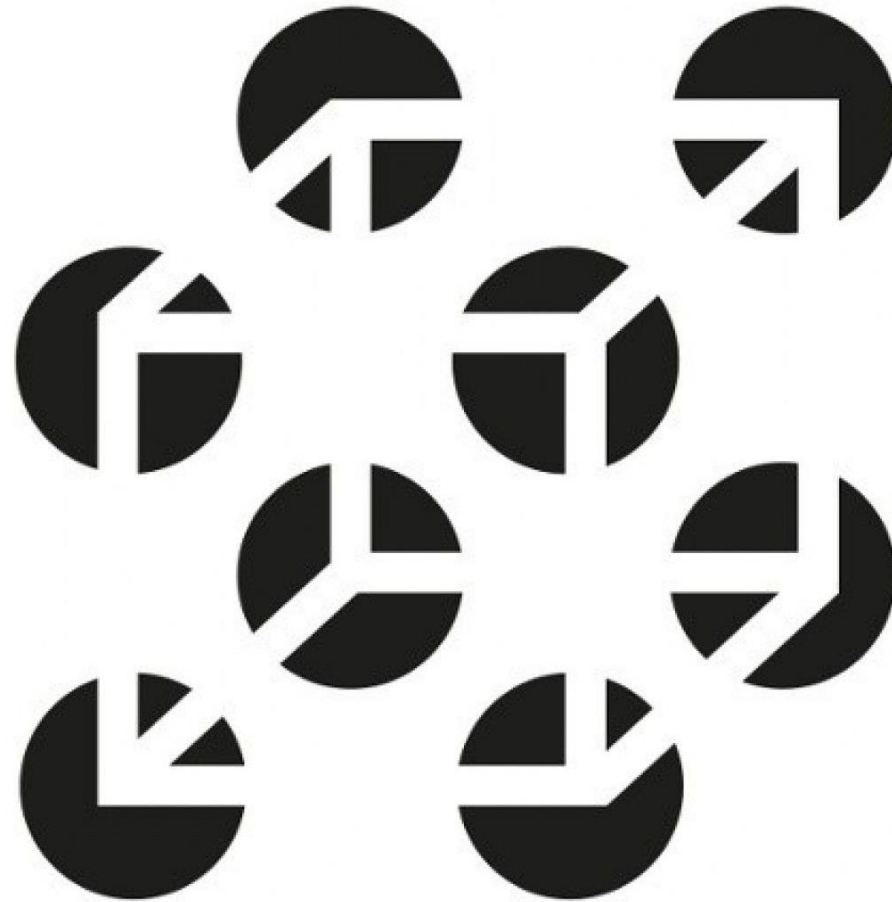


B



LEFT? = UP!

Do you *see* a cube?

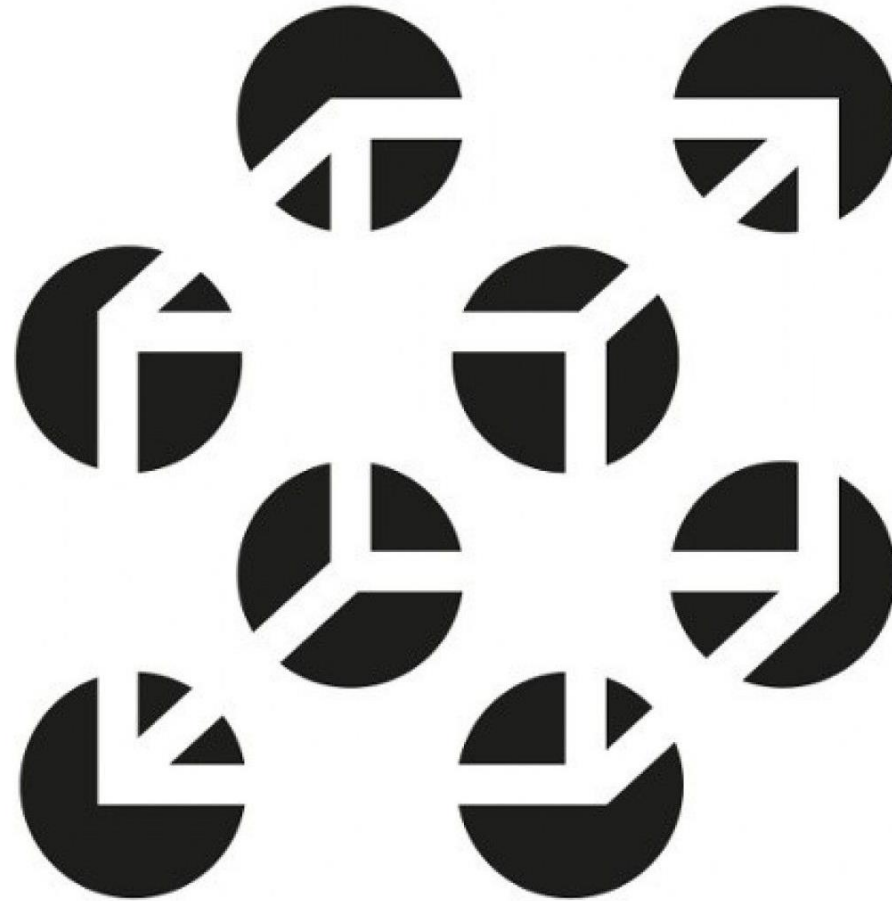


YES?

=

UP!

/s there a cube?

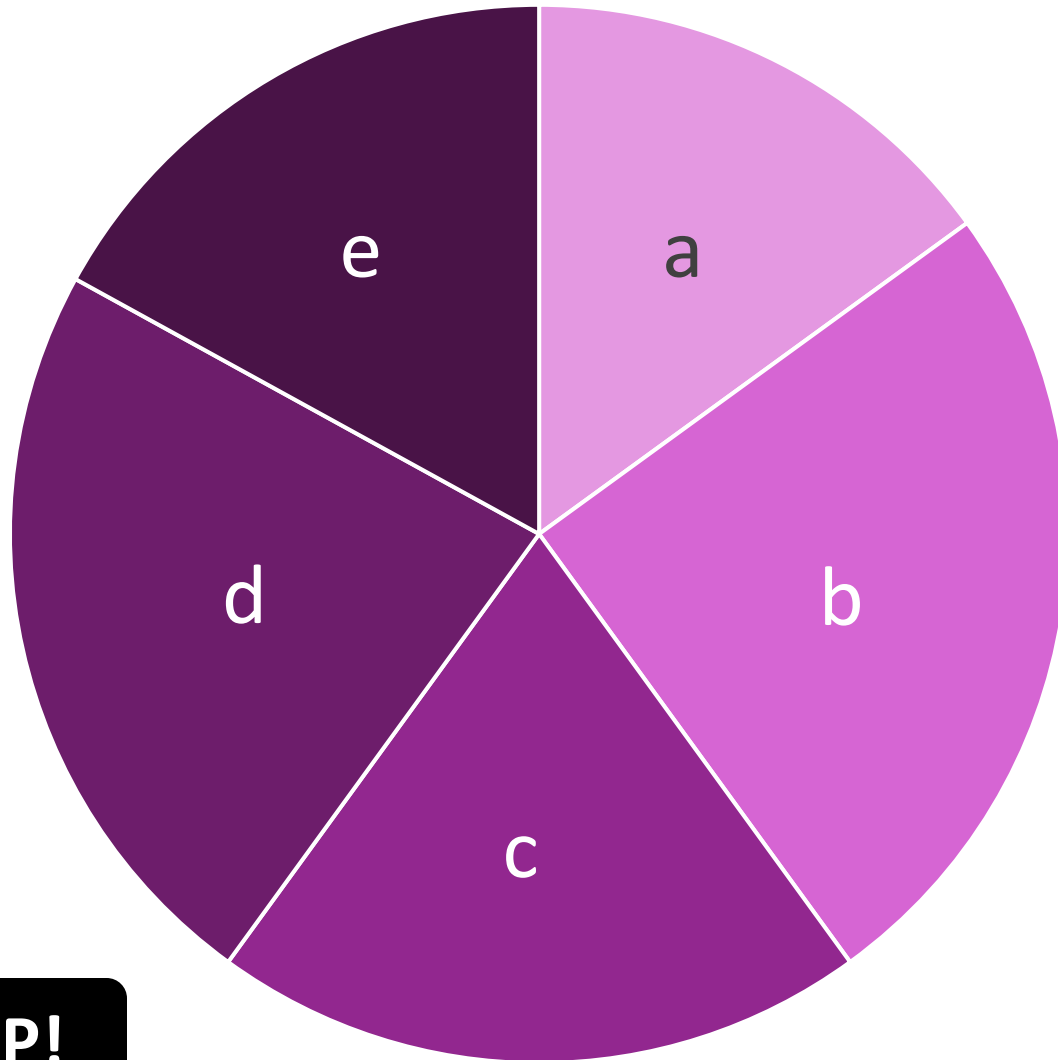


YES?

=

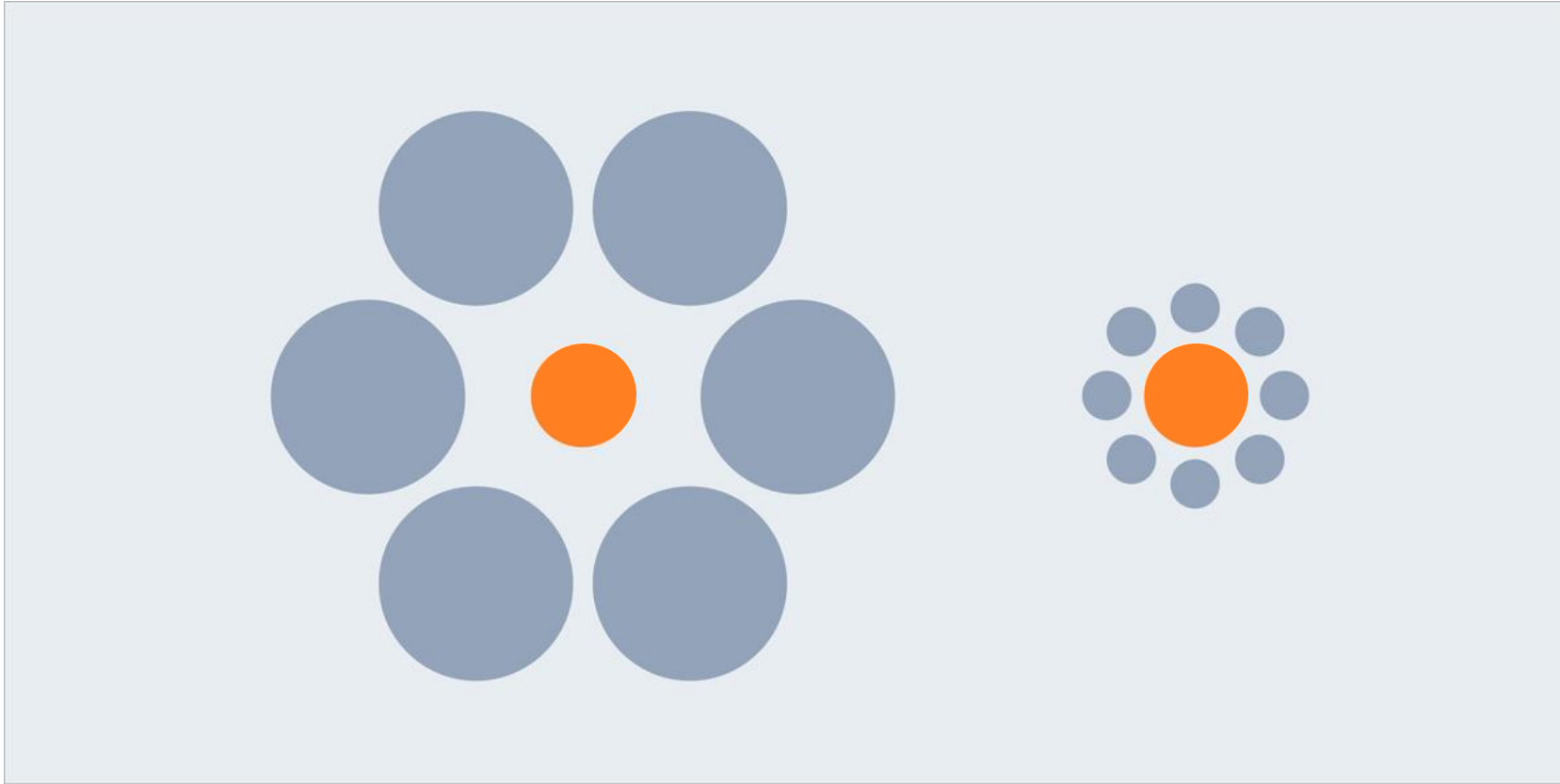
UP!

The order from high to low is $b > d > c > e > a$



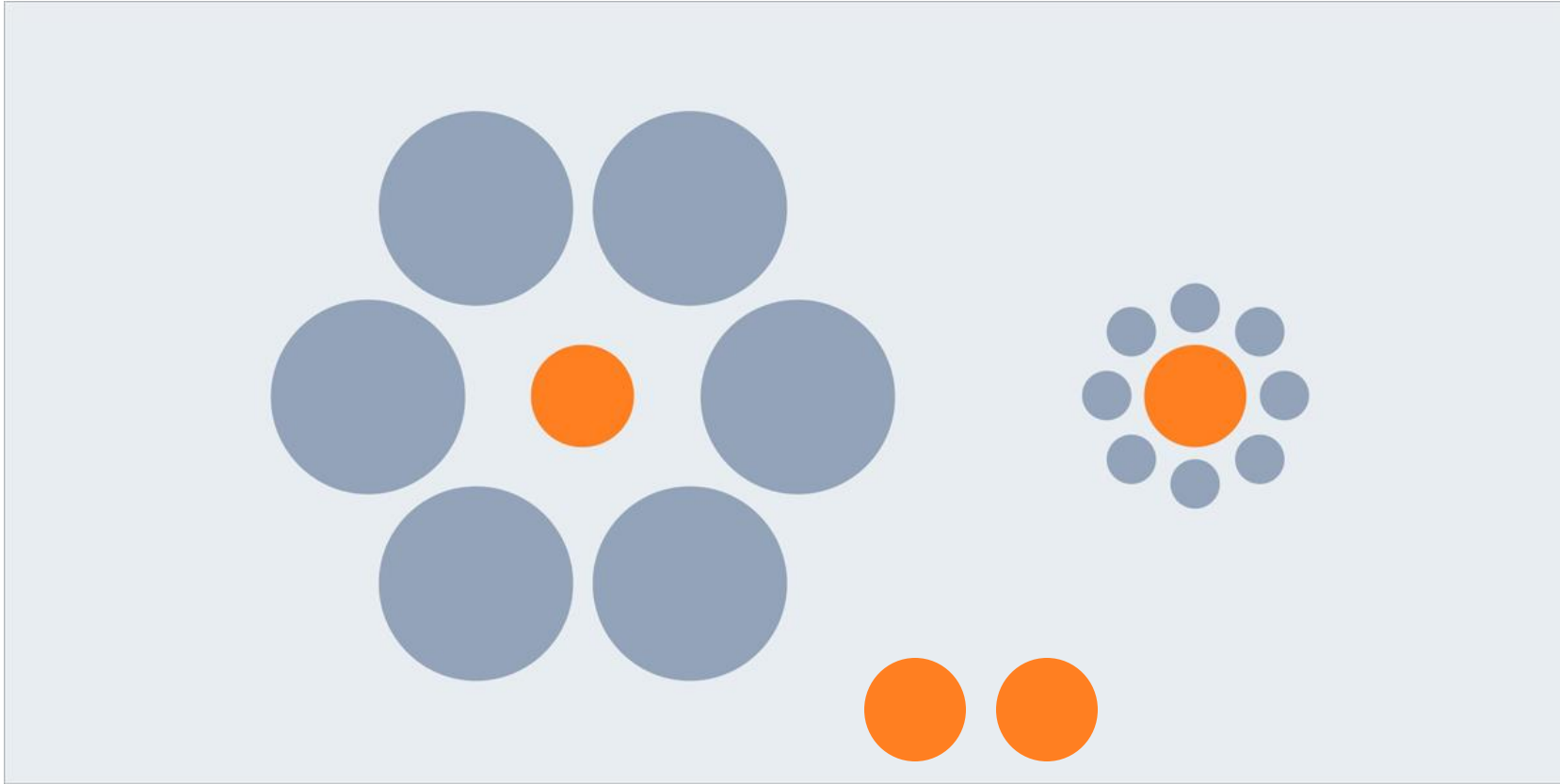
YES? = **UP!**

Which orange circle is smaller?



LEFT? = **UP!**

Which orange circle is smaller?



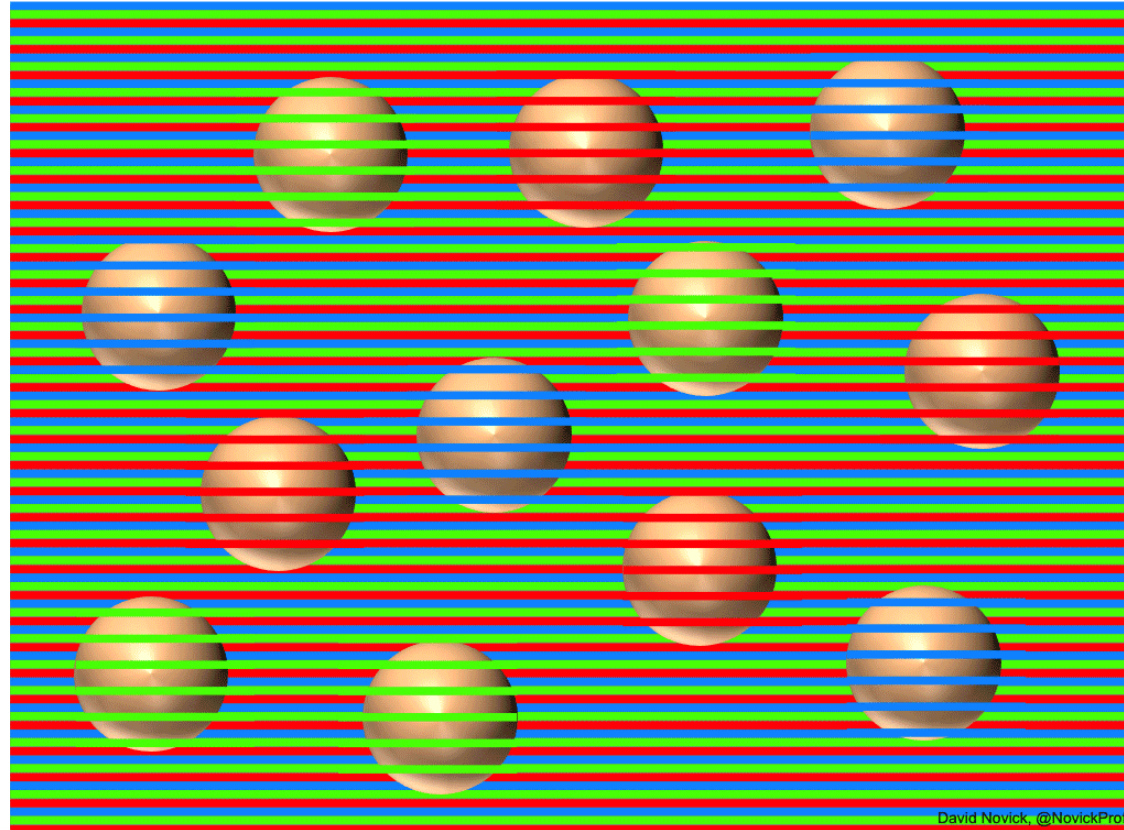
LEFT? = **UP!**

These speres have the same color



YES? = UP!

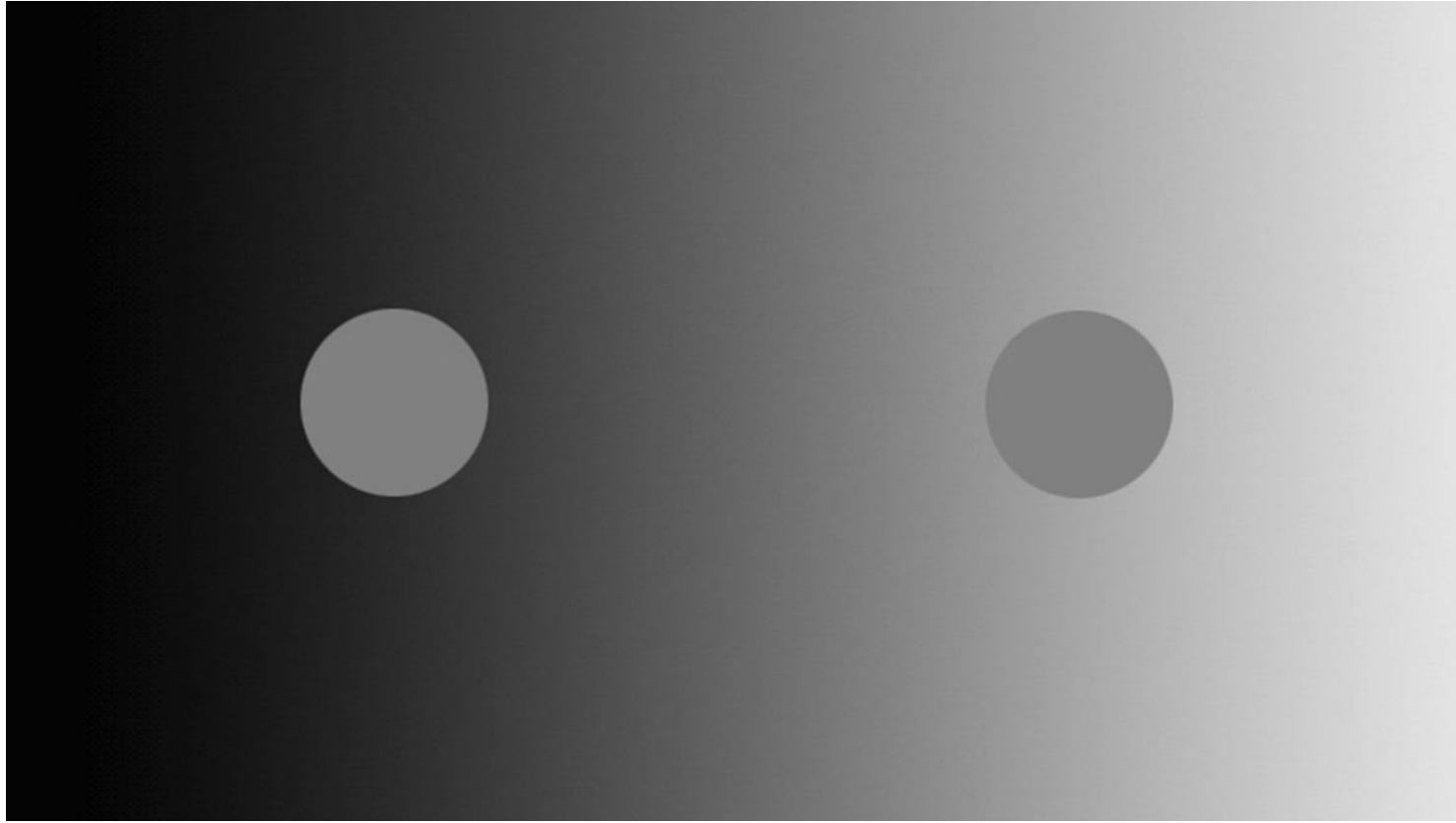
These speres have the same color



David Novick, @NovickProf

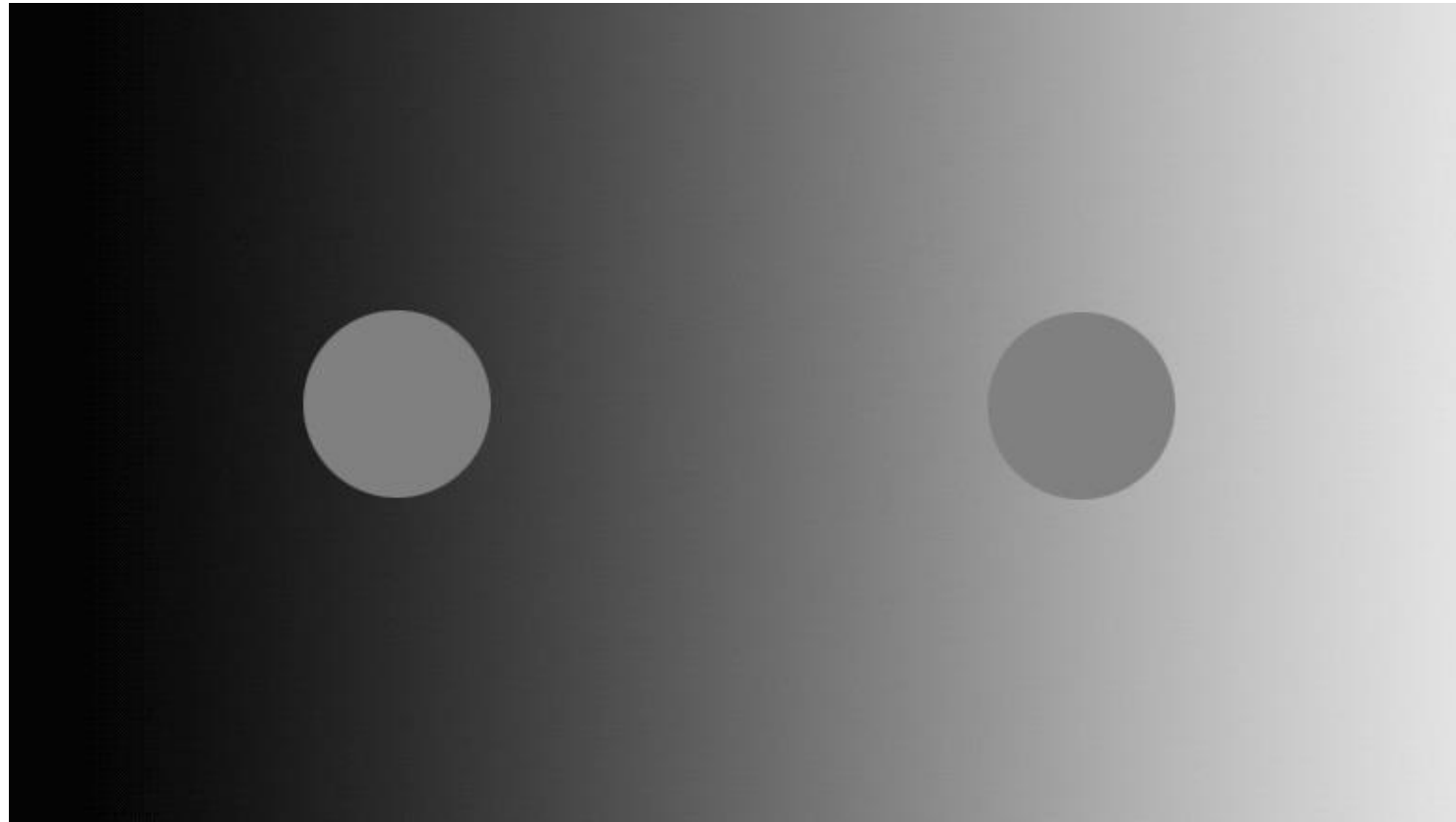
YES? = UP!

Which dot is darker?



LEFT? = **UP!**

Are you sure...?



LEFT? = **UP!**



“It’s not what you look at that matters, it’s what you see”

Henry David Thoreau



The **strengths** of our brain

“The eyes only see what the mind is prepared to comprehend.” – Henri Bergson.

How many fives do you see?

321654643216465132168461321
646432136748651349687463123
668432616266984892536496874
651687964649843616984616957
465167412323161689312439873
421432875943869234987596387

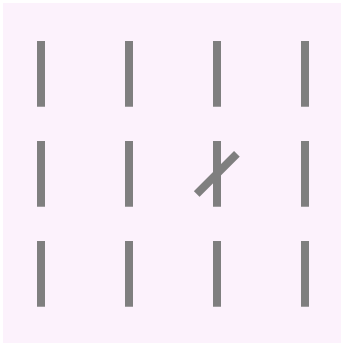
How many fives do you see?

3216**5**464321646**5**132168461321
6464321367486**5**1349687463123
668432616266984892**5**36496874
6**5**16879646498436169846169**5**7
46**5**167412323161689312439873
42143287**5**943869234987**5**96387

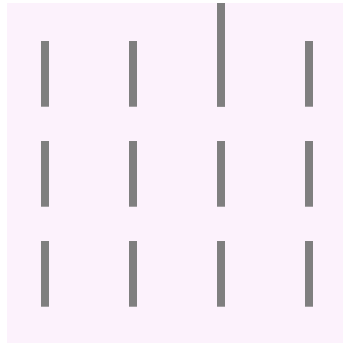
Pre-attentive attributes make your audience see what you want them to see before even seeing it!

3216**5**464321646**5**132168461321
6464321367486**5**1349687463123
668432616266984892**5**36496874
6**5**16879646498436169846169**5**7
46**5**167412323161689312439873
42143287**5**943869234987**5**96387

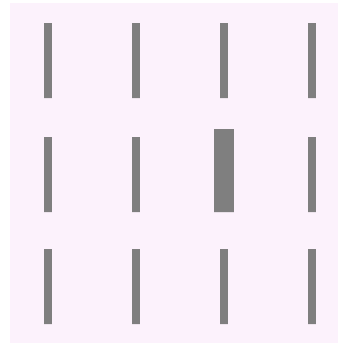
Pre-attentive attributes make your audience see what you want them to see before even seeing it!



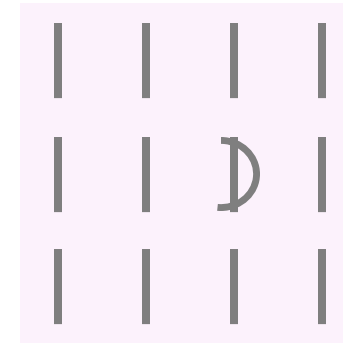
Orientation



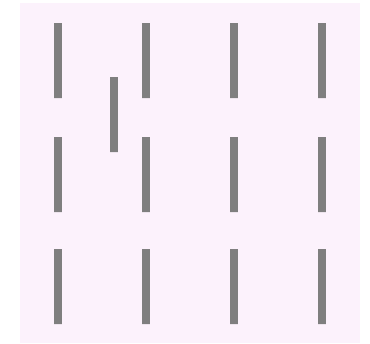
Length



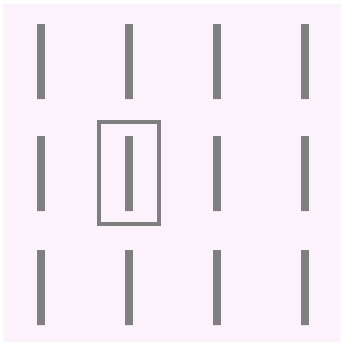
Width



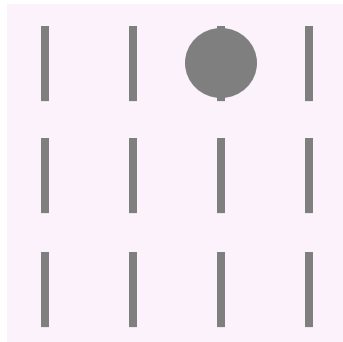
Curvature



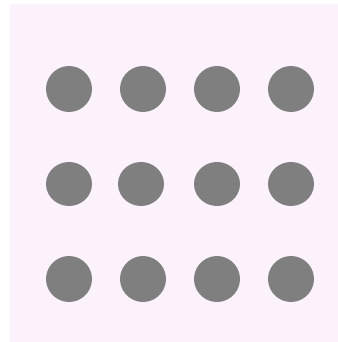
Position



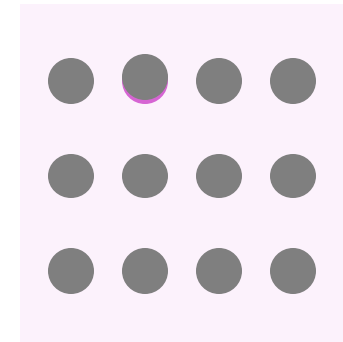
Enclosure



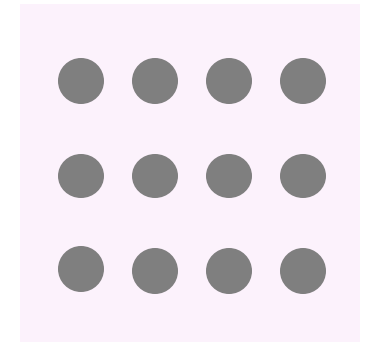
Shape



Size



Color



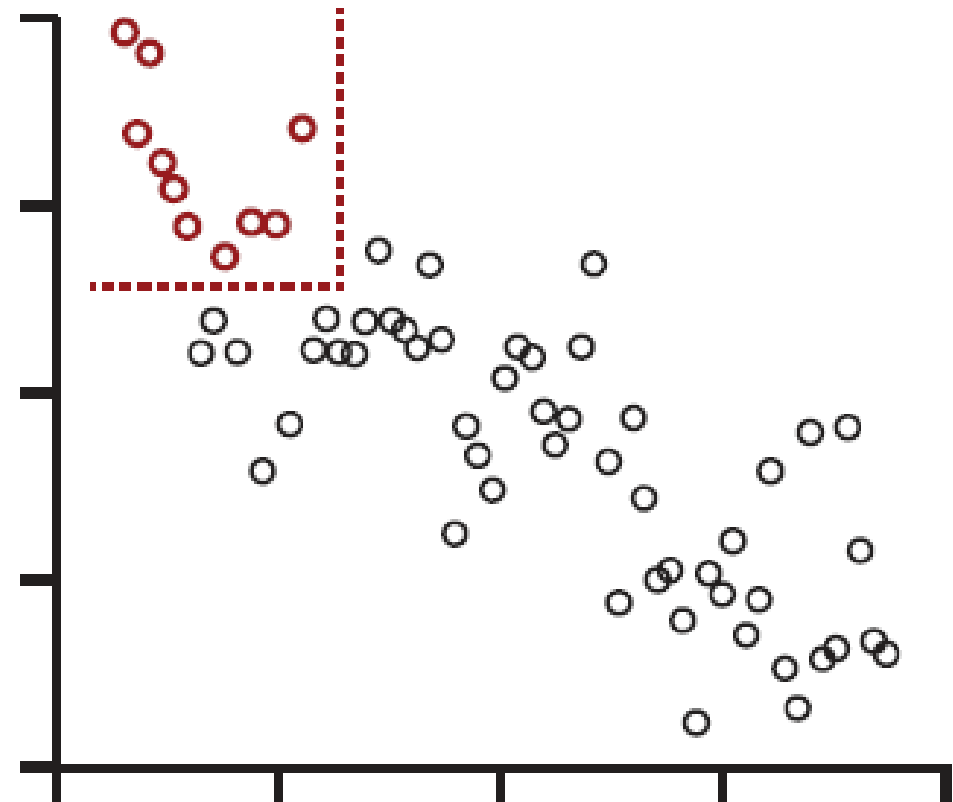
Motion

You can use pre-attentive attributes in **tables**

Color name	RGB (1–255)
Black	0, 0, 0
Orange	230, 159, 0
Sky blue	86, 180, 233
Bluish green	0, 158, 115
Blue	0, 114, 178
Vermillion	213, 94, 0

You can use pre-attentive attributes in tables and **charts**

Color name	RGB (1–255)
Black	0, 0, 0
Orange	230, 159, 0
Sky blue	86, 180, 233
Bluish green	0, 158, 115
Blue	0, 114, 178
Vermillion	213, 94, 0



You can also use pre-attentive attributes in text.

How can you leverage the use of pre-attentive attributes?

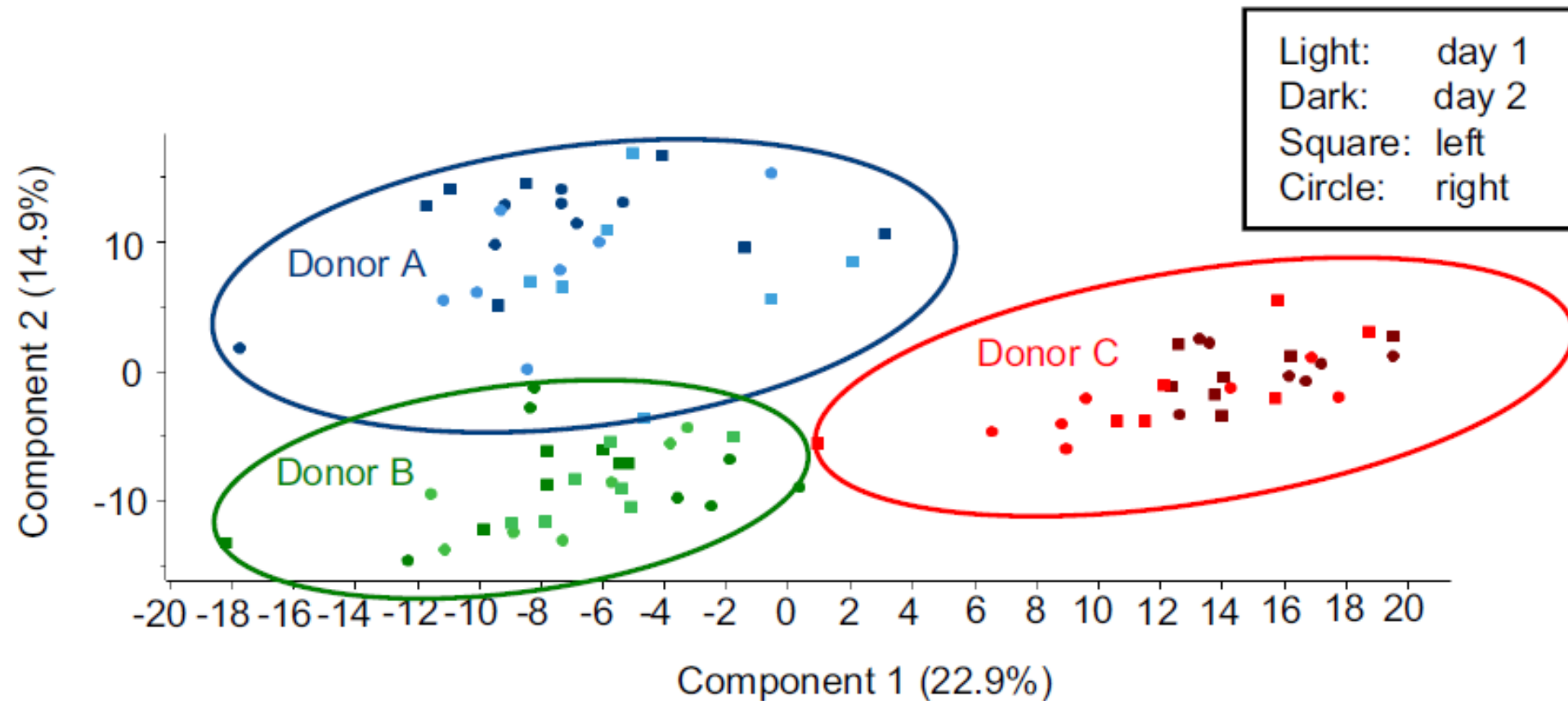
Without pre-attentive attributes you just have plain text. Or a plain figure. There is no guidance for the reader or viewer.

With pre-attentive attributes, you can:

- (1) Draw your audience attention to where you want them to look, and
- (2) Create a visual hierarchy of information

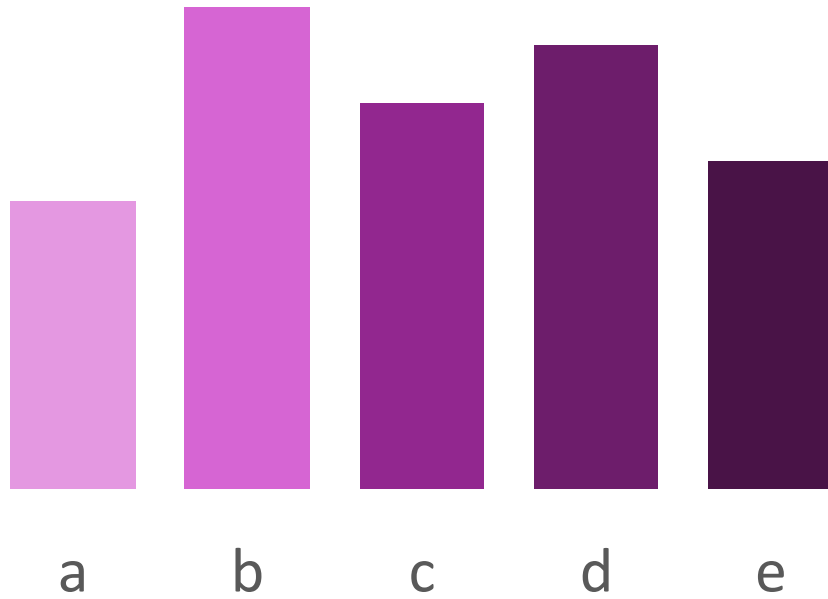
This counts for **graphs** as well as **text**!

Pre-attentive attributes can create a visual hierarchy

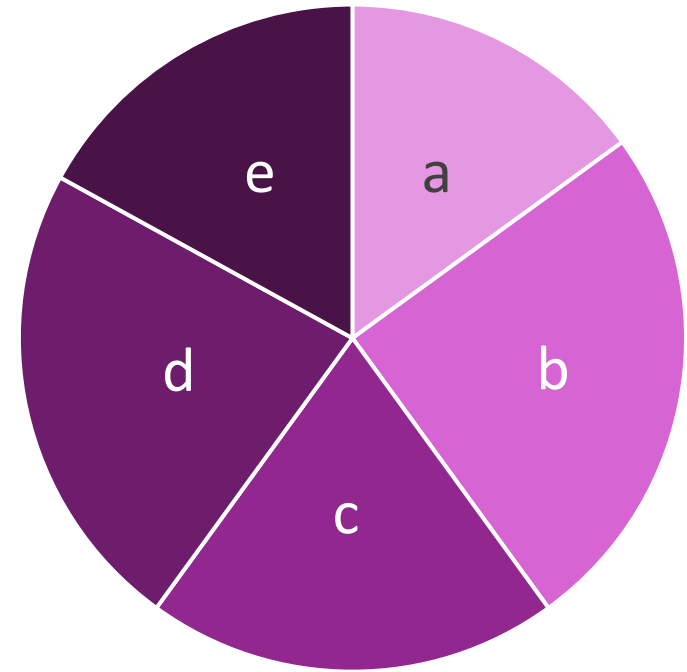


Pre-attentive attributes can be used to map quantities, e.g. by length or by angle

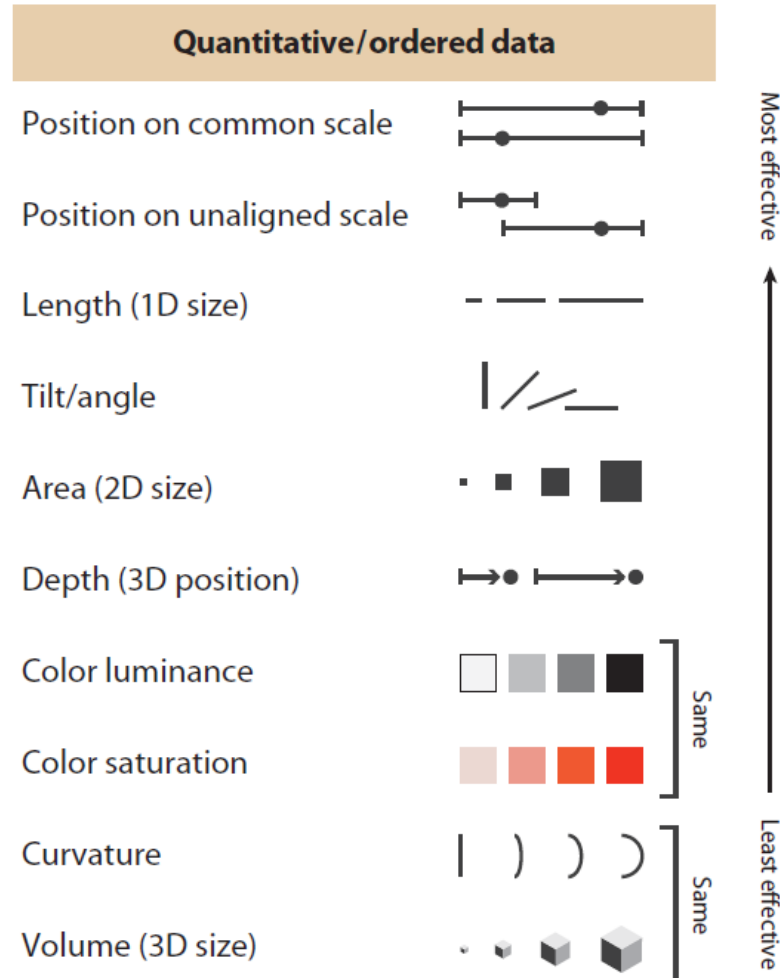
Which bar is highest?



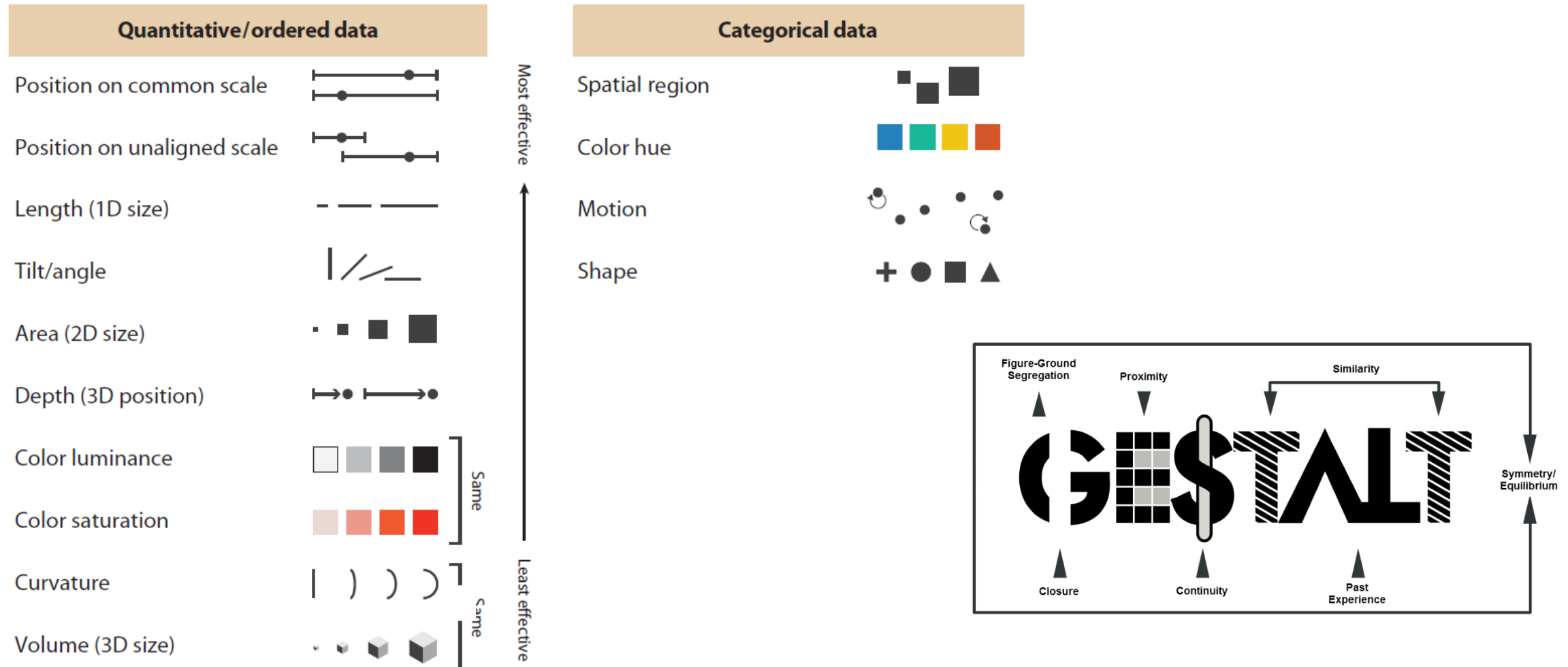
Which slice is biggest?



Some attributes are more effective to differentiate quantities



Some attributes are more effective to differentiate **quantities** or **categories**

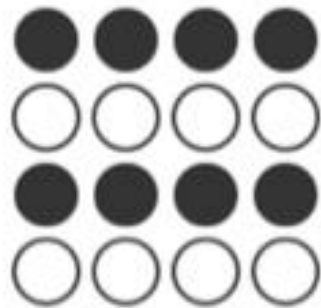


Gestalt principles: The whole is other than the sum of the parts.

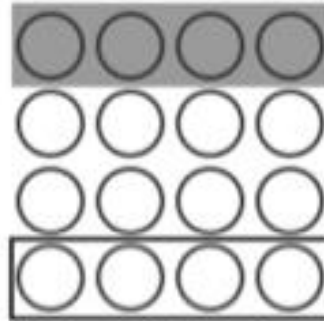
Proximity



Similarity



Enclosure



Symmetry

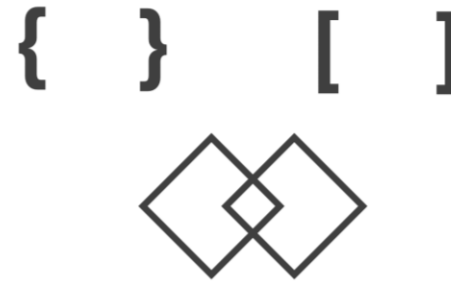
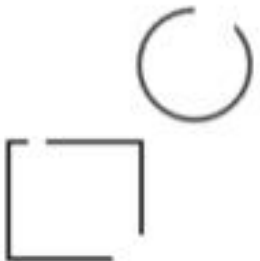


Figure-ground



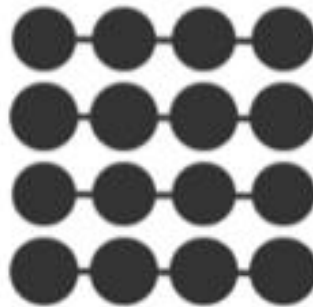
Closure



Continuity



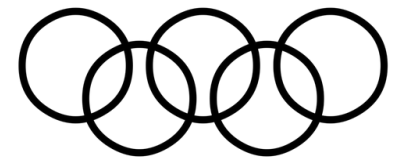
Connection



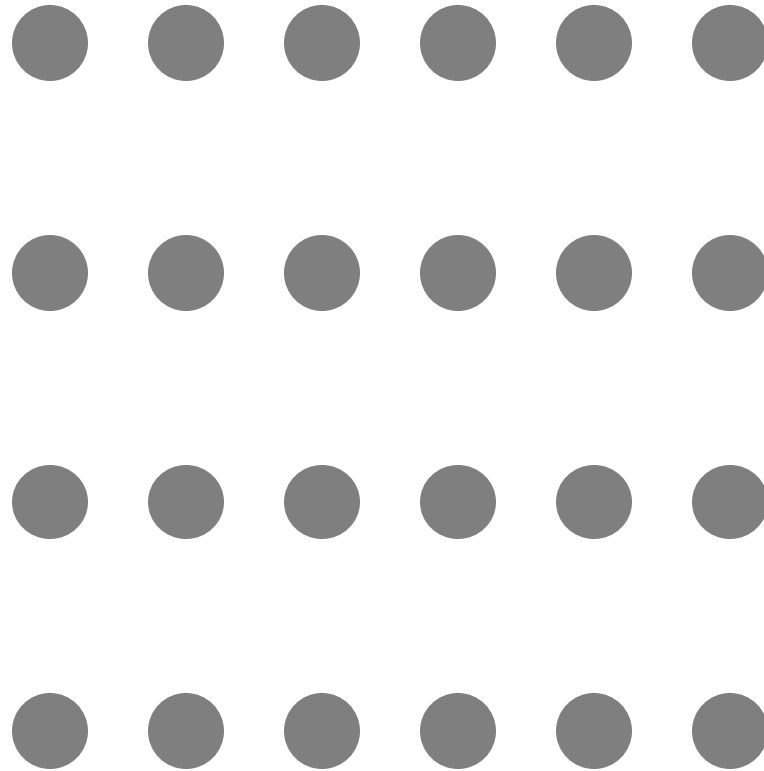
Common Fate



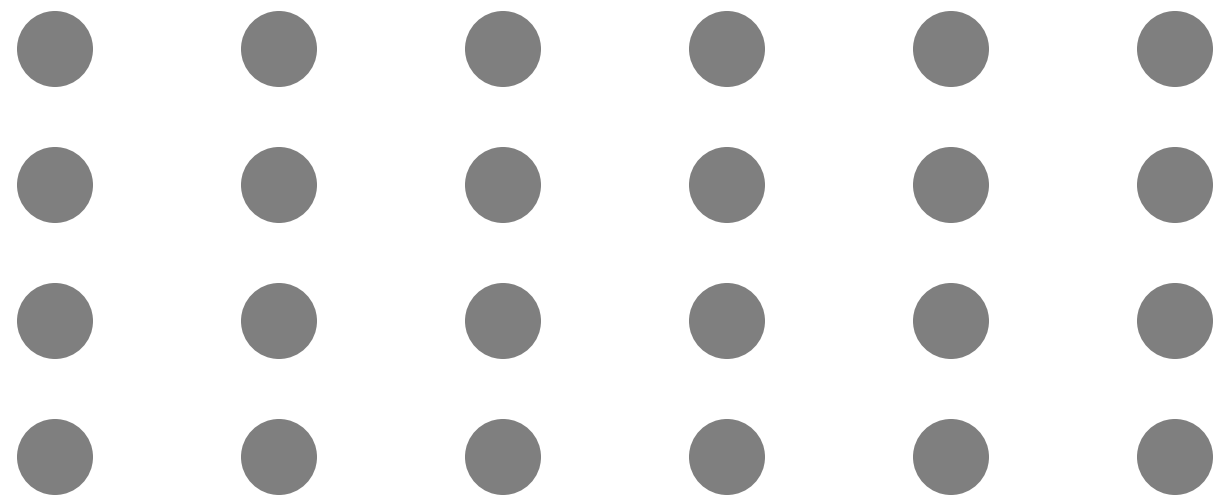
Simplicity



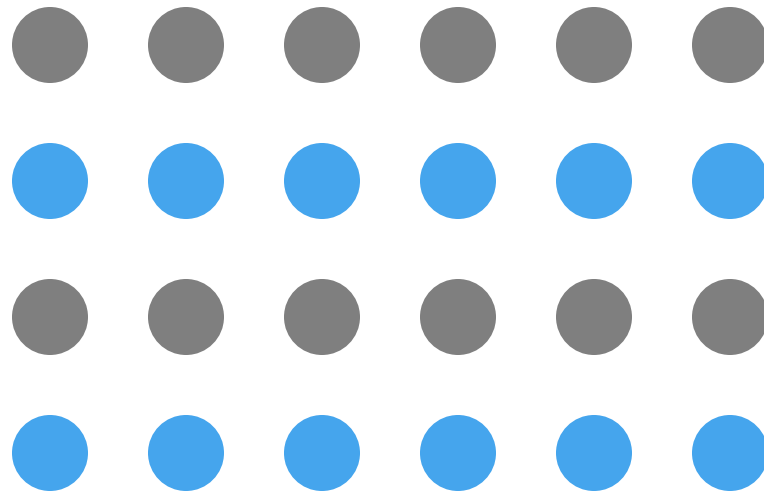
Do you see 4 rows or 6 columns?



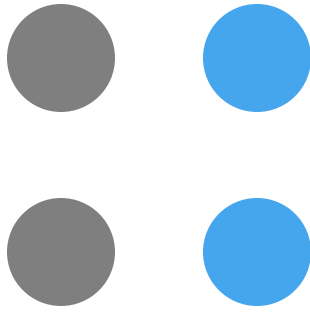
And now?



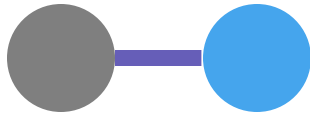
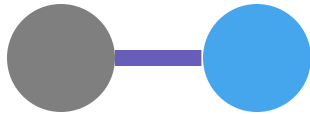
Do you see 4 rows or 6 columns?



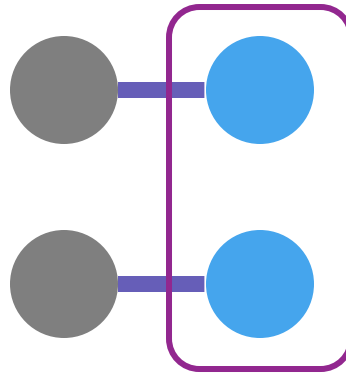
Which dots belong together?



And now...?



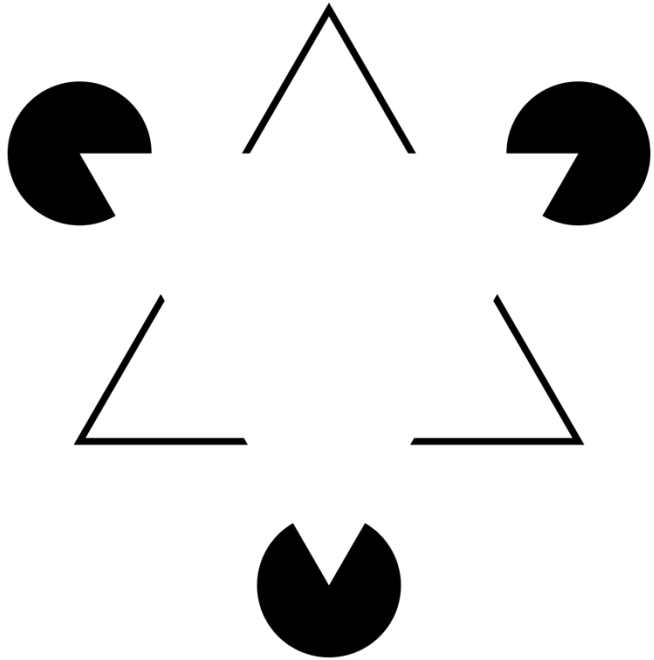
And now...?



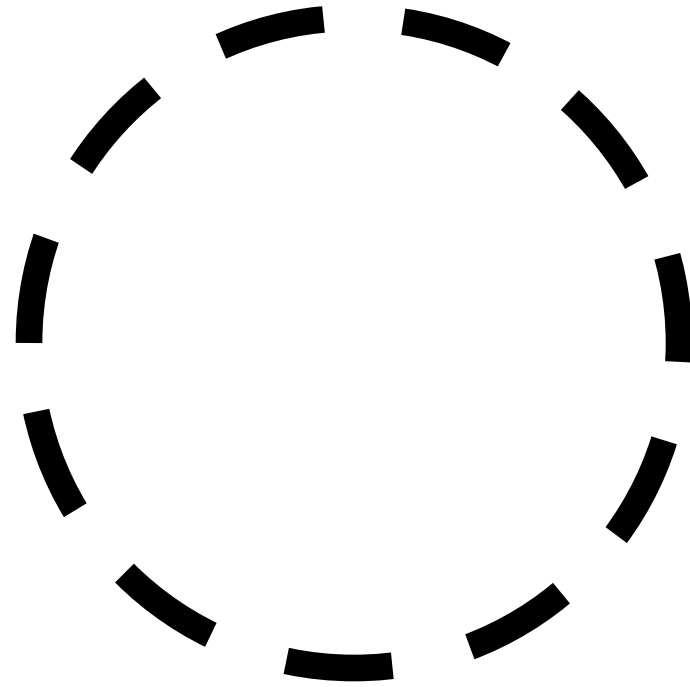
Can you read this?

According to a research at Cambridge University, it doesn't matter in what order the letters in a word are, the only important thing is that the first and last letter be at the right place. The rest can be a total mess and you can still read it without problem. This is because the human mind does not read every letter by itself, but the word as a whole.

We're good at seeing things that may not explicitly exist!

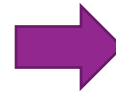
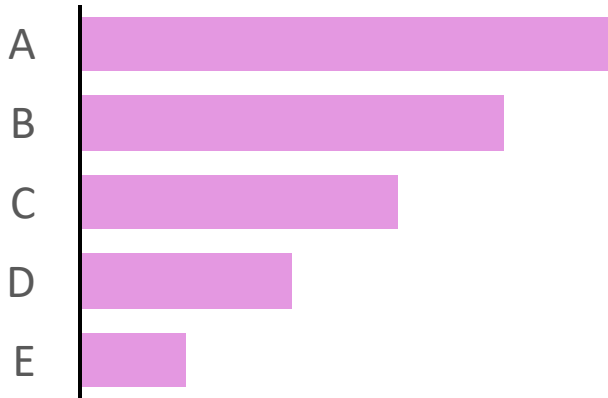
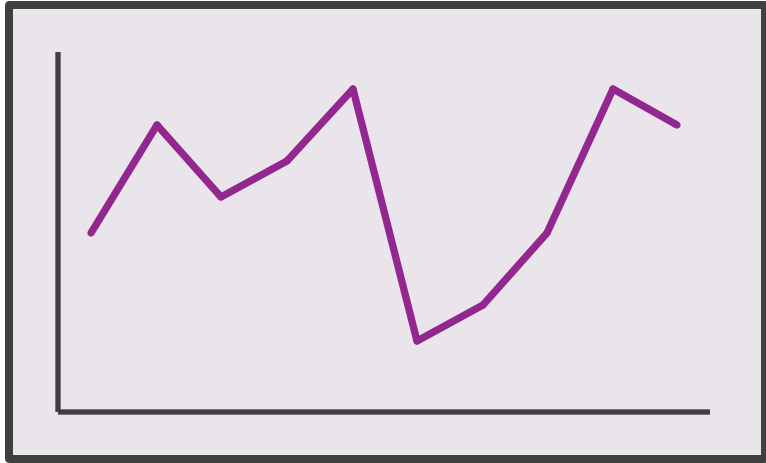


Continuity

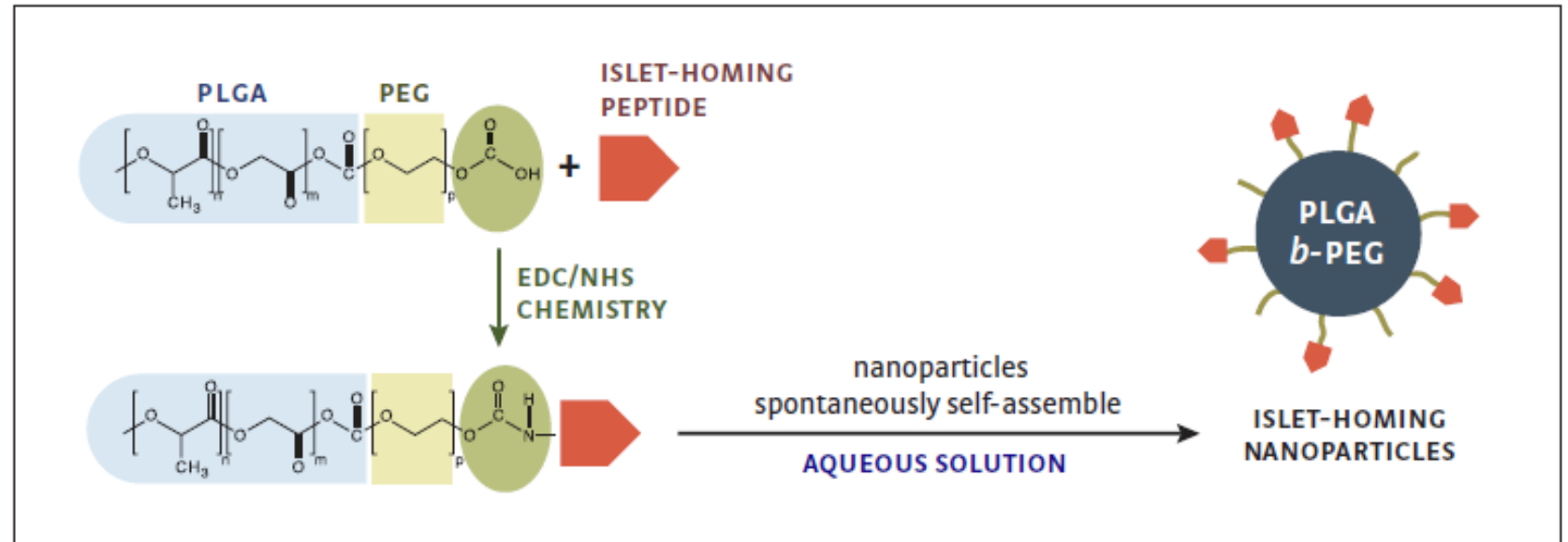
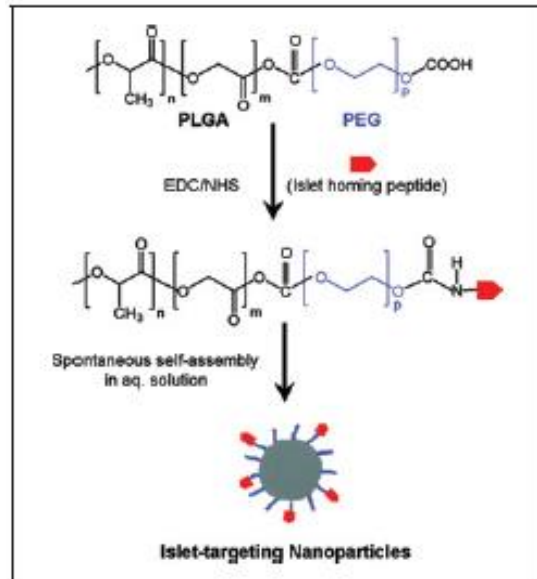


Closure

You can use the **Continuity** and **Closure** principle to remove unnecessary elements in your chart



The **Gestalt principles** also apply to graphical abstracts



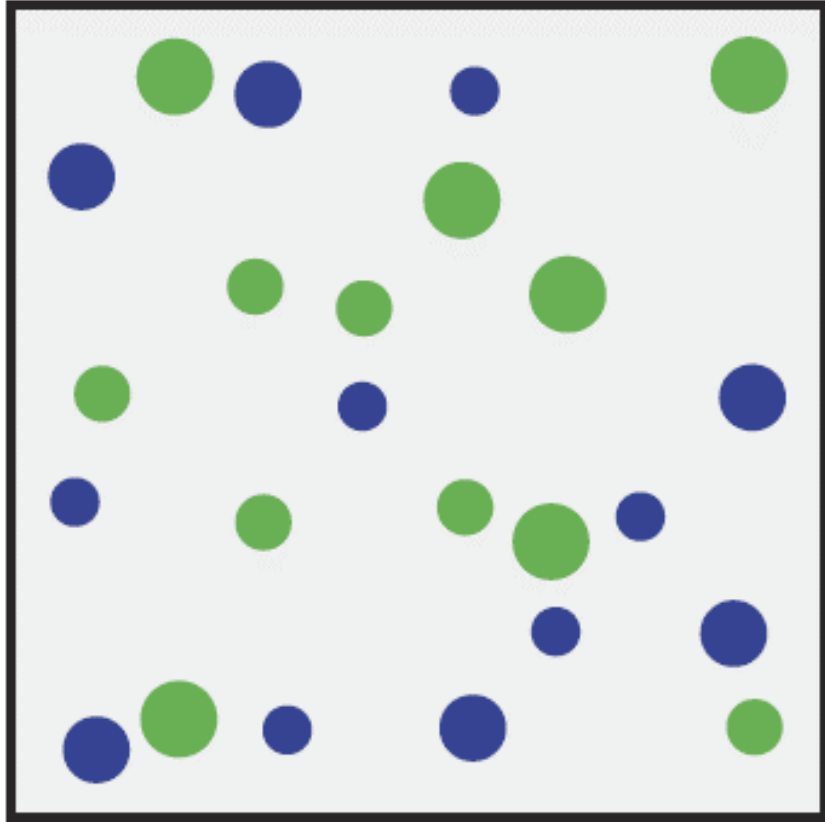
Shapes and colors used to identify similar structures.

But...

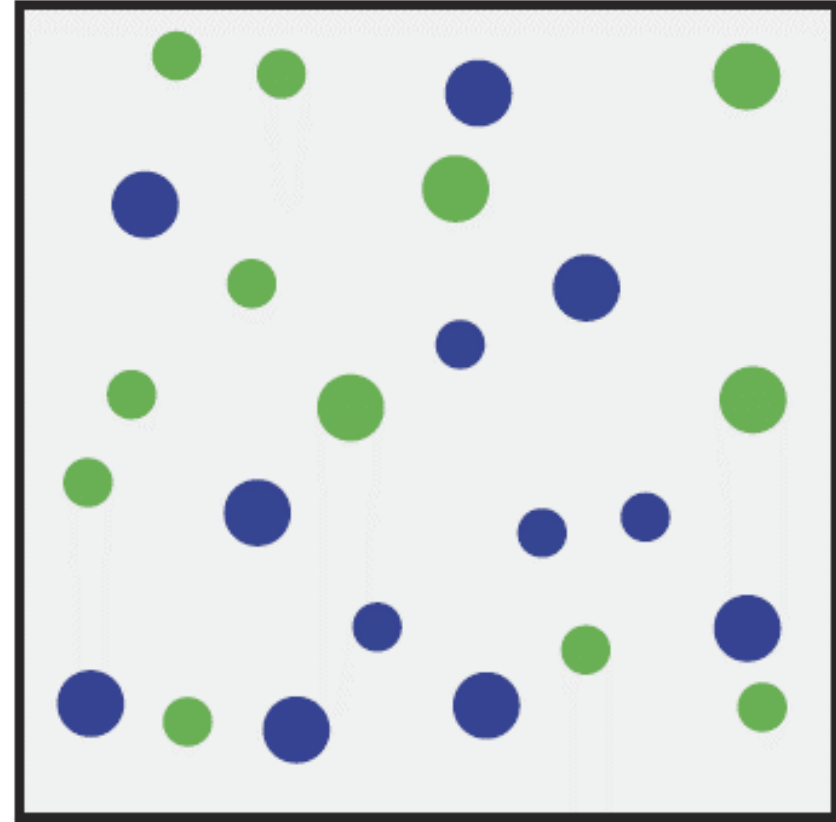


Which panel has the biggest **green** dots?

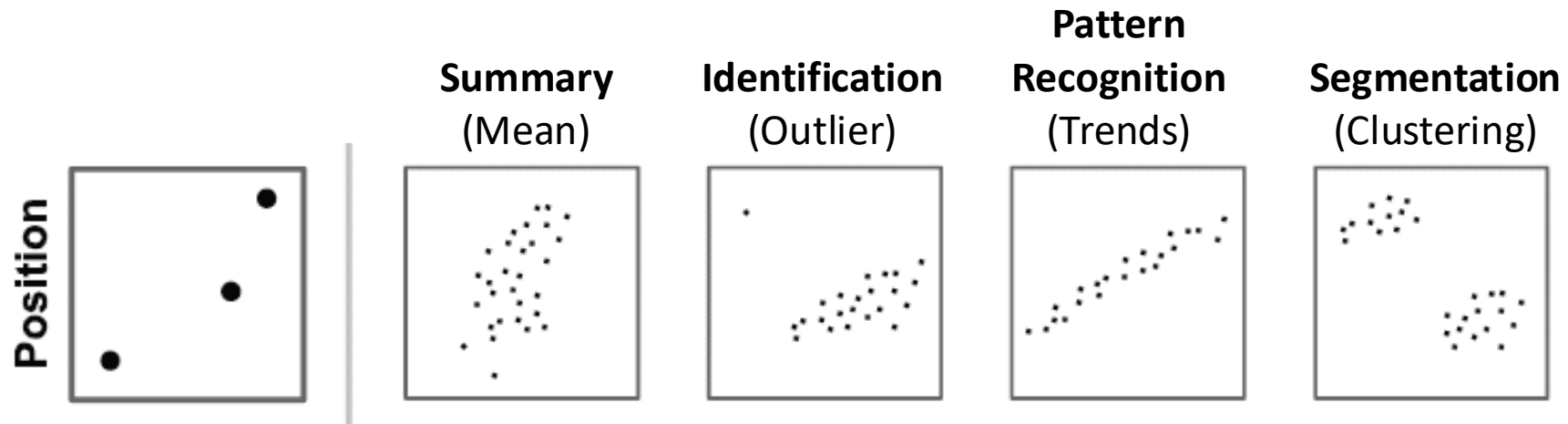
A



B



We can quickly deduce basic descriptive statistics from visuals!

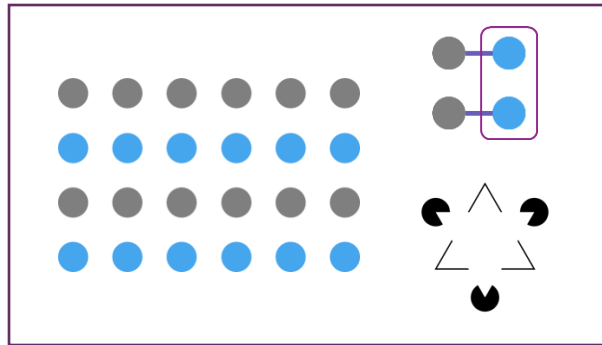


The **strengths** of our brain - summary

321654643216465132168461321
646432136748651349687463123
668432616266984892536496874
651687964649843616984616957
465167412323161689312439873
421432875943869234987596387

1.

See before seeing it
(Pre-attentive attributes)



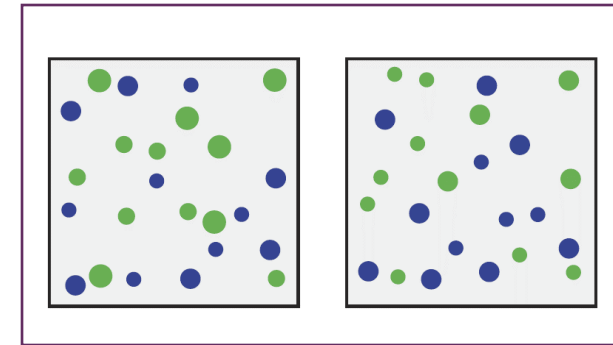
2.

See what belongs
together
(Gestalt Principles)



3.

Deduce quantitative
information from visual
shapes



4.

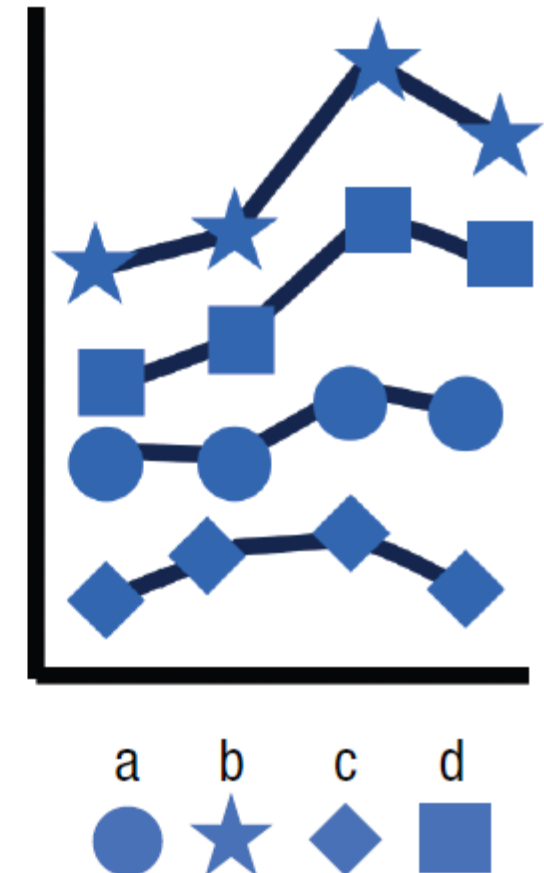
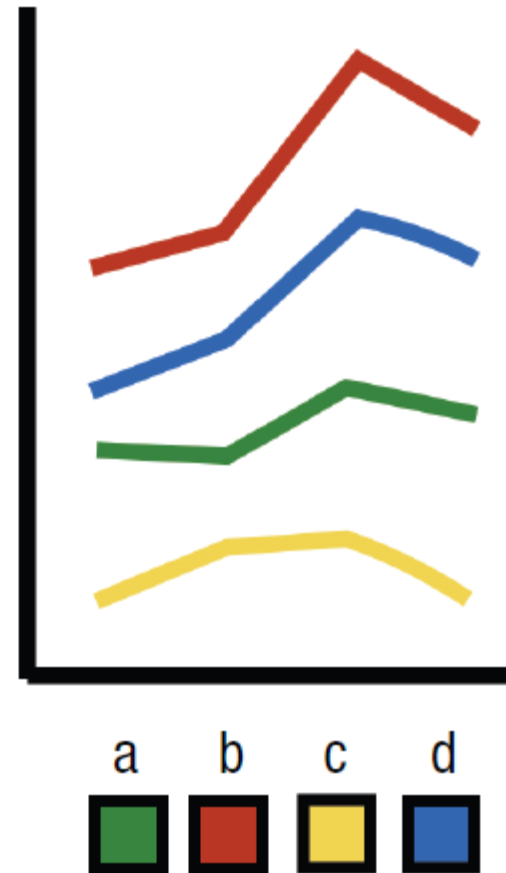
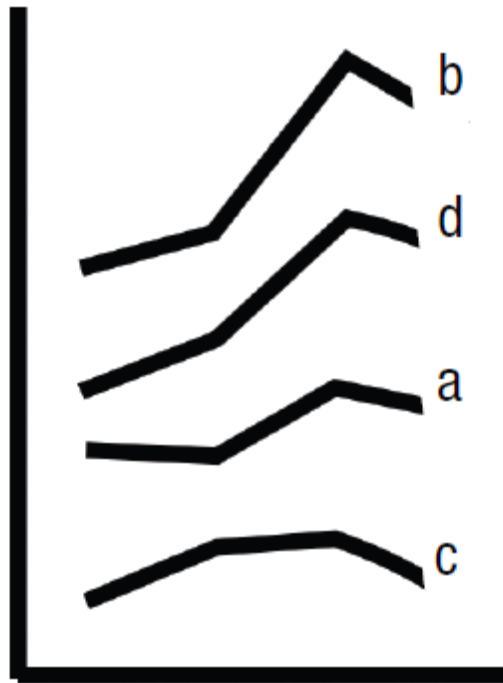
Perform basal descriptive
statistics



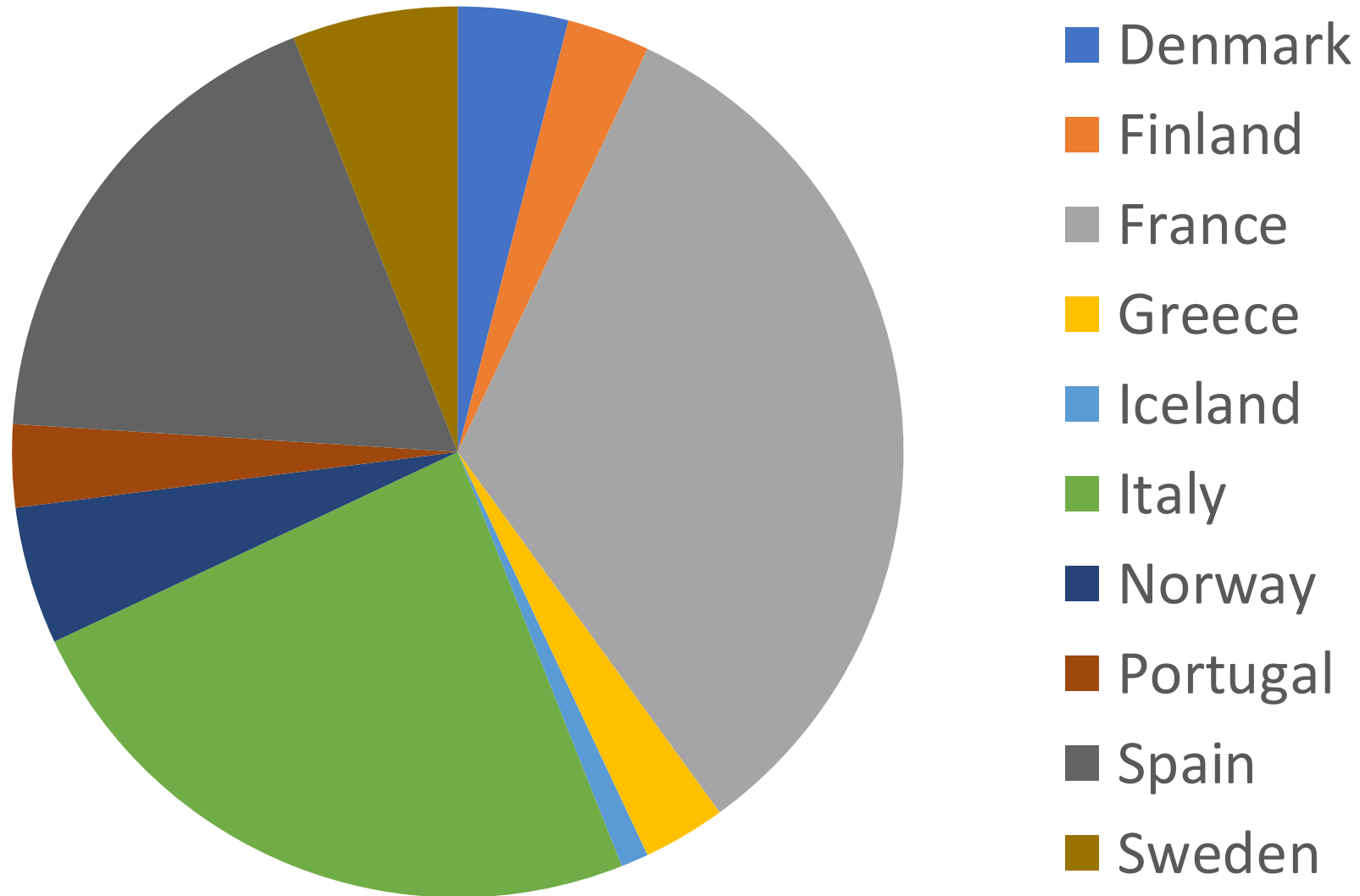
The **weaknesses** of our brain

“The eyes only see what the mind is prepared to comprehend.” – Henri Bergson.

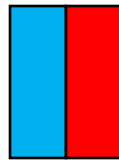
What is the order from top-to-bottom?



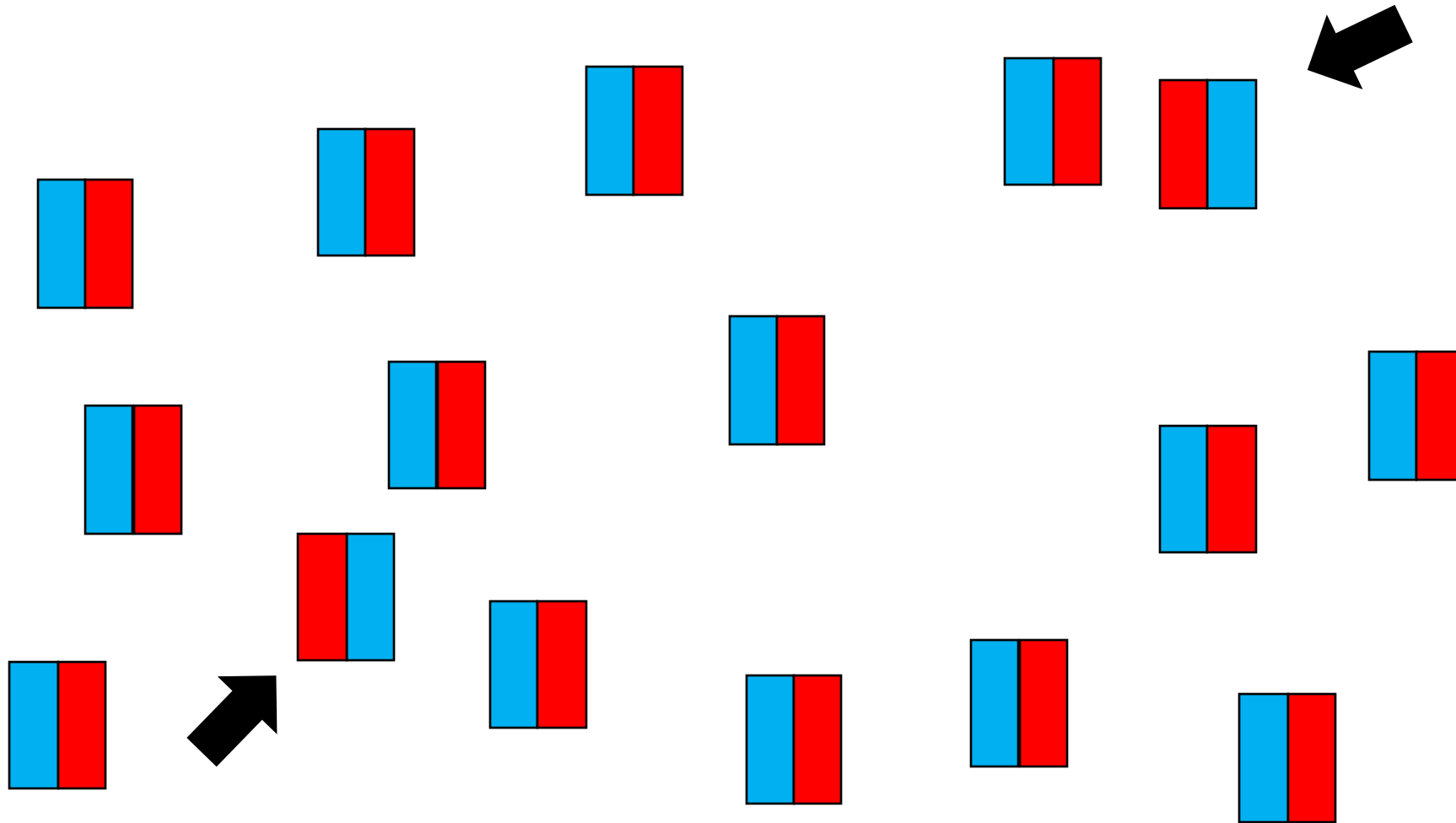
Norway has more widgets than Finland?



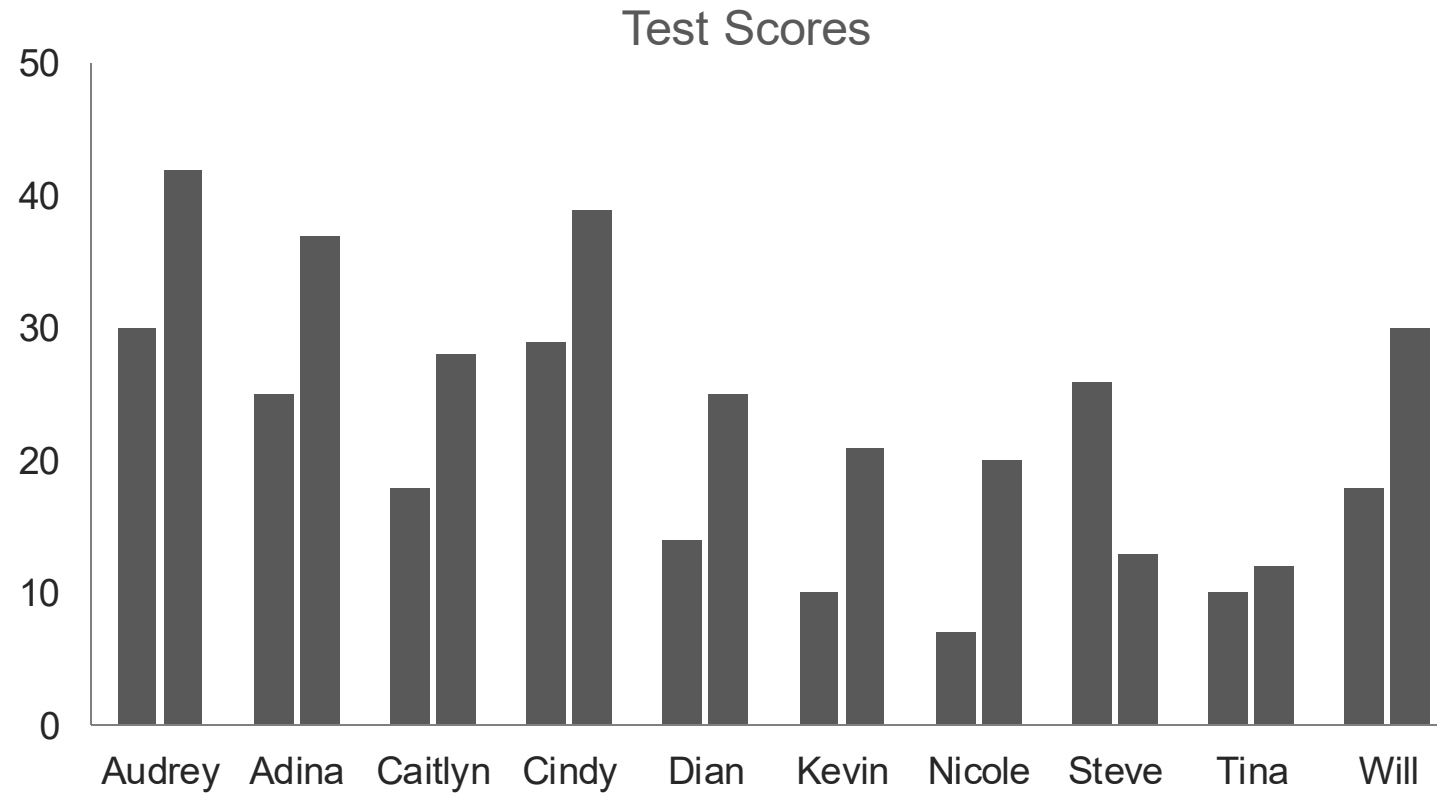
Comparing one set of colors is easy...



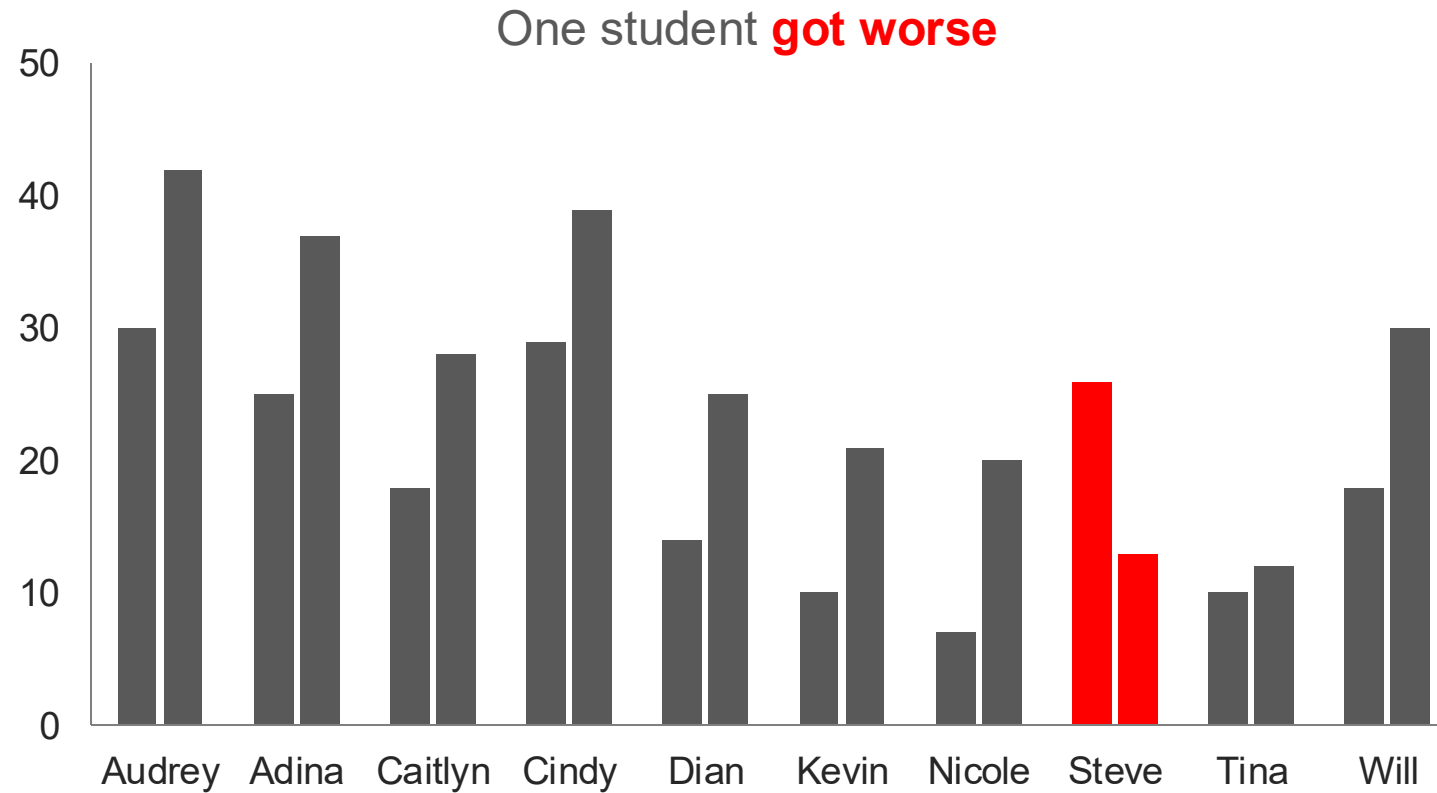
But: which one is not like the others?



Which student got worse?



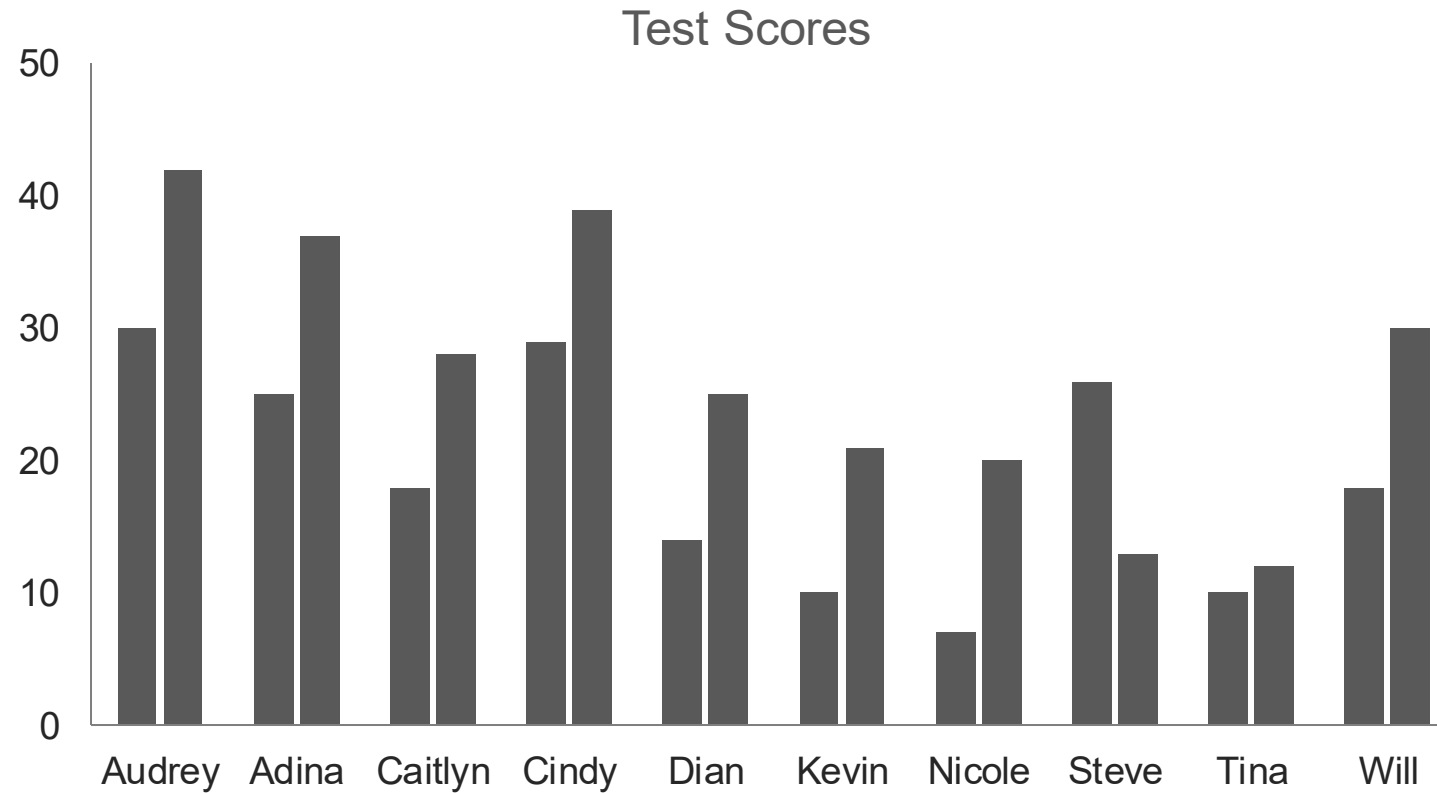
Which student got worse?



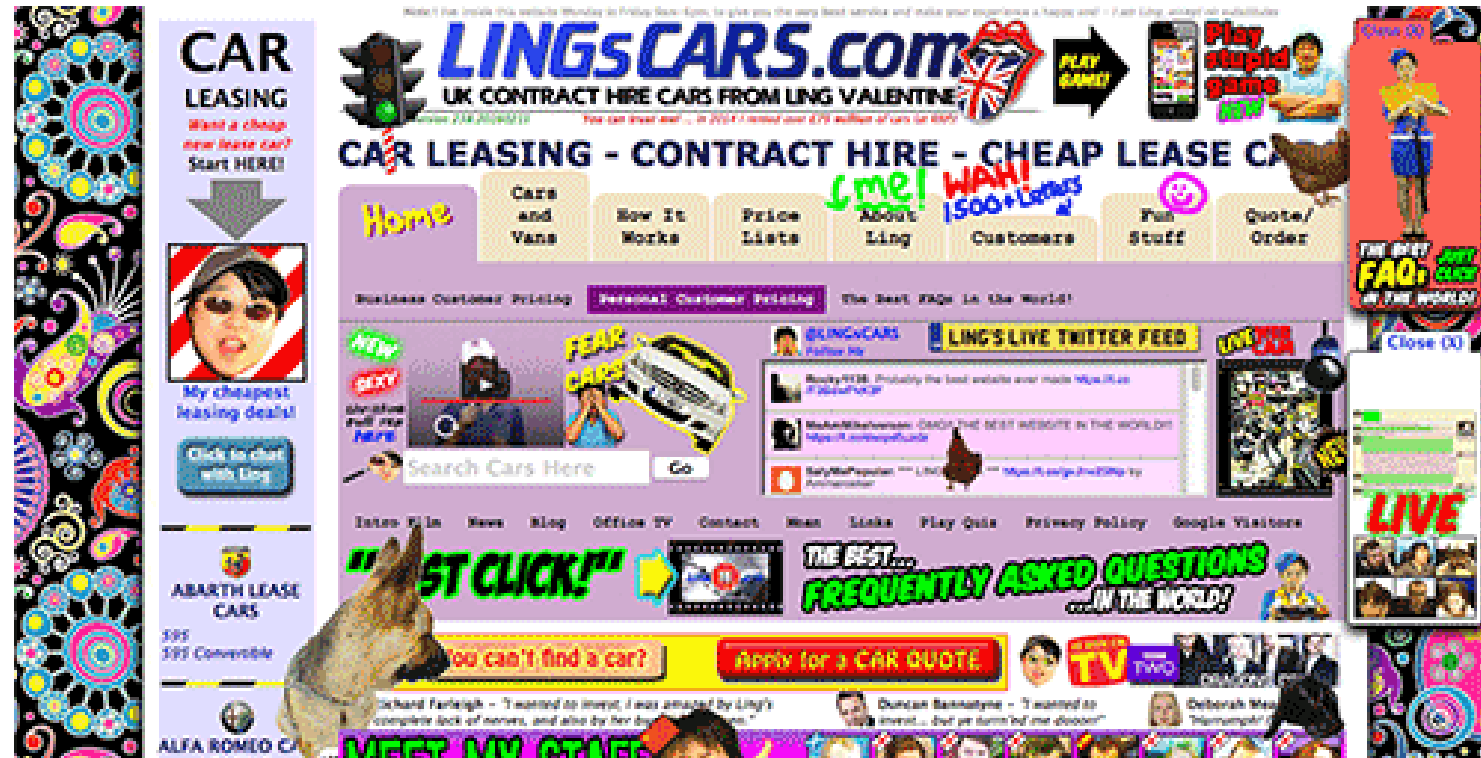
What is the average improvement?



What is the average improvement?

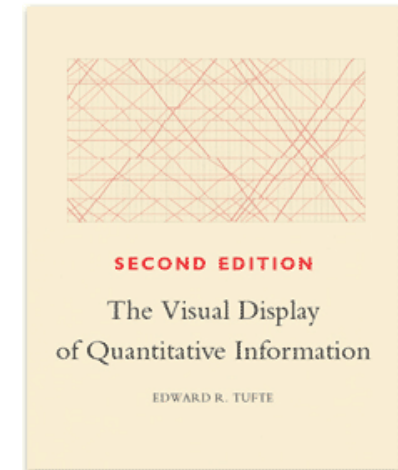


Where to look?

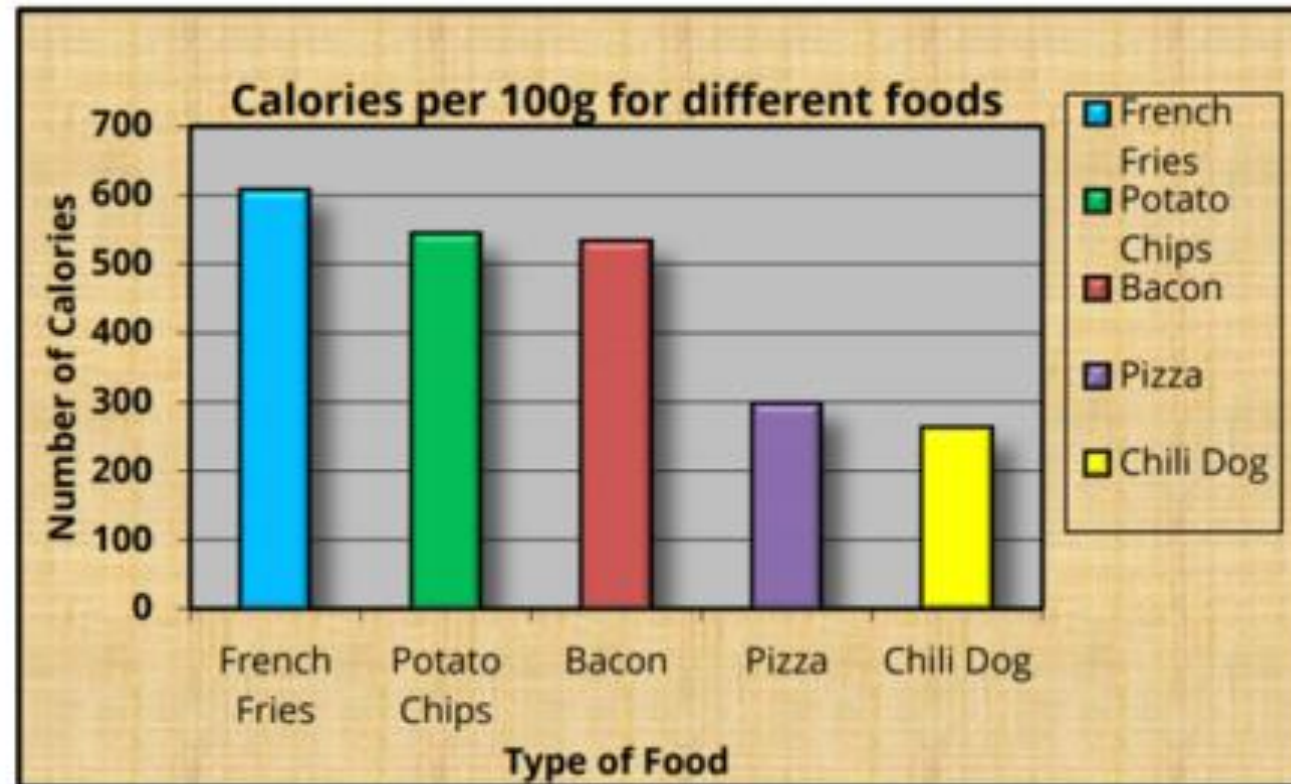


The data-ink ratio

$$\begin{aligned}\text{Data-ink ratio} &= \frac{\text{Data-ink}}{\text{Total ink used to print the graphic}} \\ &= \text{proportion of a graphic's ink devoted to the} \\ &\quad \text{non-redundant display of data-information} \\ &= 1.0 - \text{proportion of a graphic that can be erased}\end{aligned}$$



Minimize distraction: maximize **data-ink ratio**



The data-ink ratio: remove to improve

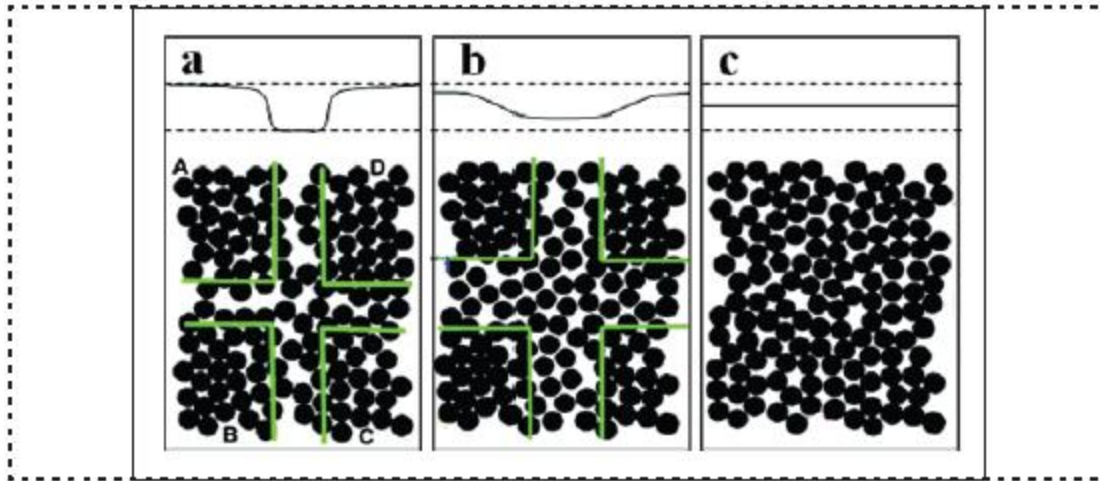
Remove
to improve
(the **data-ink** ratio)

Data-ink ratio also applies to tables

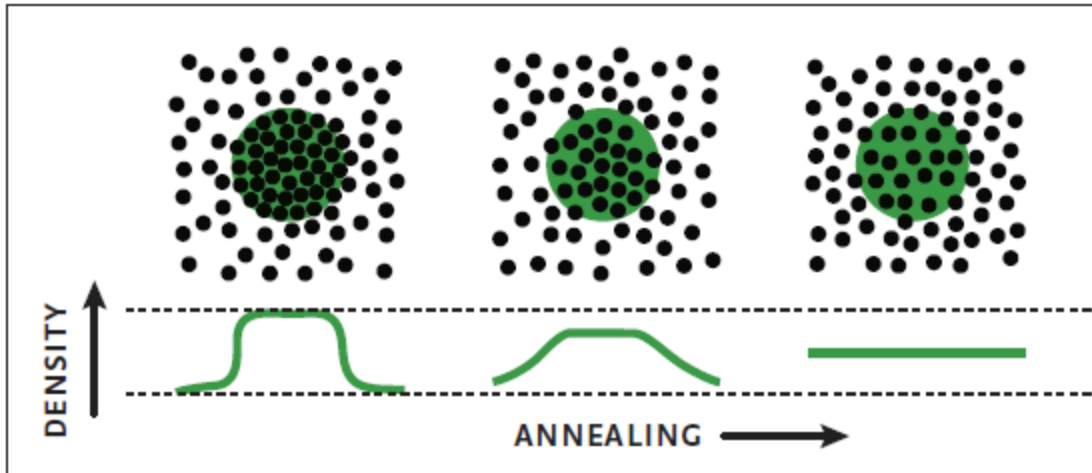
Student	Test1	Test2	Difference
Audrey	30.00	42.00	12.00 pts
Adina	25.00	37.00	12.00 pts
Caitlyn	18.00	28.00	10.00 pts
Cindy	29.00	39.00	10.00 pts
Dian	14.00	25.00	11.00 pts
Kevin	10.00	21.00	11.00 pts
Nicole	7.00	20.00	13.00 pts
Steve	26.00	13.00	-13.00 pts
Tina	10.00	12.00	2.00 pts
Will	18.00	30.00	12.00 pts

Student	Test1	Test2	Difference (pts)
Audrey	30	42	12
Adina	25	37	12
Caitlyn	18	28	10
Cindy	29	39	10
Dian	14	25	11
Kevin	10	21	11
Nicole	7	20	13
Steve	26	13	-13
Tina	10	12	2
Will	18	30	12

And to graphical abstracts



Too many interior boxes clutter the layout.

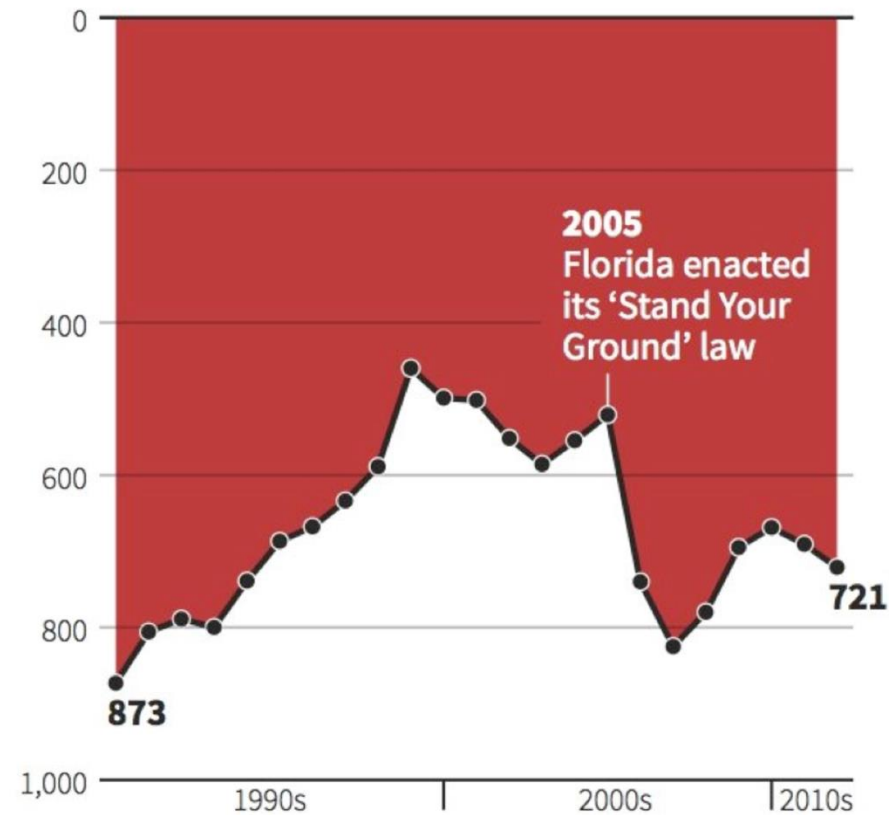


Boxes removed and arrows simplified.

Do gun deaths increase or decrease?

Gun deaths in Florida

Number of murders committed using firearms



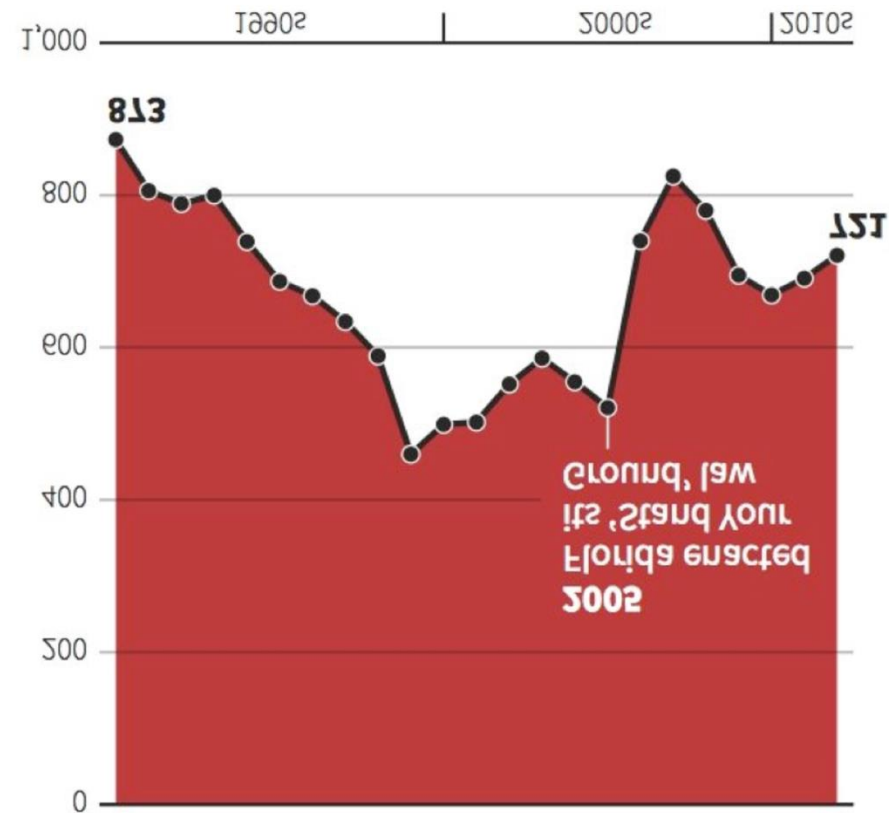
Source: Florida Department of Law Enforcement

Do gun deaths increase or decrease?

C. Chan 12/03/2014

REUTERS

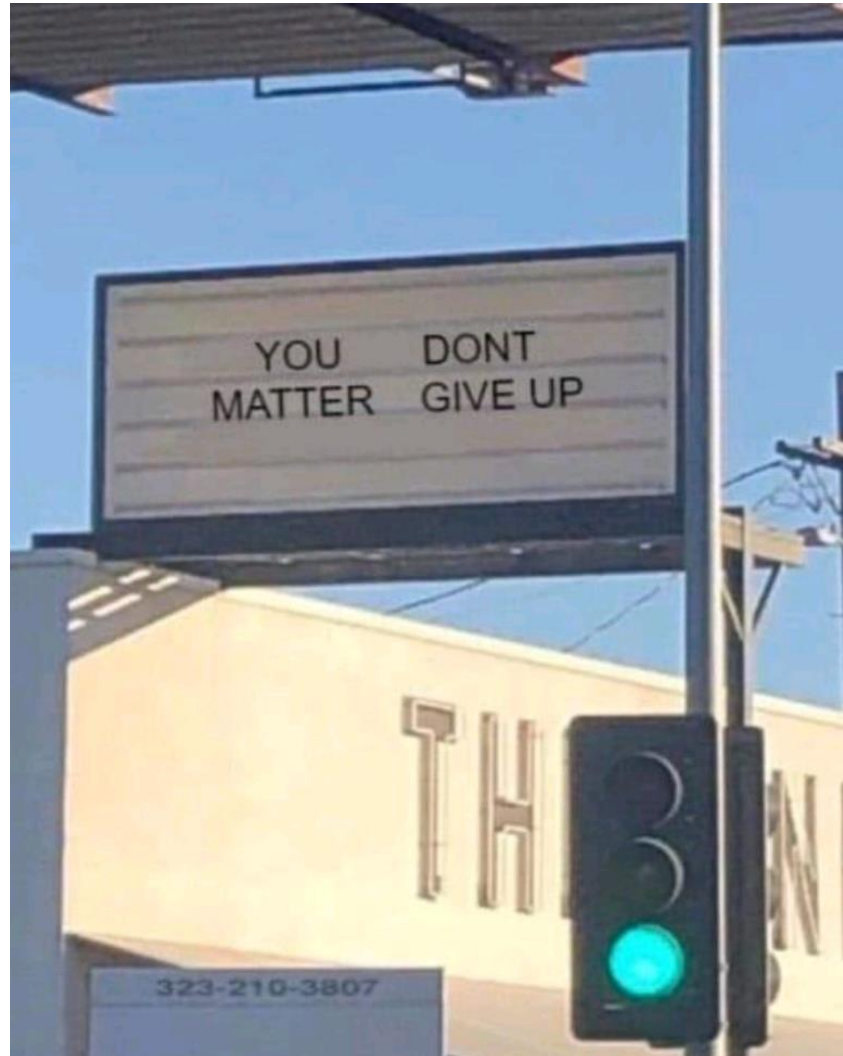
Source: Florida Department of Law Enforcement



Number of murders committed using firearms

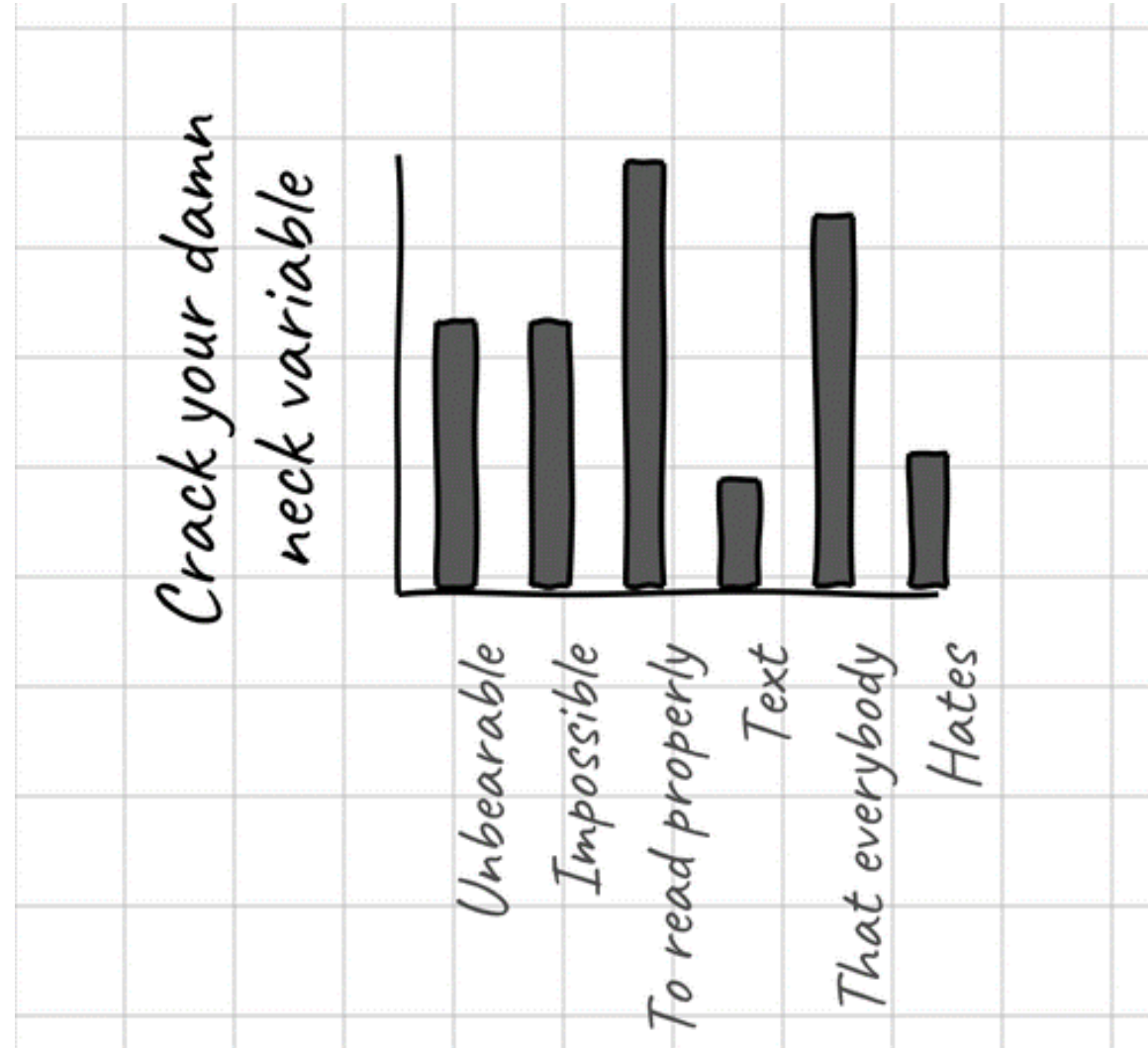
gun deaths in Florida

We read from left-to-right...

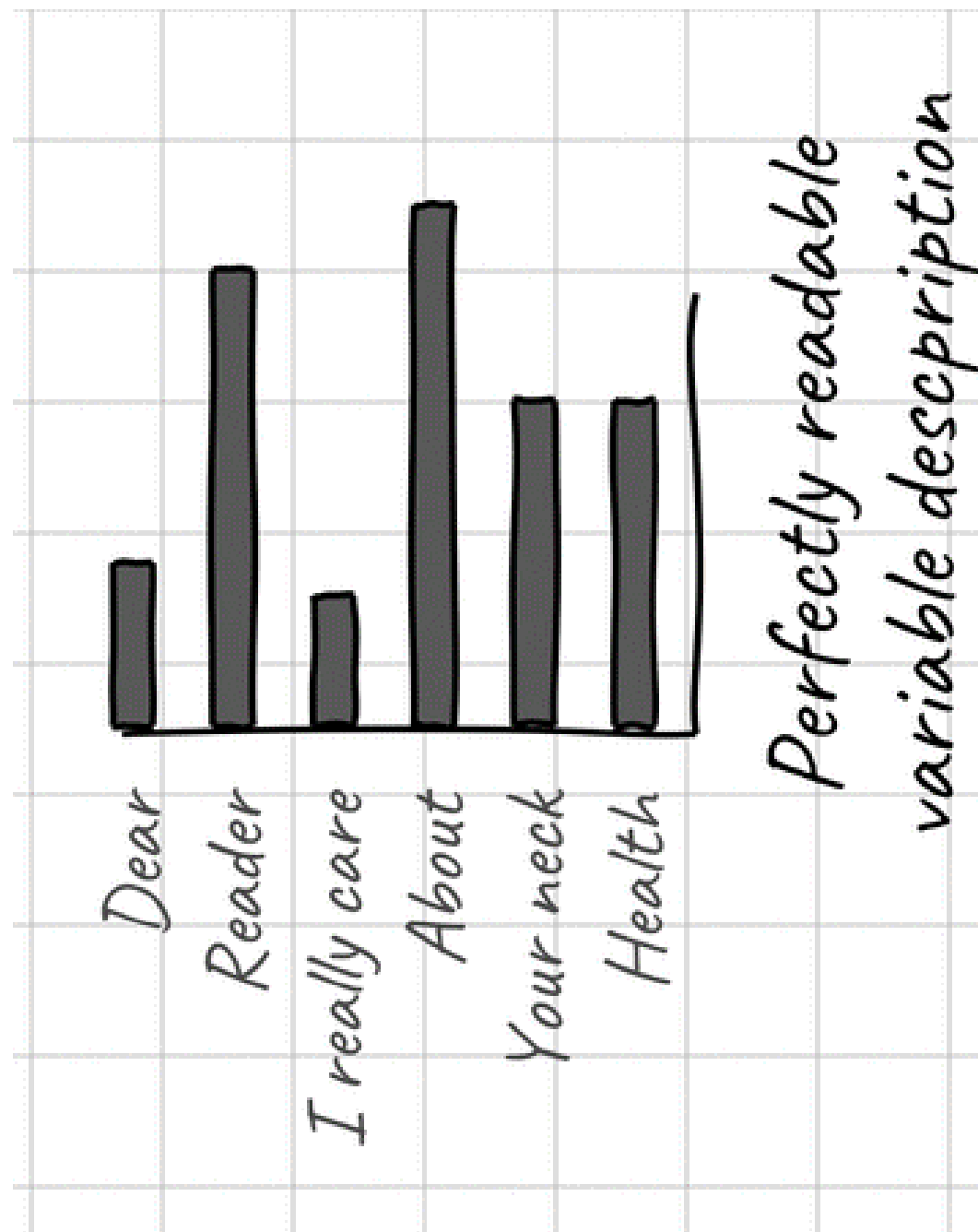


..right?

How's reading these labels?



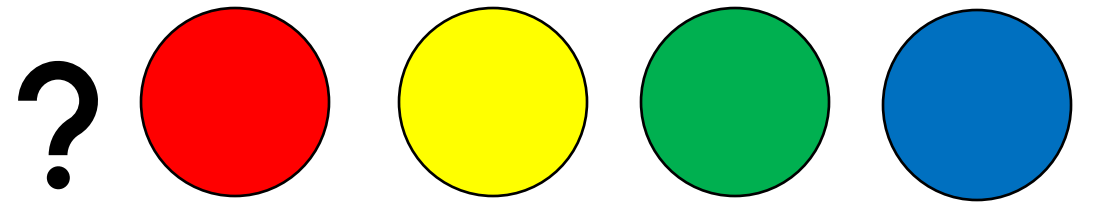
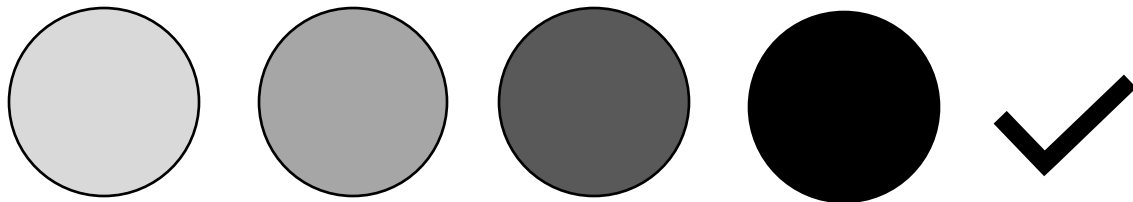
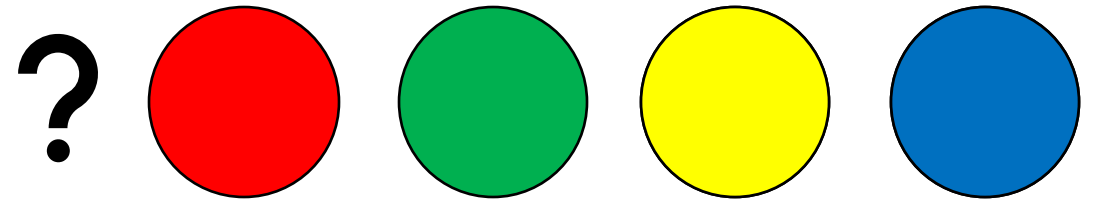
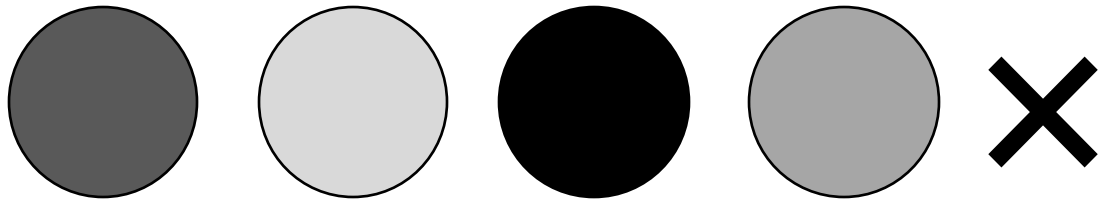
Better?



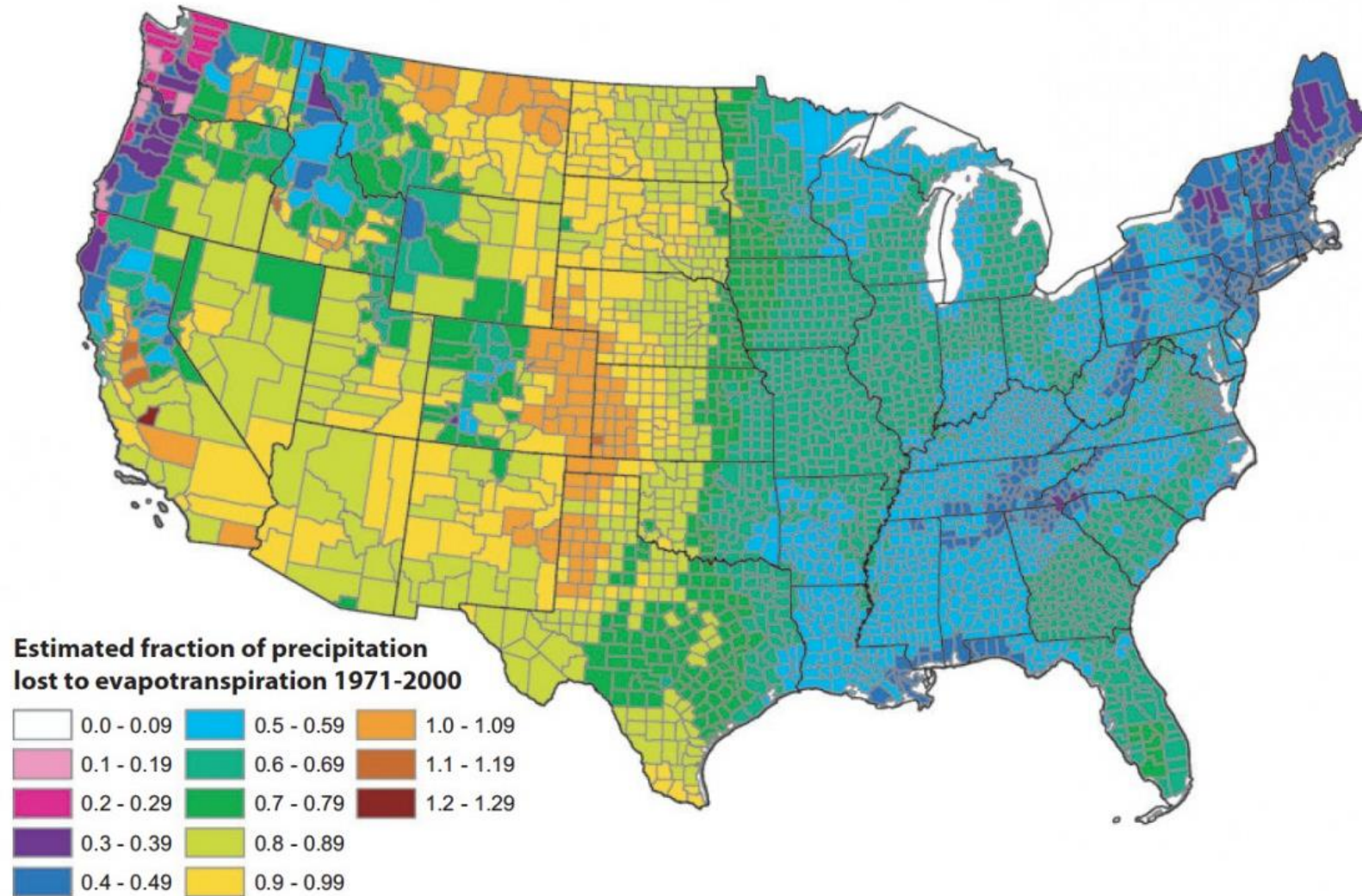
How to rank those circles from low to high?



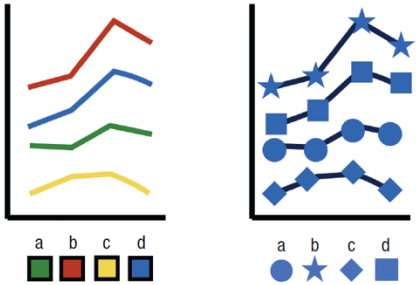
How to rank those circles from low to high?



Avoid rainbow color maps for quantities!



The **weaknesses** of our brain - Summary



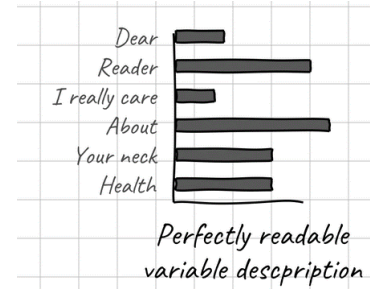
1.

Use our short-term
memory



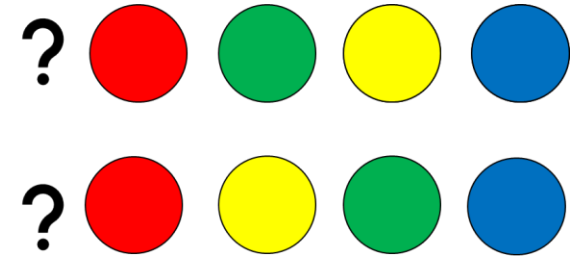
2.

Process too much
information at once



3.

Go against the natural
flow



4.

Correct quantification of
color

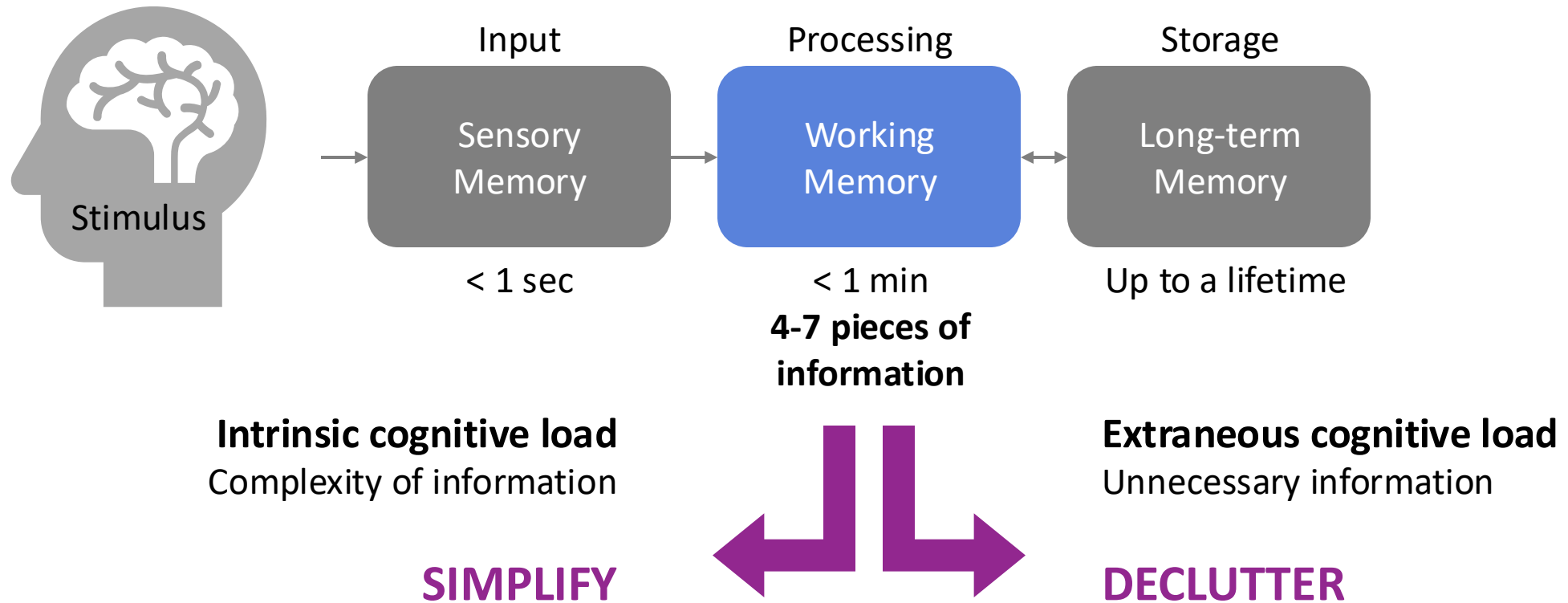


<https://declutterme.london>

Simplify and Declutter

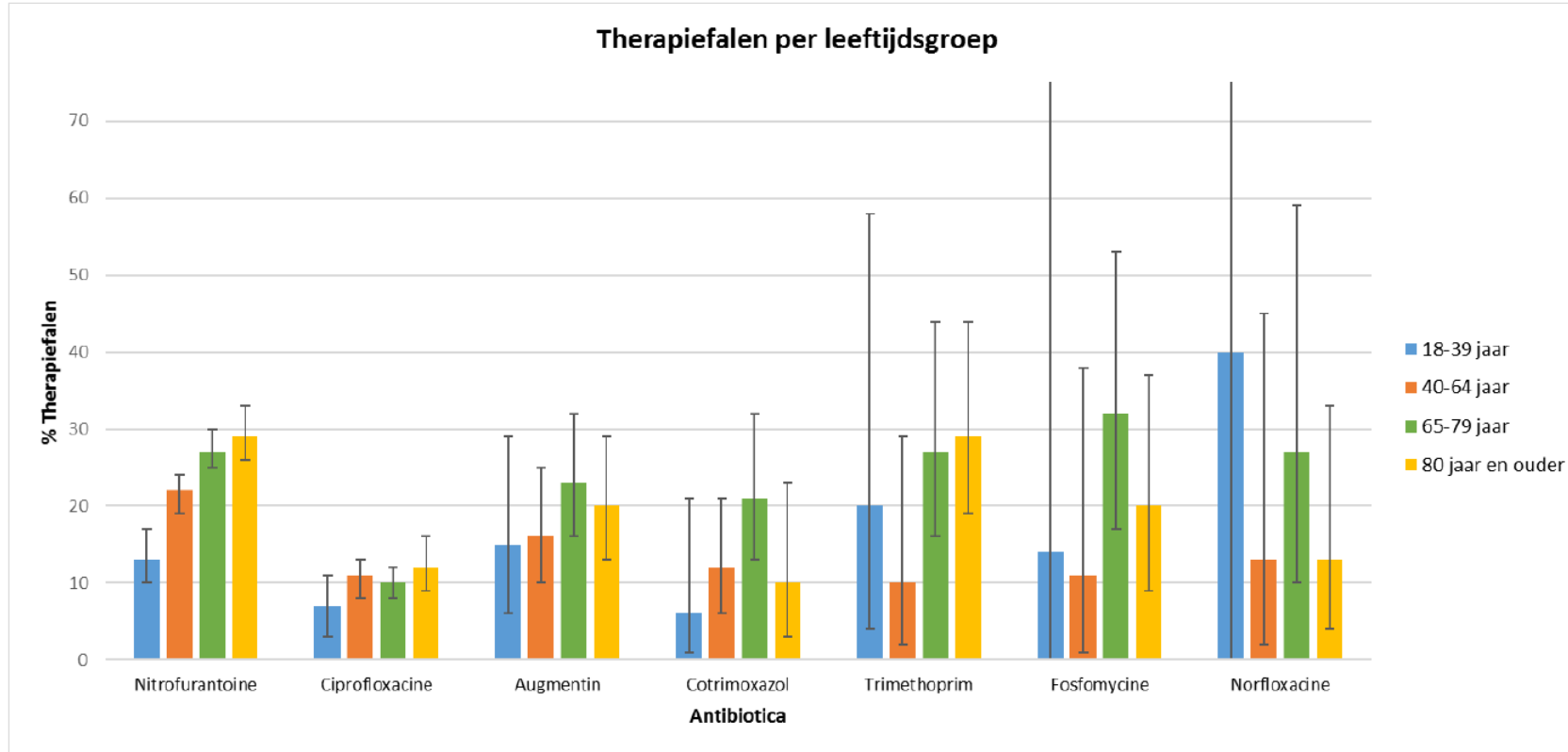


Optimize visual perception: minimize *intrinsic* and *extraneous* cognitive load



Simplify: Show differences directly and deprioritize unnecessary information

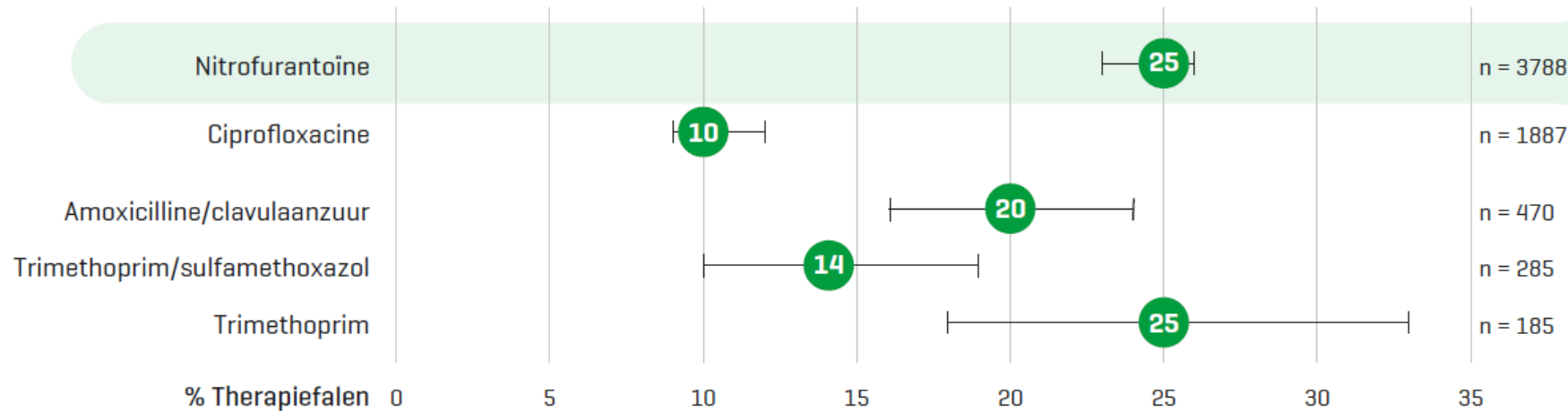
Figuur 1 Therapiefalen voor verschillende antibiotische regimes per leeftijdsgroep



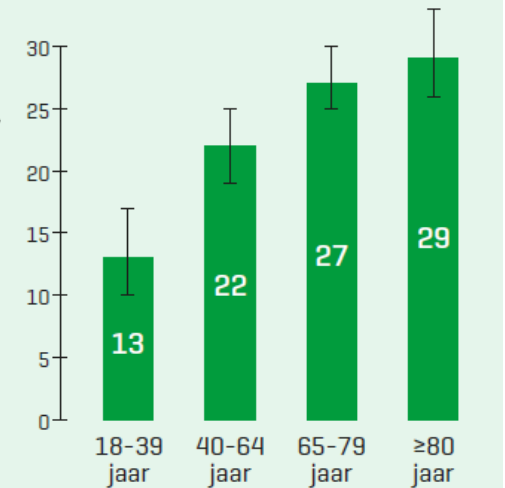
Simplify: Show differences directly and deprioritize unnecessary information

Voorschriften en faalpercentages van verschillende antibiotica bij mannen met cystitis

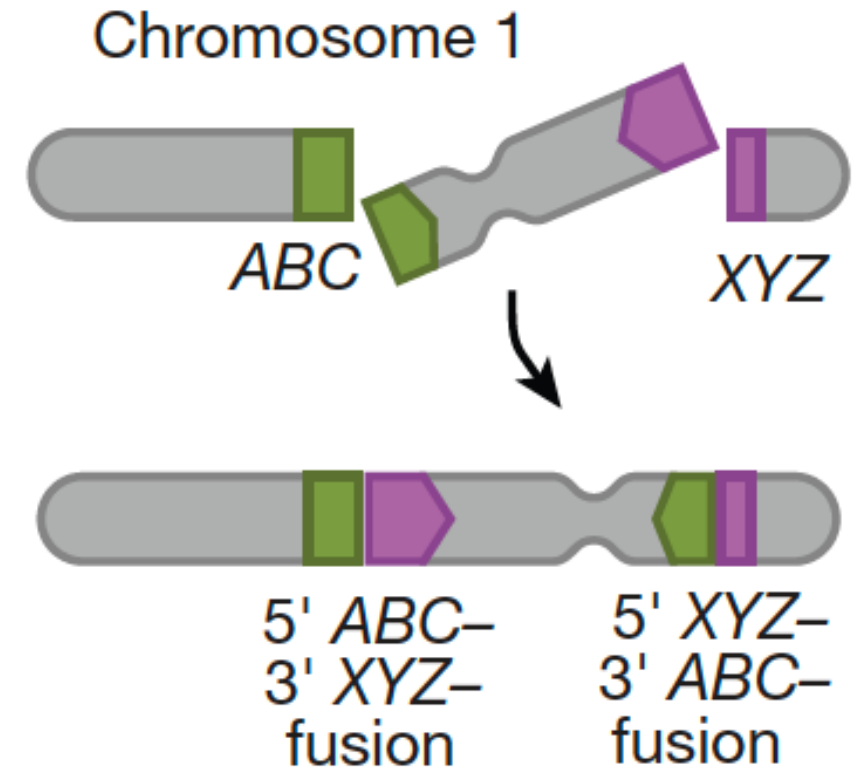
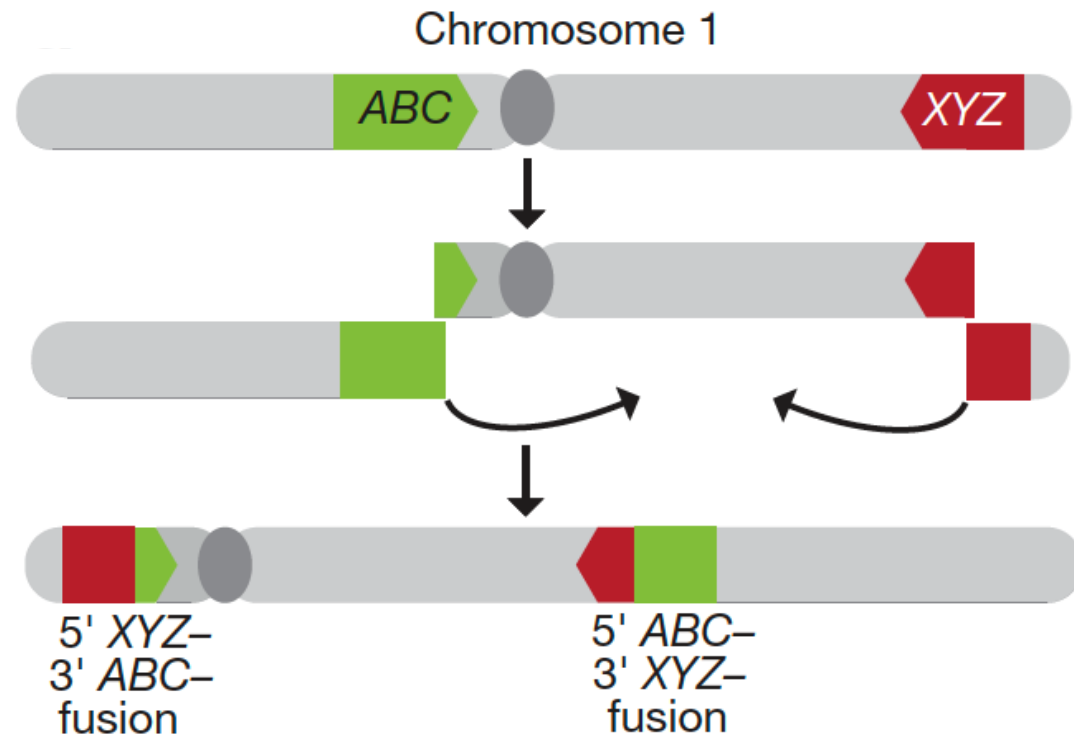
*'Nitrofurantoïne,
de eerste keus volgens de NHG-Standaard,
faalde in 25% van de episodes'*



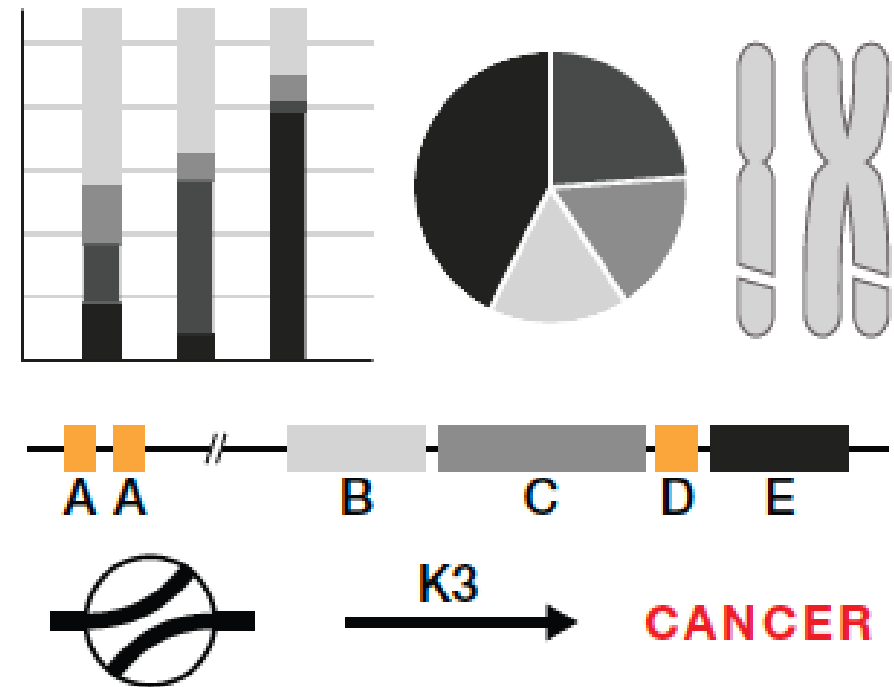
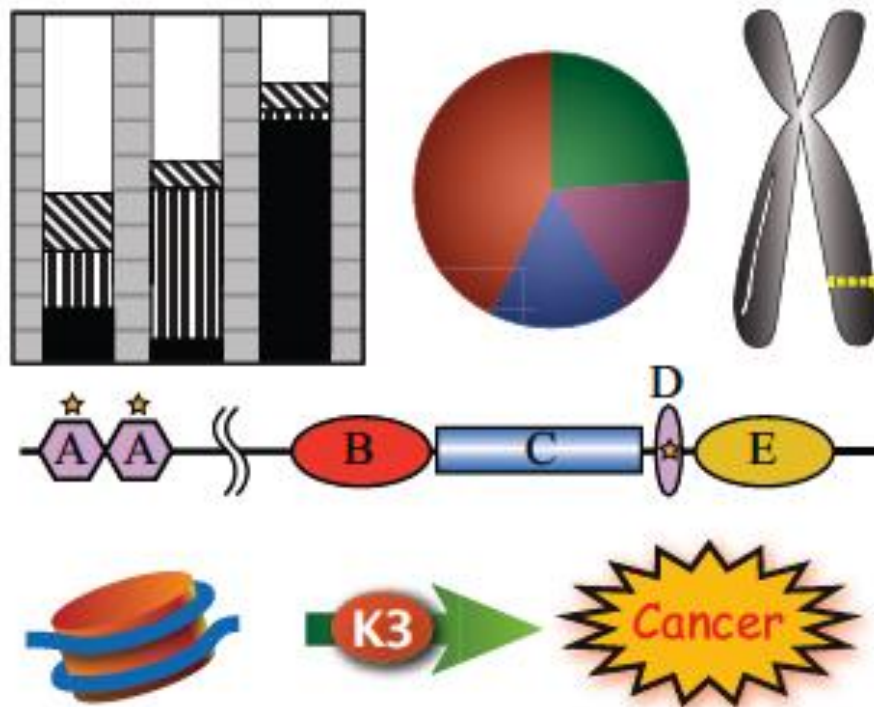
*'Therapiefalen voor
nitrofurantoïne neemt toe
met de leeftijd'*



Simplify to clarify: simple $\neq$ simplistic



Simplify: Use the **simplest** visual representation and reserve the use of color for **emphasis**



Declutter: Minimize distractions



Effective. Not optimal.



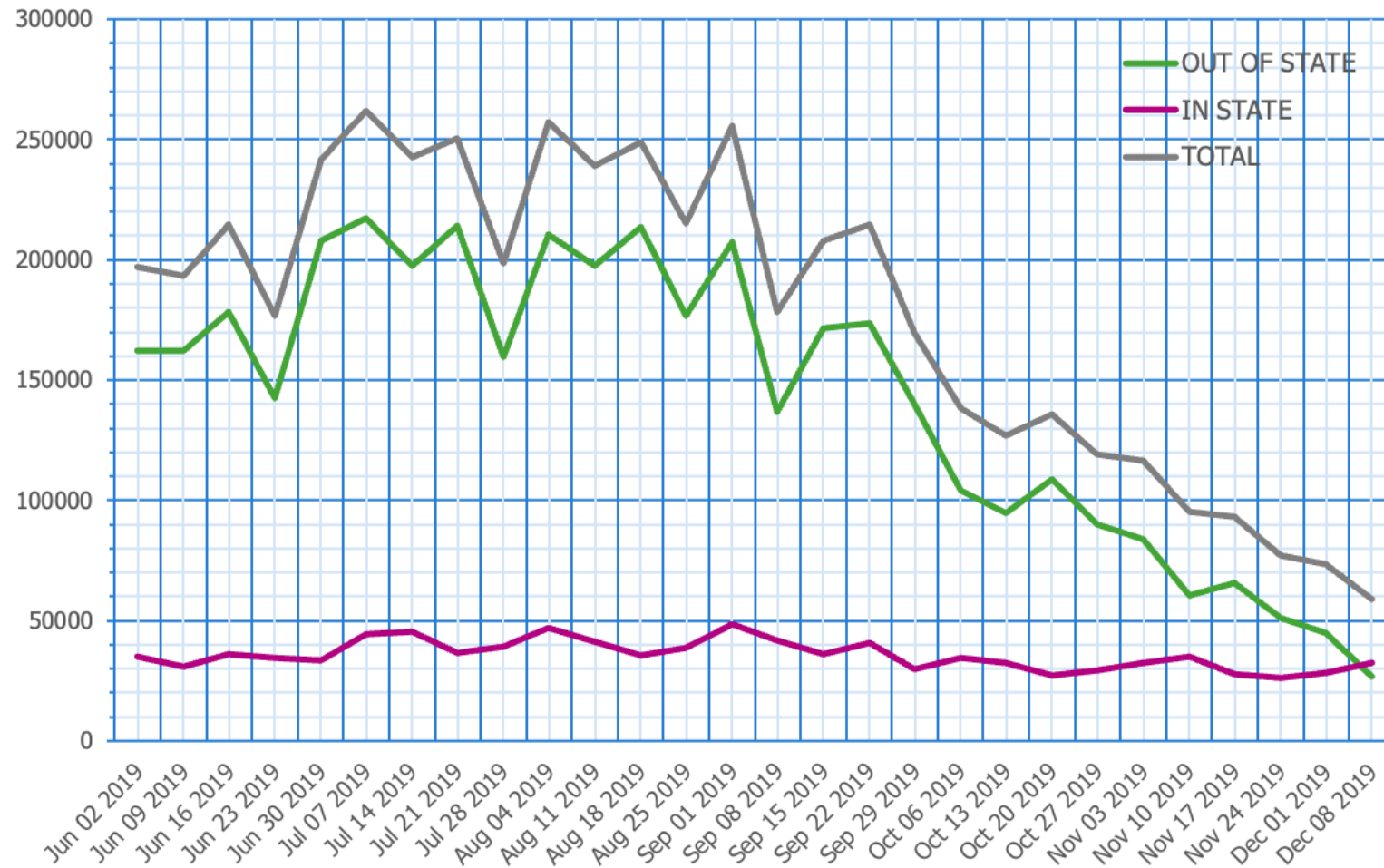
Simple, effective, optimal.



Bad.

Declutter: Minimize distractions

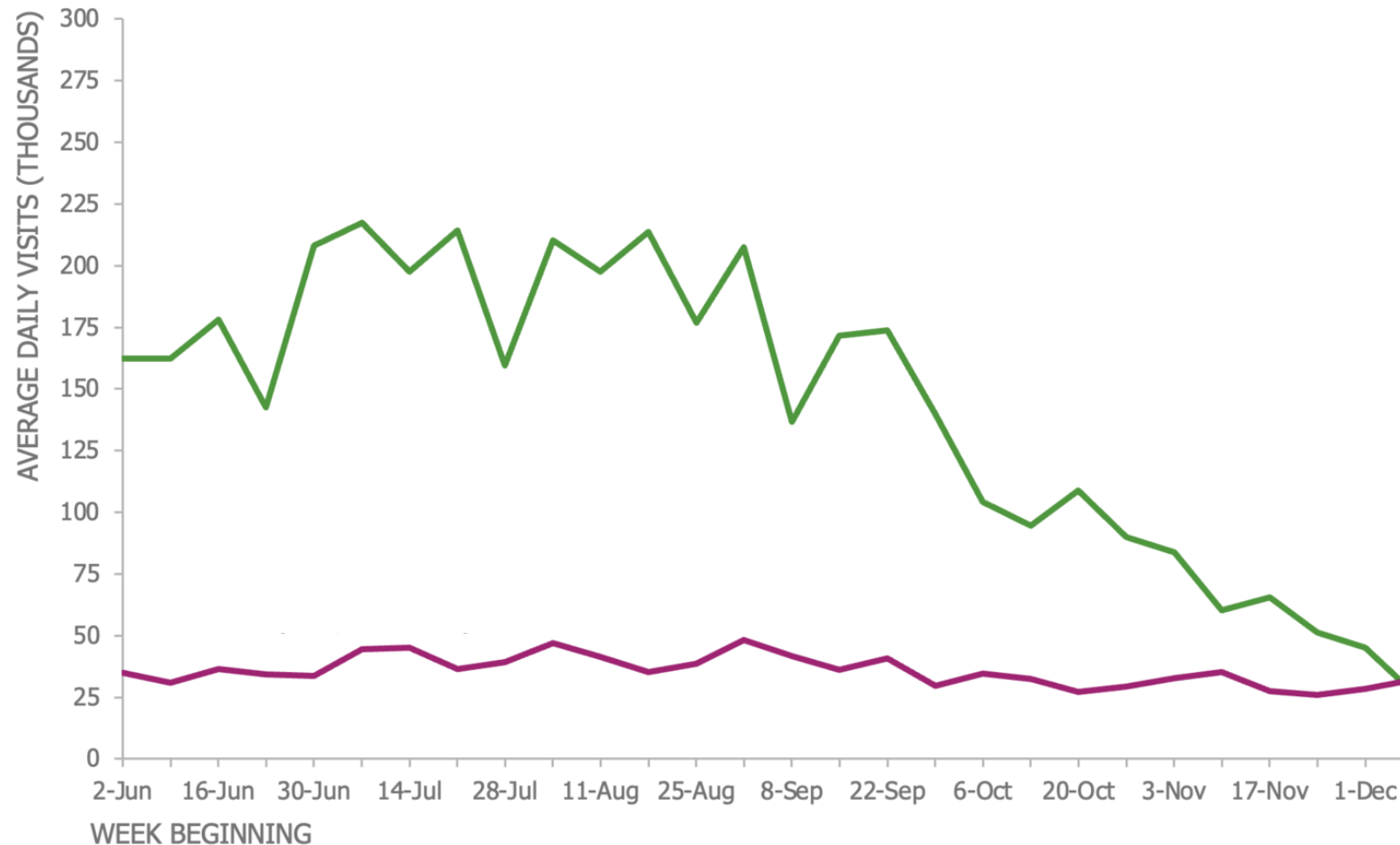
Daily Average Park Visitors By Week



Declutter: Minimize distractions

Daily visitors to Aurora Park in summer/fall 2019

VALUES ARE CALCULATED WEEKLY AS A 7-DAY AVERAGE



Exercise: Simplify and Declutter



Simplify: Enable direct (fast) comparisons

Declutter: Eliminate irrelevant distractions



<https://declutterme.london>

In groups - look at each others visual and discuss:

- ☐ What is your first thought, feeling or observation when you look at the visual?
- ☐ What takes most time to process? Could this process be simplified?
- ☐ **Are there any unnecessary, redundant or distracting elements that can be removed/reduced?**
 - E.g., boxes, borders, lines, text, colors, shades, shapes, etc...
- ☐ How could the message be more emphasized (or clarified)?
- ☐ Are all text and labels legible?

Part 3: **The Narrative**

“Maybe stories are just data with a soul” – Brené Brown

From data to **datastory**

1 Data Collection



2 Data Preparation



3 Data Visualization



4 Data Analysis



5 Data Storytelling



3 essential elements of data **STORIES**



3 essential elements of data **STORIES**

Explain

- What's happening?
- Why is it important?
- Add structure and guide audience

Enlighten

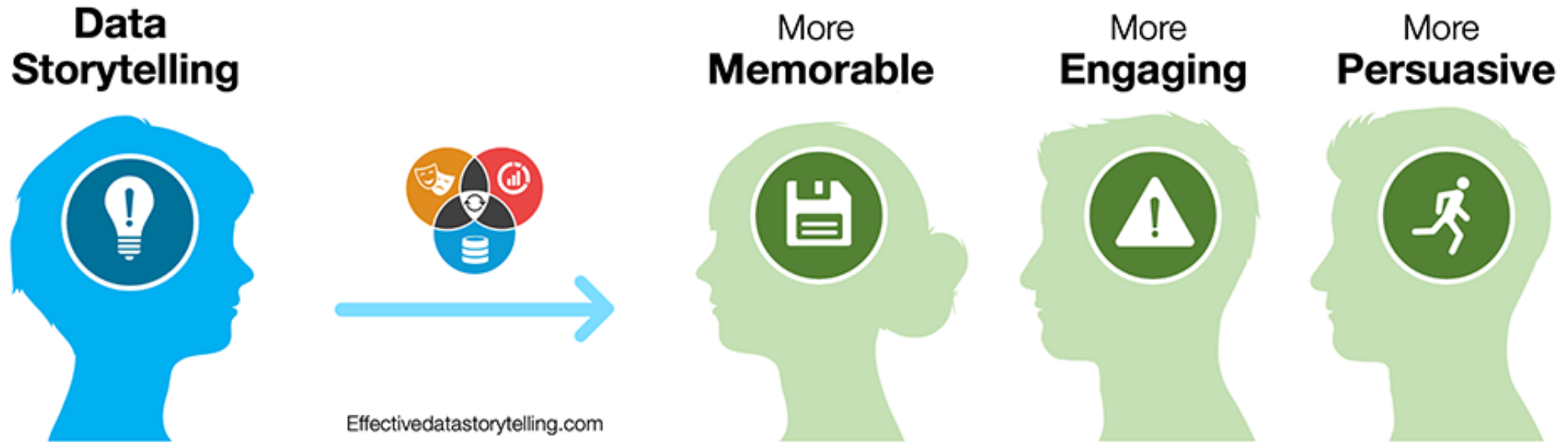
- Provide insights that would remain unseen without graphs
- Interesting patterns or outliers

Engage

- Offers different perspectives



We hear statistics, but we *feel* stories...



Words are how we think, stories are how we link – Christina Baldwin

Notice and Wonder

Notice (< 15 sec):

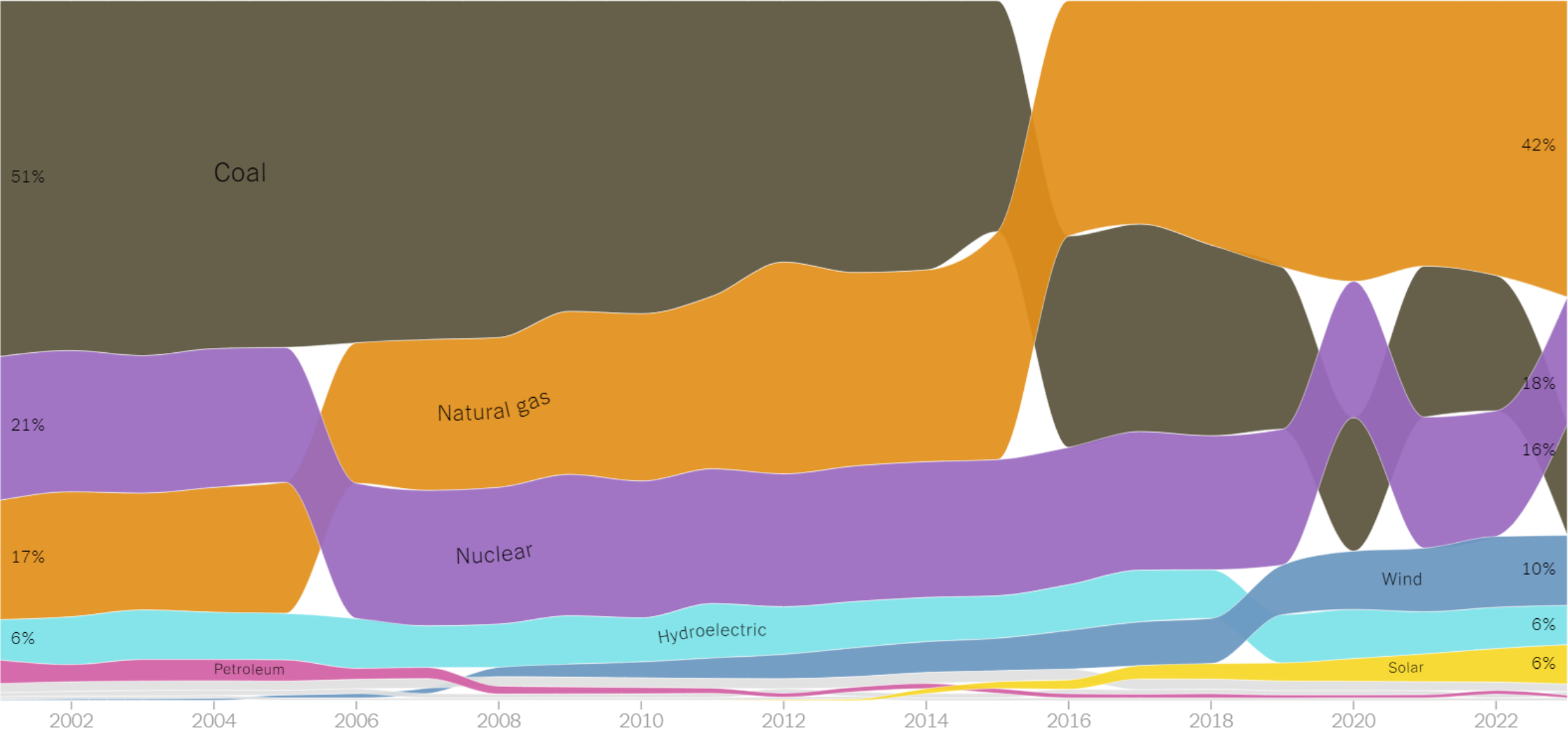
- What do you see? What stands out? What is the first thing that comes to your mind?
- This can be a **thought**, a **color**, a **data point**, a **feeling**, an **impression**, a **word**, an **element**, etc.
- It can be anything: Don't think, just say it out loud!

Wonder

- What are you curious about?
- What needs clarification?

How **the United States** made electricity from 2001 to 2023

Percentage of power produced from each energy source

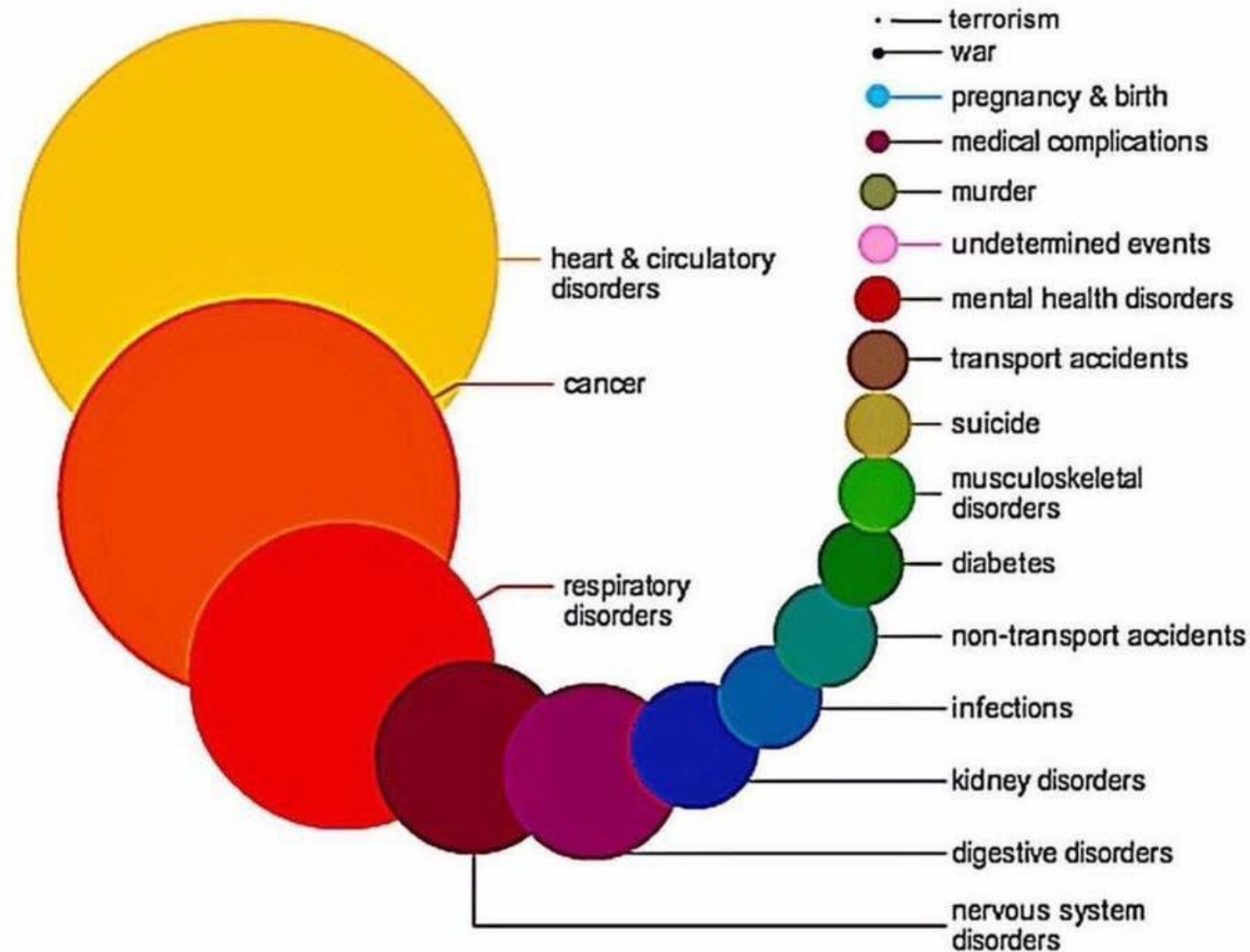


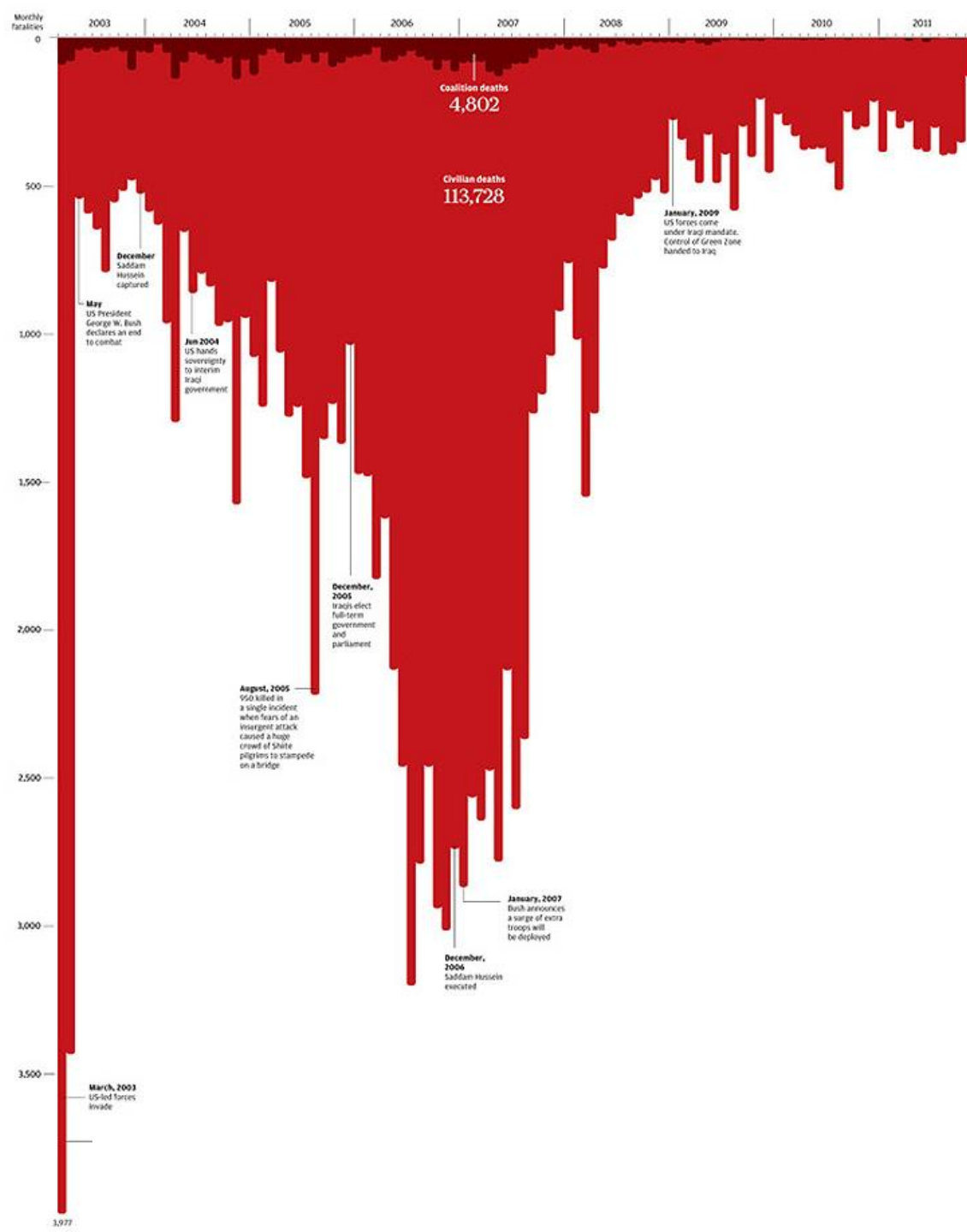
Technomass
1,300 Gt

Biomass
1,100 Gt

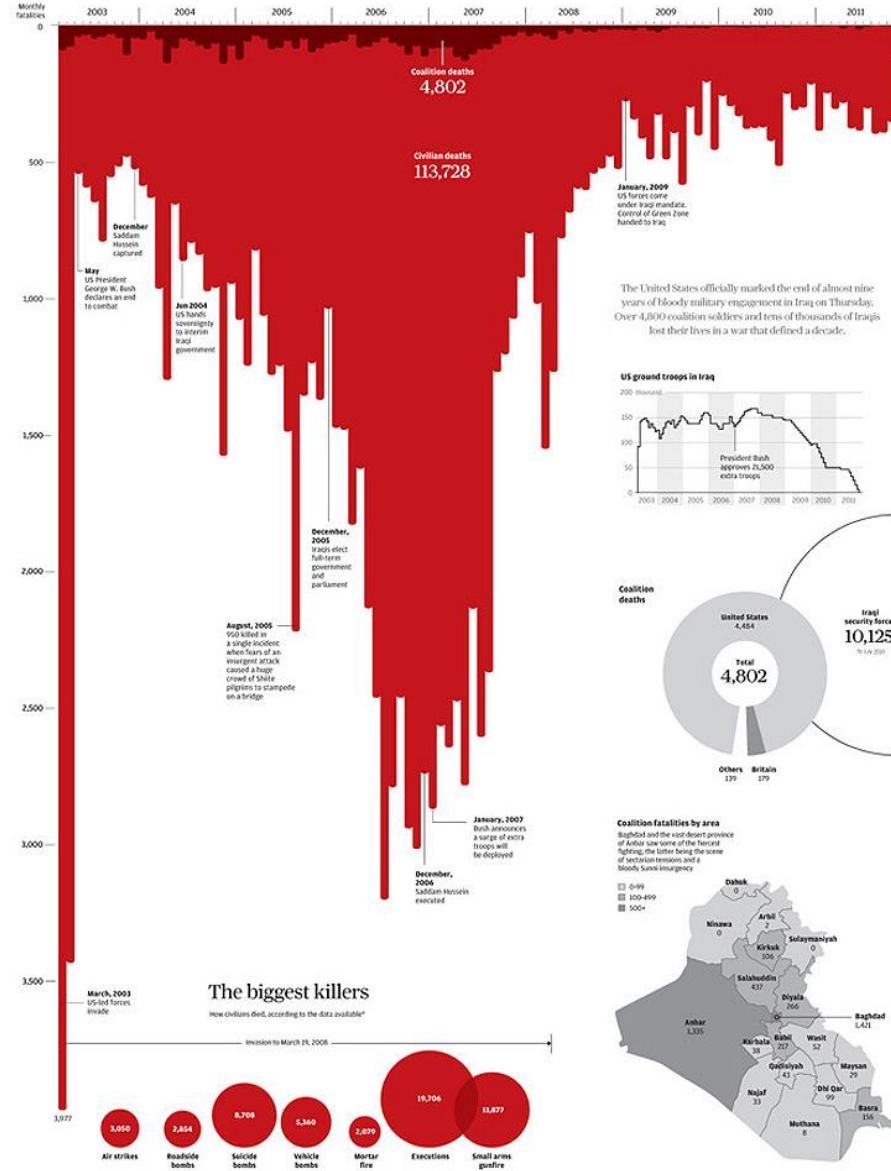


Leading causes of death in perspective





Iraq's bloody toll



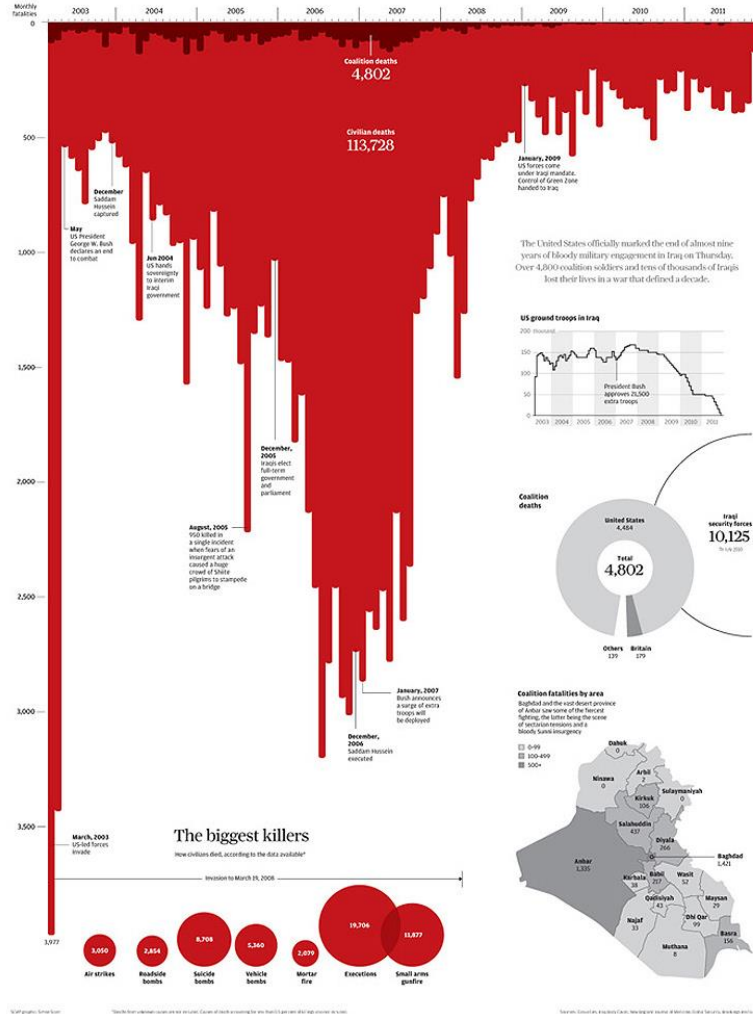
*AFP graphics, Simon Scarr

*Specs from unconfirmed sources are not included. Causes of death accounting for less than 10 per cent of total deaths are not shown.

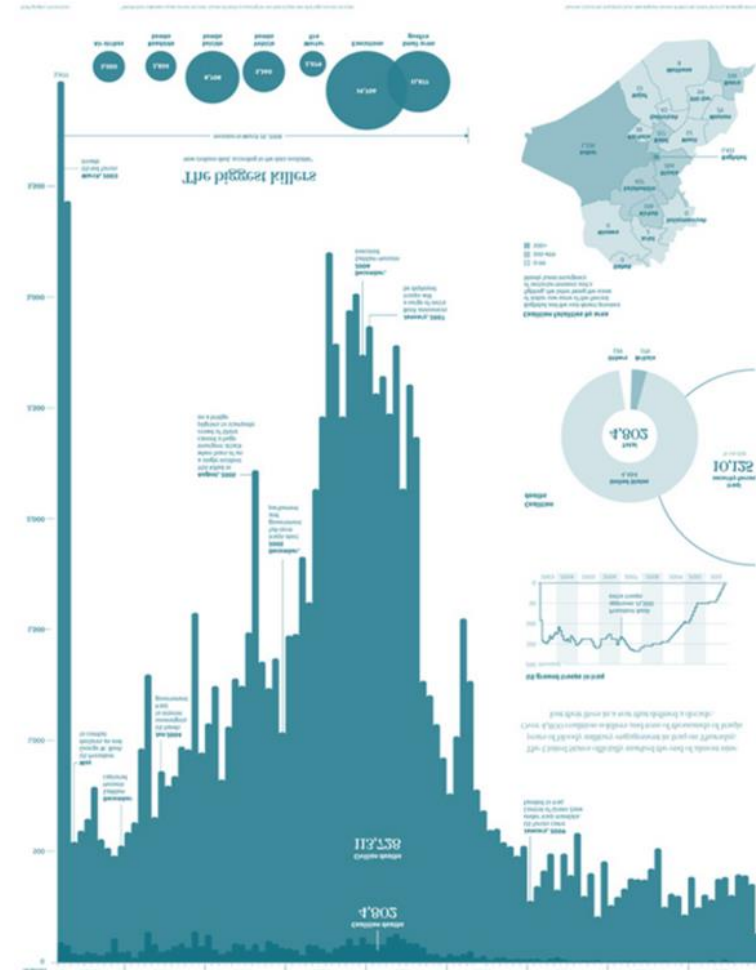
Sources: Reuters, Associated Press, CNN, New England Journal of Medicine, British Medical Journal, Washington Post

Same data, different story

Iraq's bloody toll



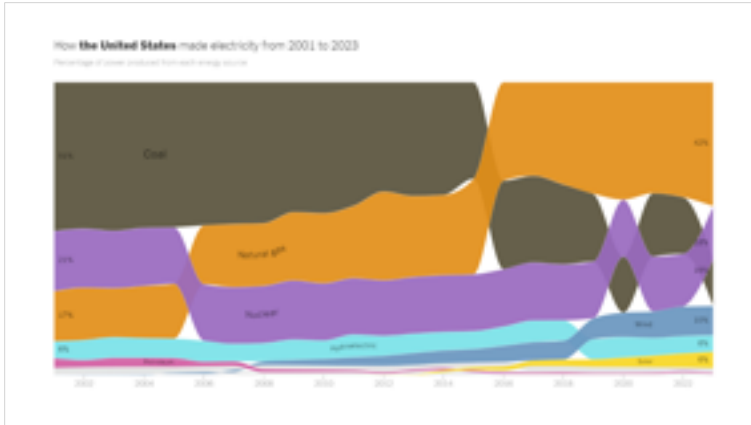
Iraq: Deaths on the decline



Simon Scarr, December 17, 2011
South China Morning Post

Andy Cotgreave, October 16, 2014
<https://gravyanecdote.com/uncategorized/should-you-trust-a-data-visualisation/>

Exercise: think of a catchy headline



“Shifting Winds: The Current of Energy Production in America”

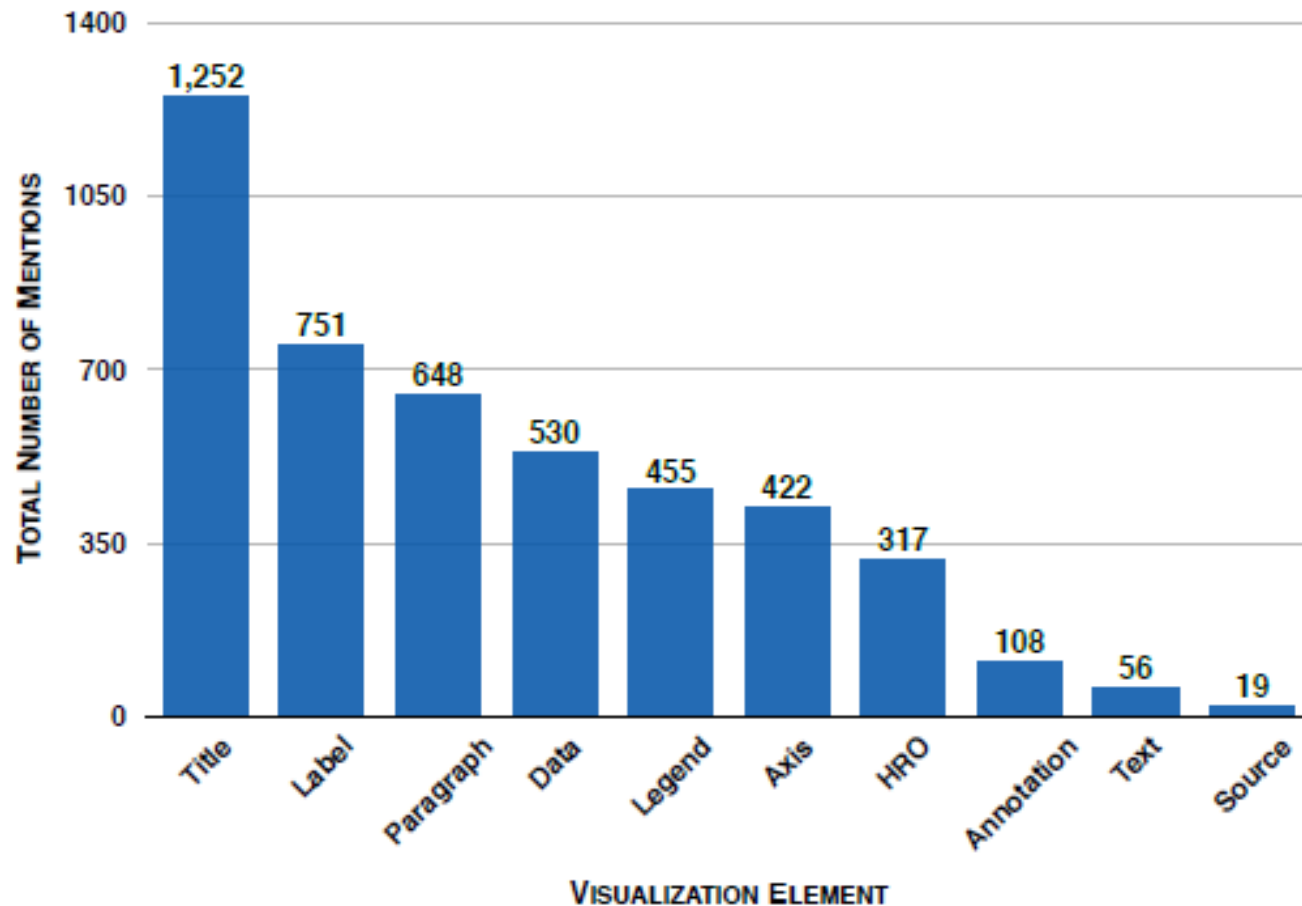


“Humans Are Small, but Our Footprint Is Massive”



“The things most likely to kill you”

A strong title makes your visual more memorable

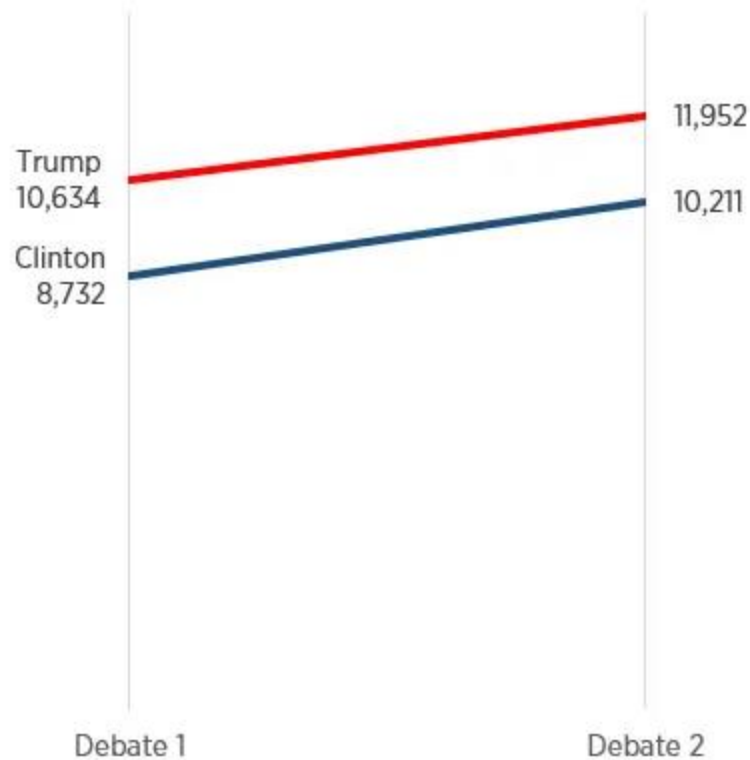


Rank:	1st	2nd	3rd
Overall	Title (46%)	Label (27%)	Paragraph (24%)
Infographics	Title (72%)	Label (28%)	Paragraph (24%)
News	Paragraph (45%)	Title (43%)	Label (33%)
Government	Title (55%)	Legend (26%)	Data (21%)
Science	Label (27%)	Axis (14%)	Legend (13%)

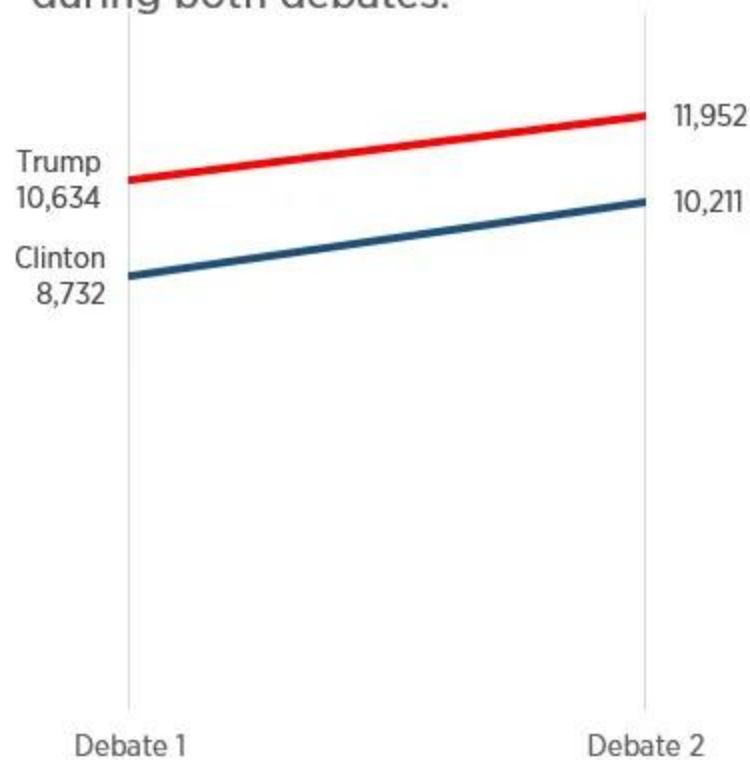
The title is the reader's first impression

Describe ≠ Judge

Words Spoken at Debate

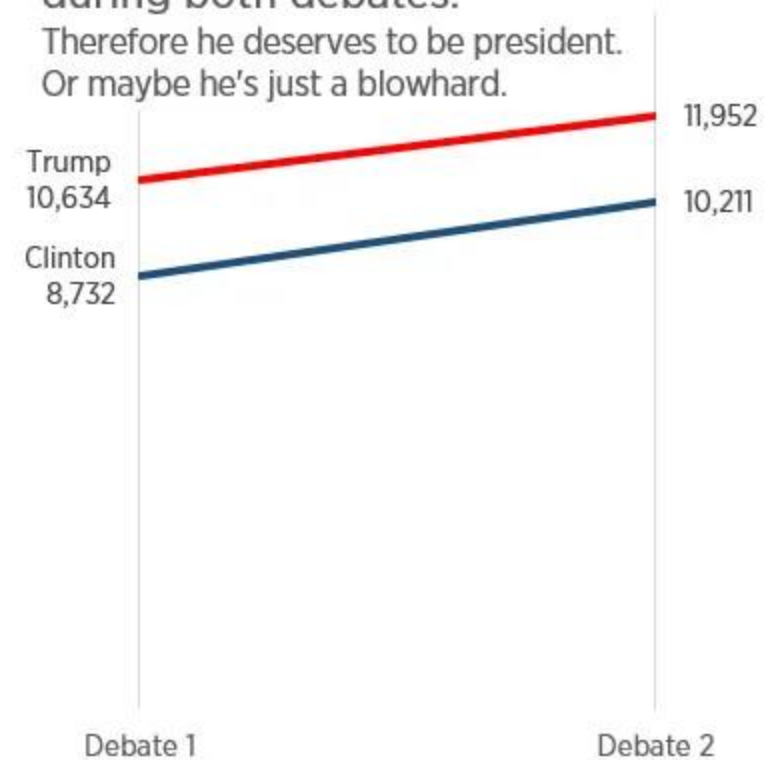


Trump spoke more words during both debates.



Trump spoke more words during both debates.

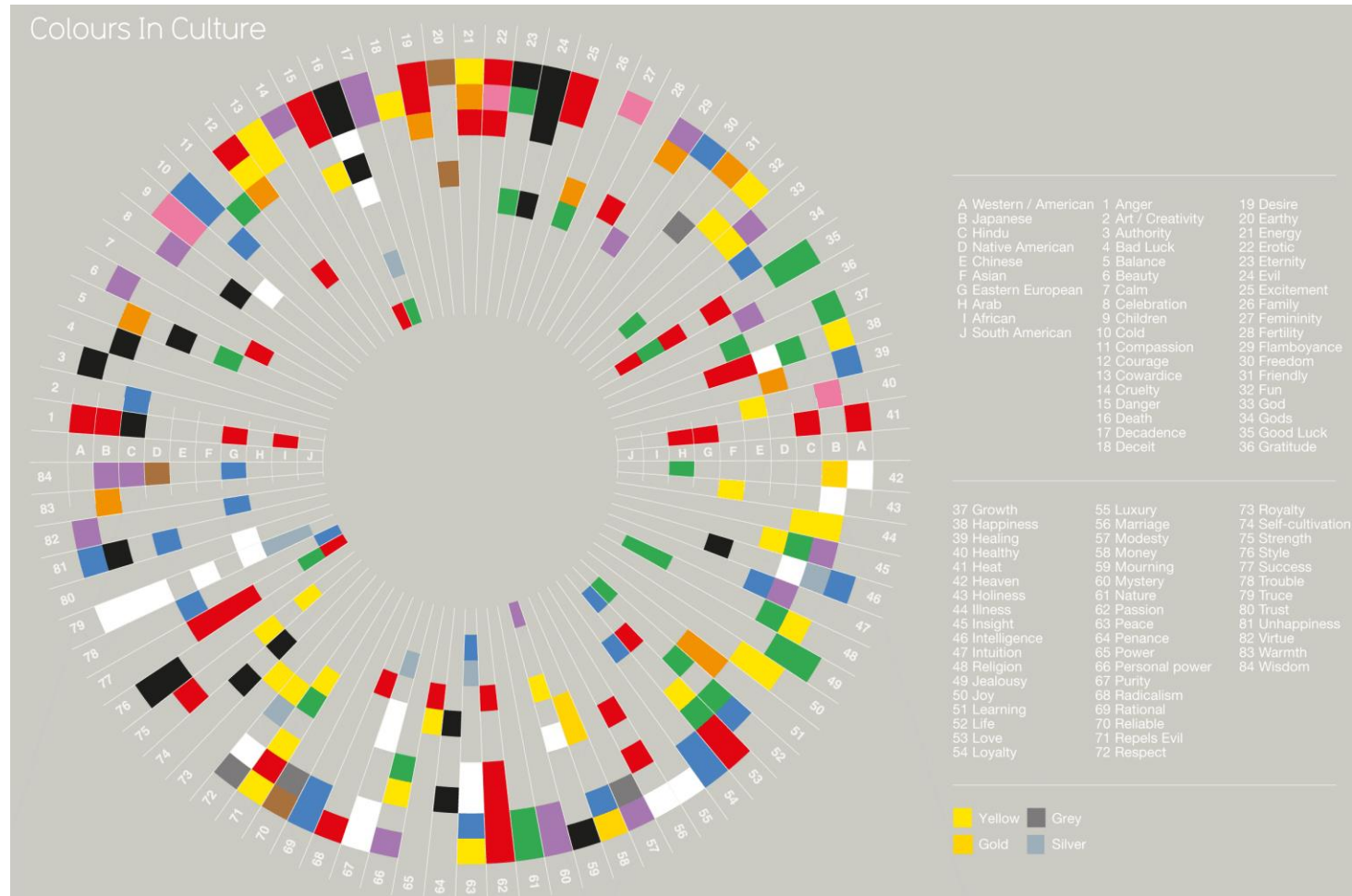
Therefore he deserves to be president.
Or maybe he's just a blowhard.



Color is associated to feelings...



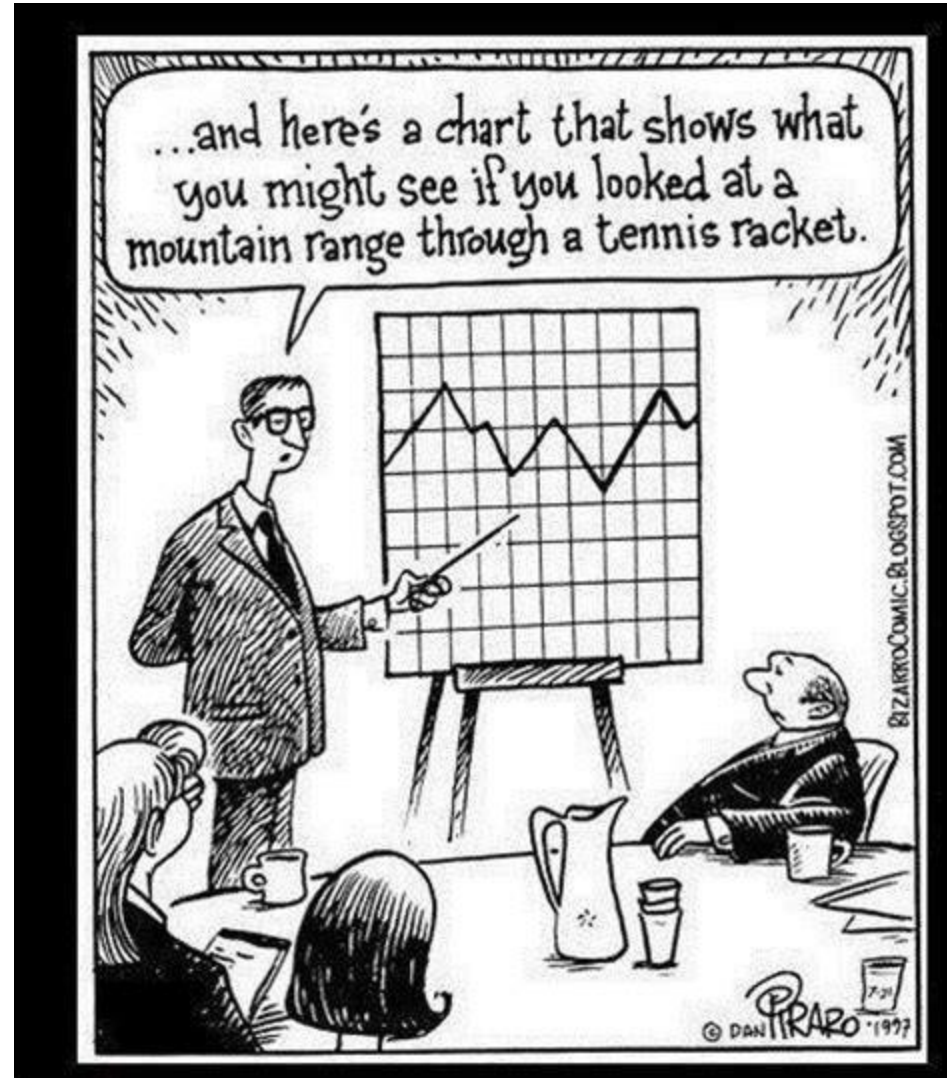
...but differ between cultures!



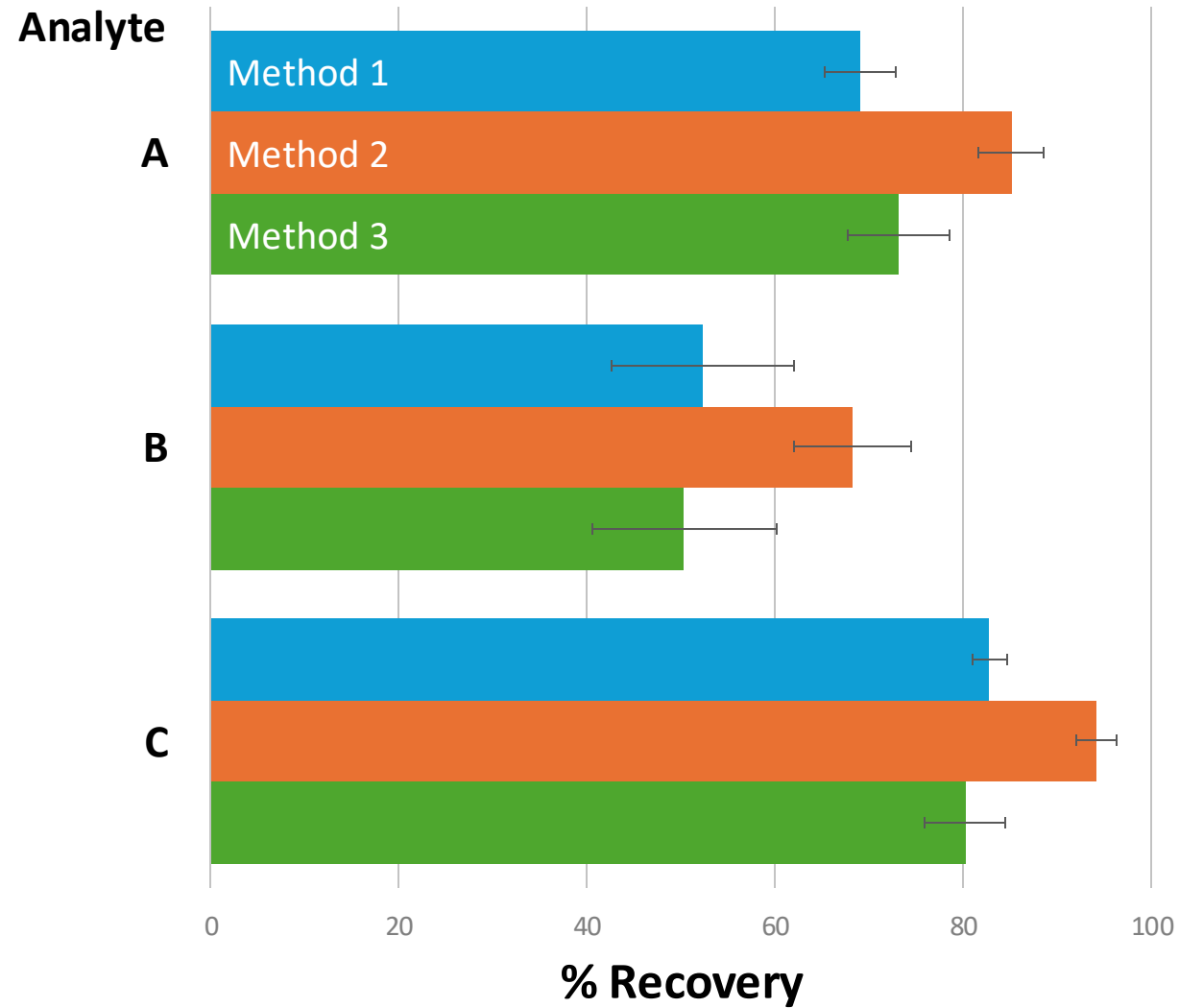
David McCandless, Colours in Culture

<https://www.informationisbeautiful.net/visualizations/colours-in-cultures/>

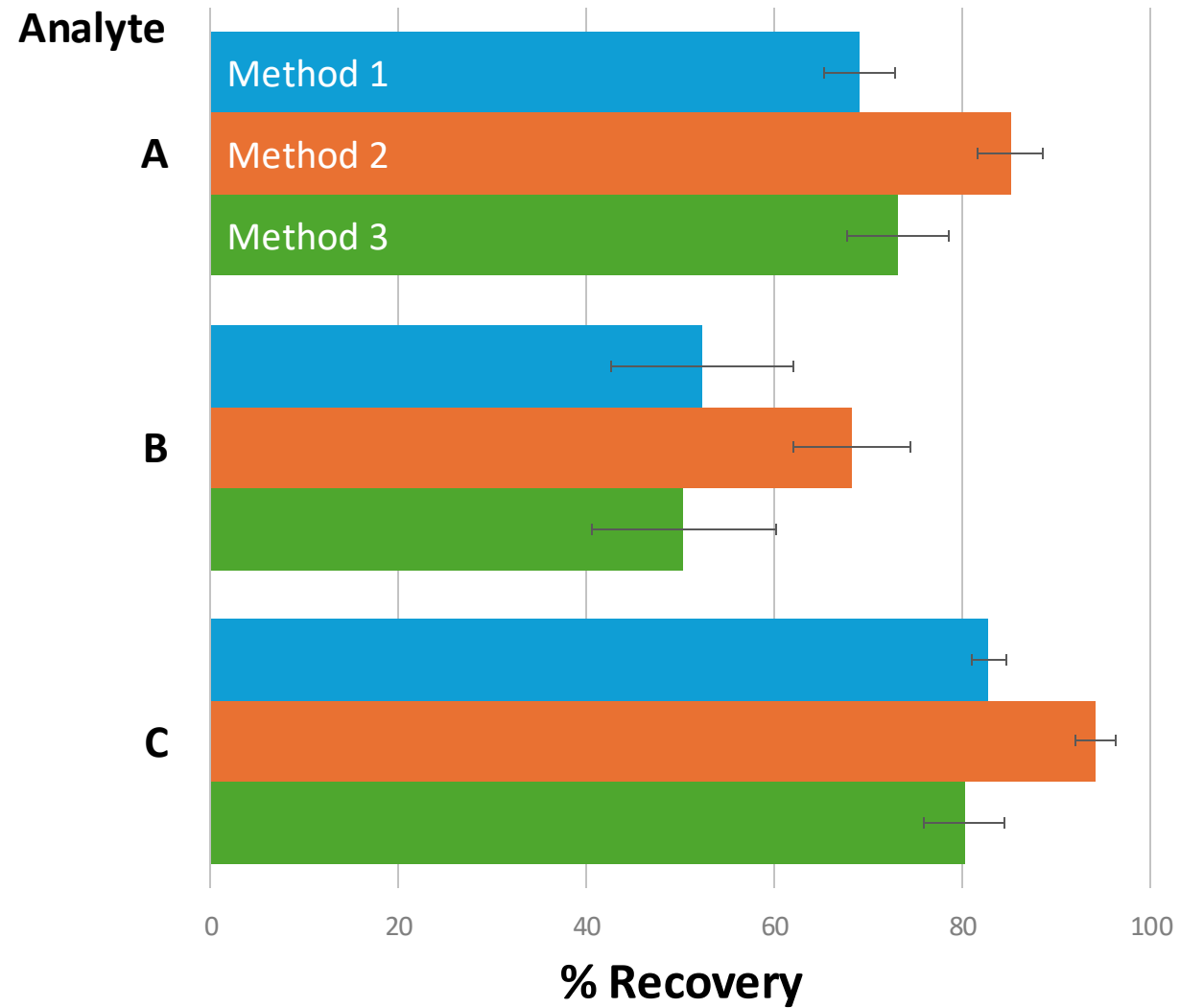
Explaining is much easier when presenting



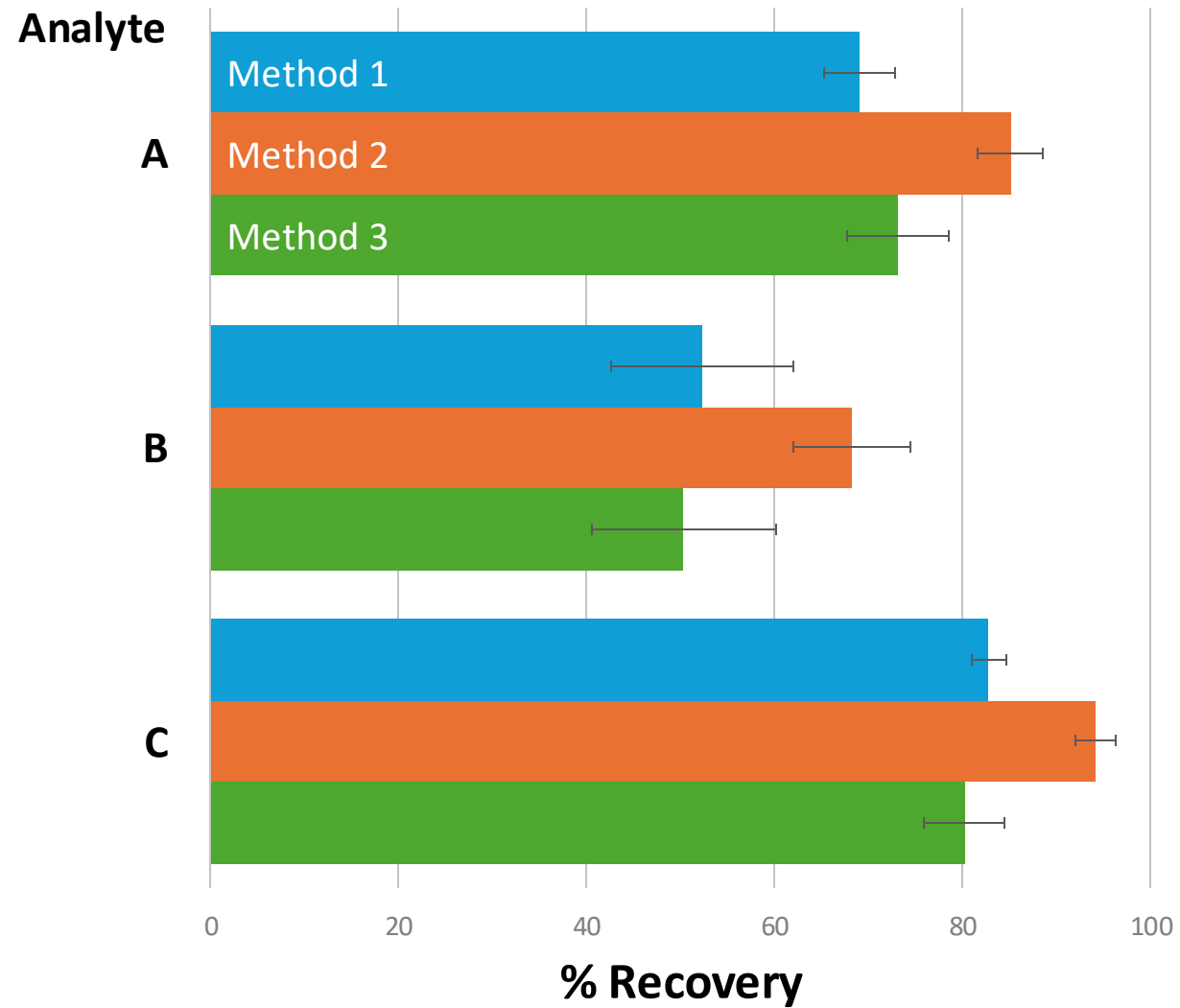
Avoid overwhelming your audience with data



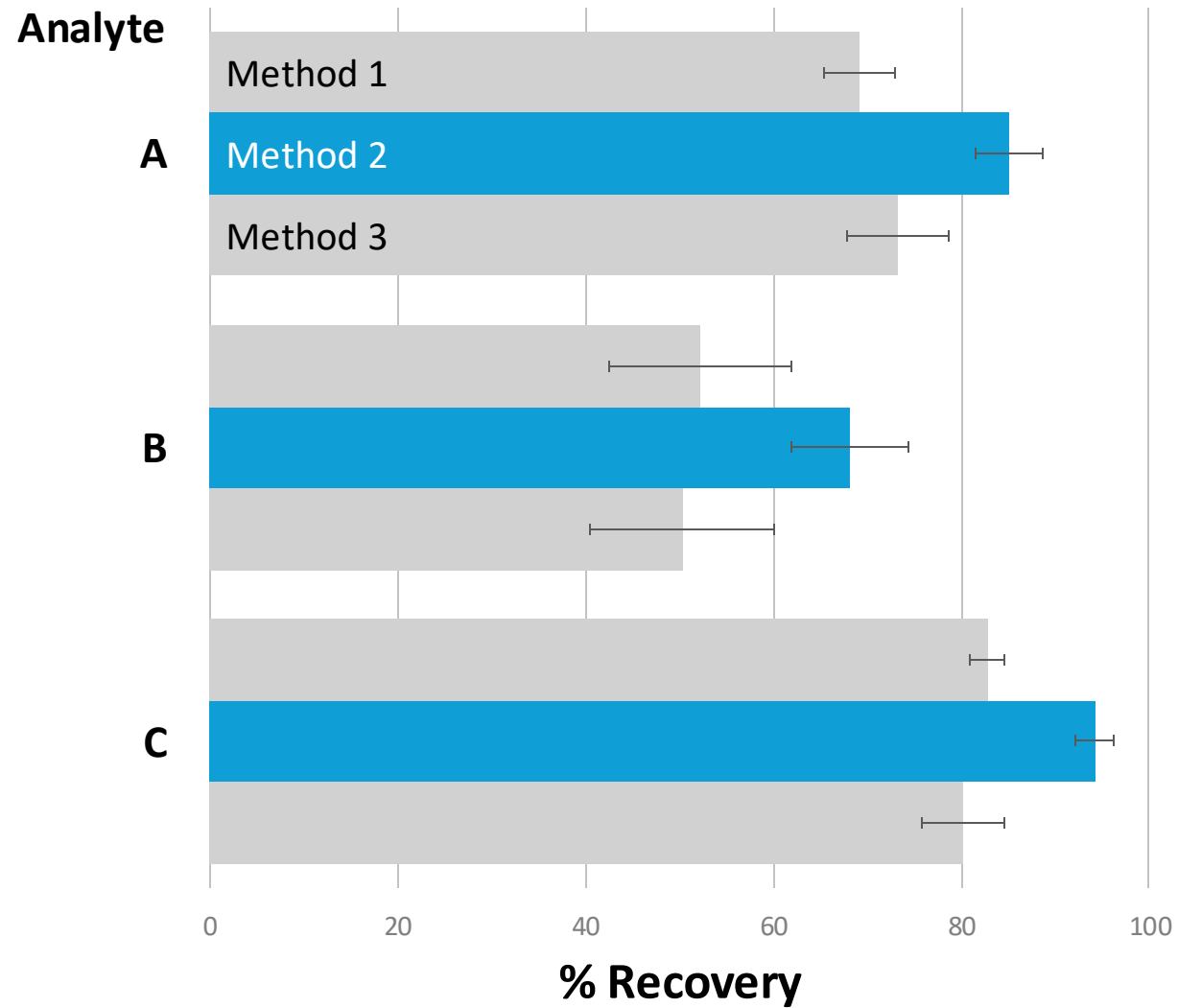
Animate your graph - by series



Animate your graph - by category



Highlight what is important

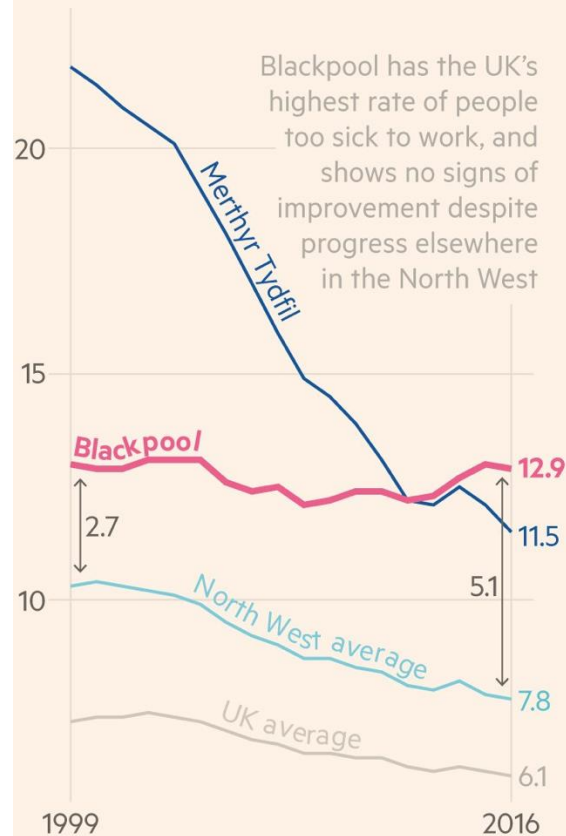


Method 2
showed the
highest recovery
for all analytes

Annotations help to clarify what's important

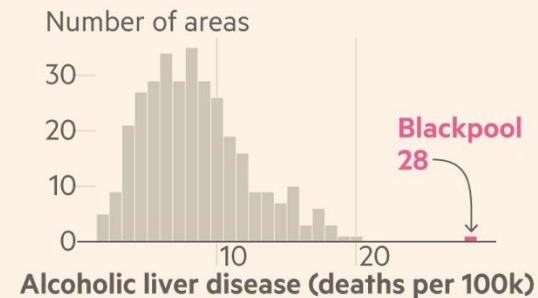
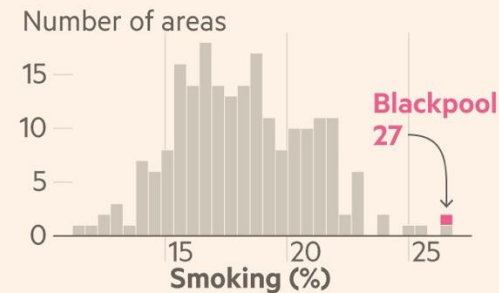
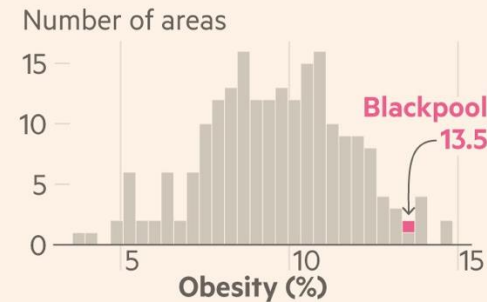
Economically and physically, **Blackpool** is unwell

Percentage of working-age population in receipt of incapacity benefits (IB) or employment support allowance (ESA)



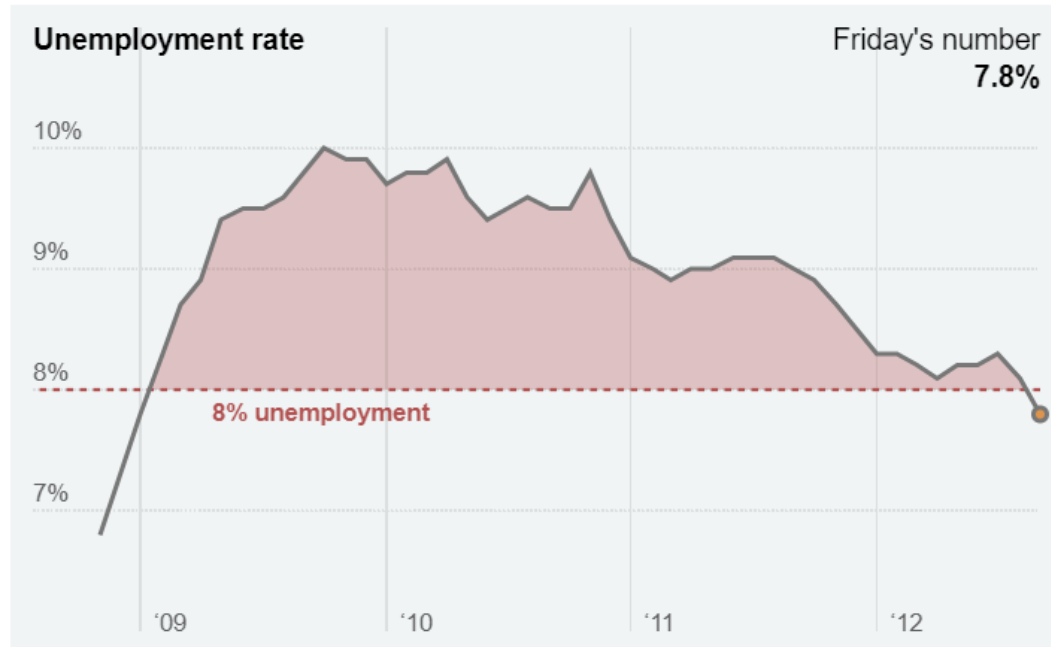
Source: Department for Work and Pensions, Public Health England
Graphic by John Burn-Murdoch / @jburnmurdoch
© FT

Blackpool has some of England's highest rates of obesity and smoking, and the top mortality rate from alcoholic liver disease

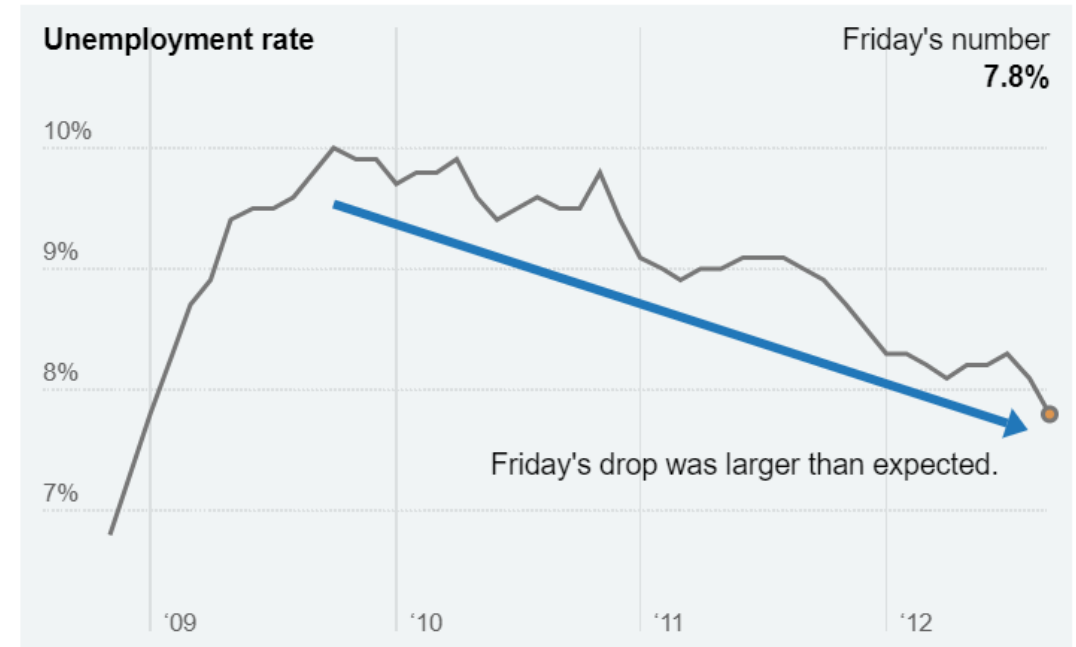


Annotations can emphasize your point...

The rate was above 8 percent for 43 months.



The rate has fallen more than 2 points since its recent peak.



Exercise: Highlight your message

How can you highlight your message in (one of) the following situations?



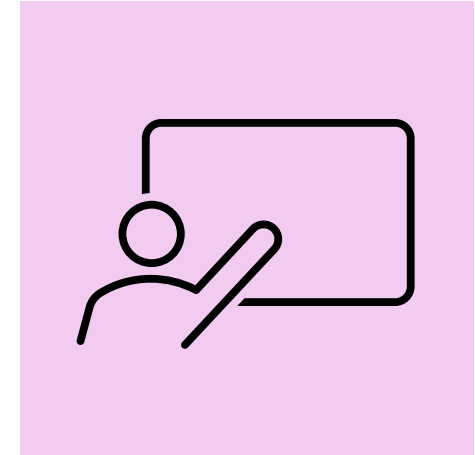
Scientific publication:

- Can you think of a (more) **descriptive title**?
- Can you add **focus** to highlight your key message?



News article:

- Can you come up with a **catchy headline**?
- What would you change or leave out?
- How can you attract (more) attention?



Presentation:

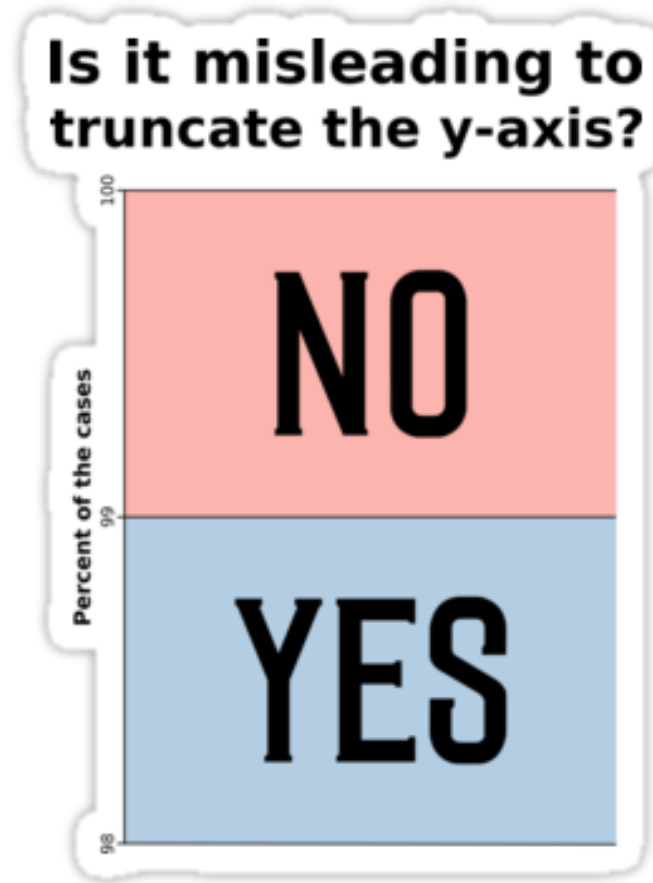
- Can you **animate** (parts of) your visual?
- Can you **highlight** (parts of) your visual?

Do's and Don'ts

Show the *right* data, show the *data* right!

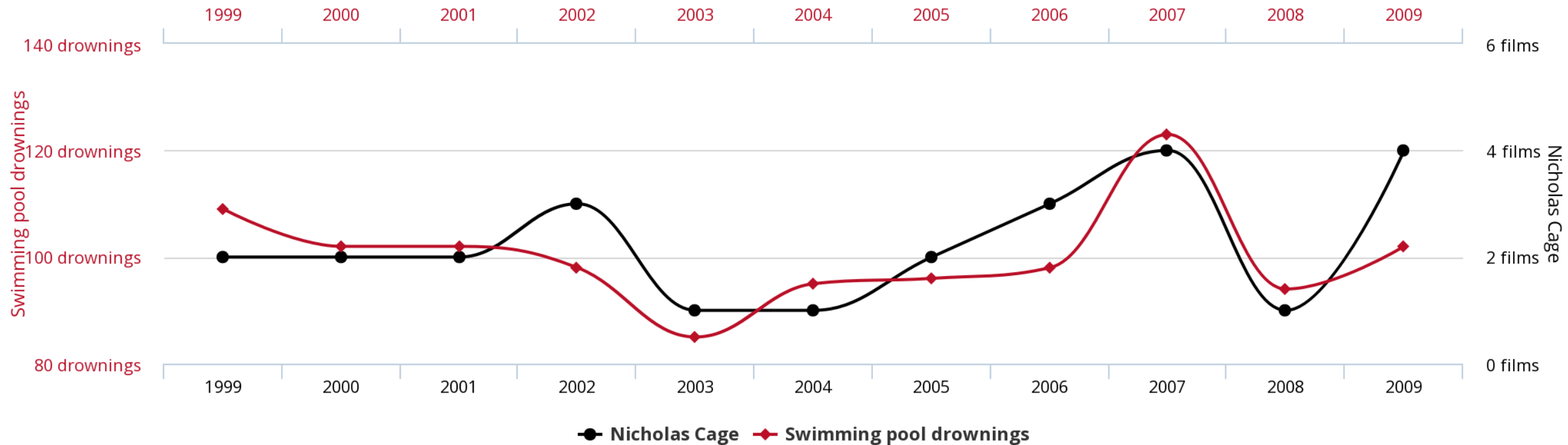


Don't: Truncate the y-axis

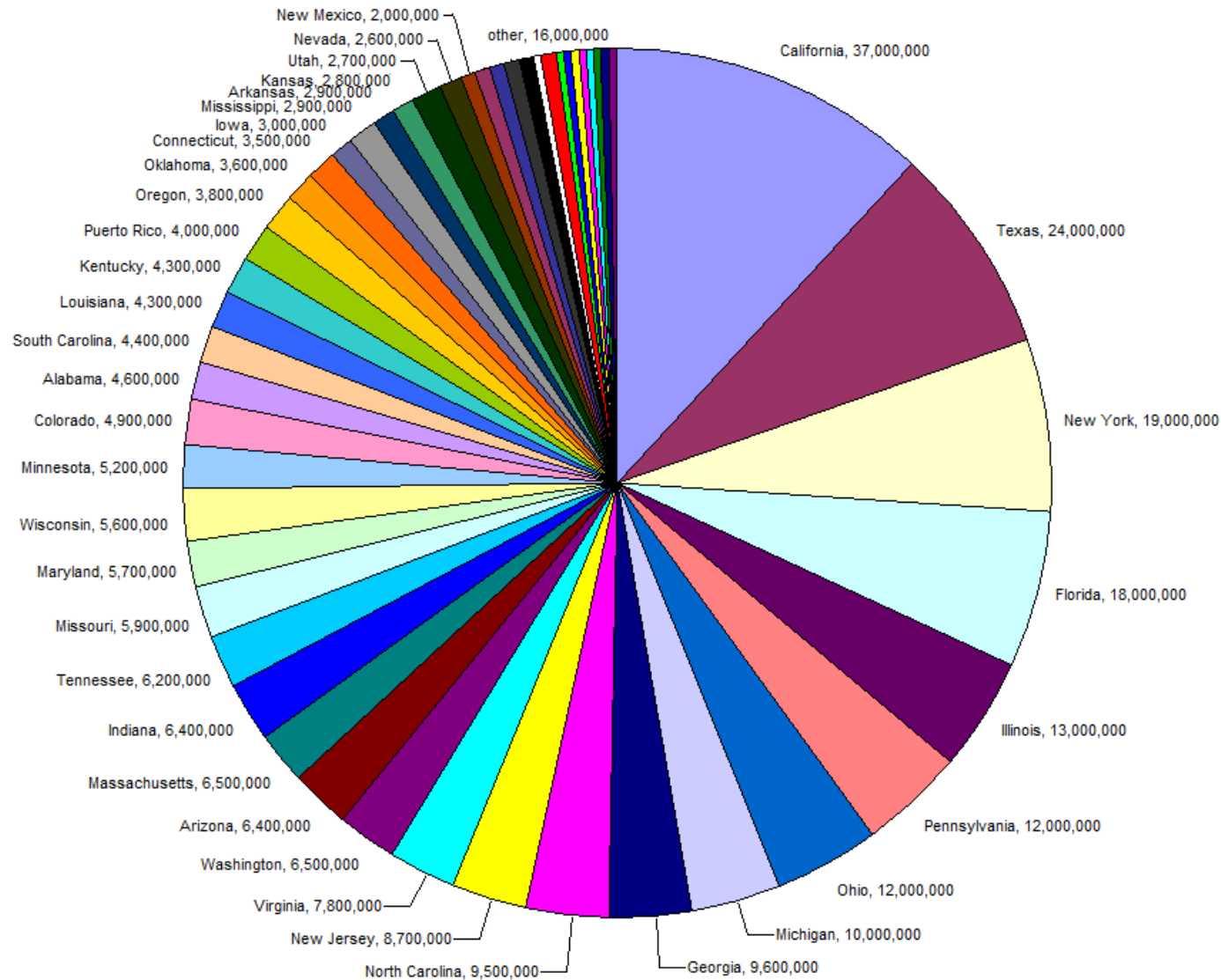


Don't: use double axes

Number of people who drowned by falling into a pool
correlates with
Films Nicolas Cage appeared in

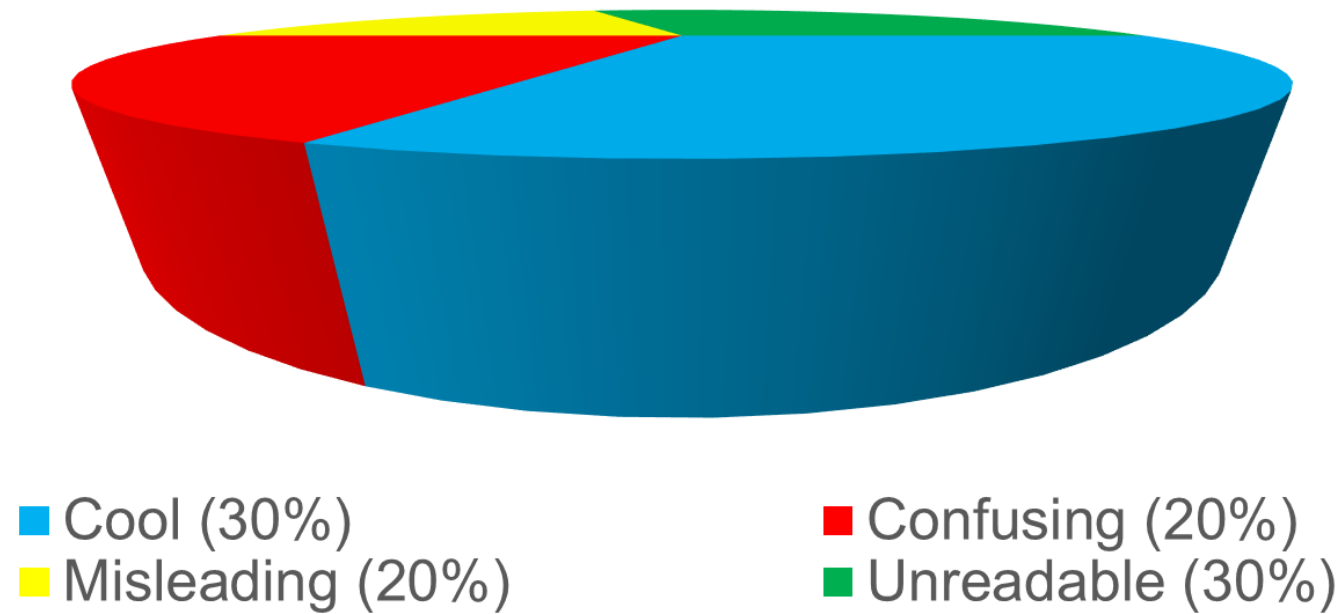


Don't: Use pie charts with >5 categories

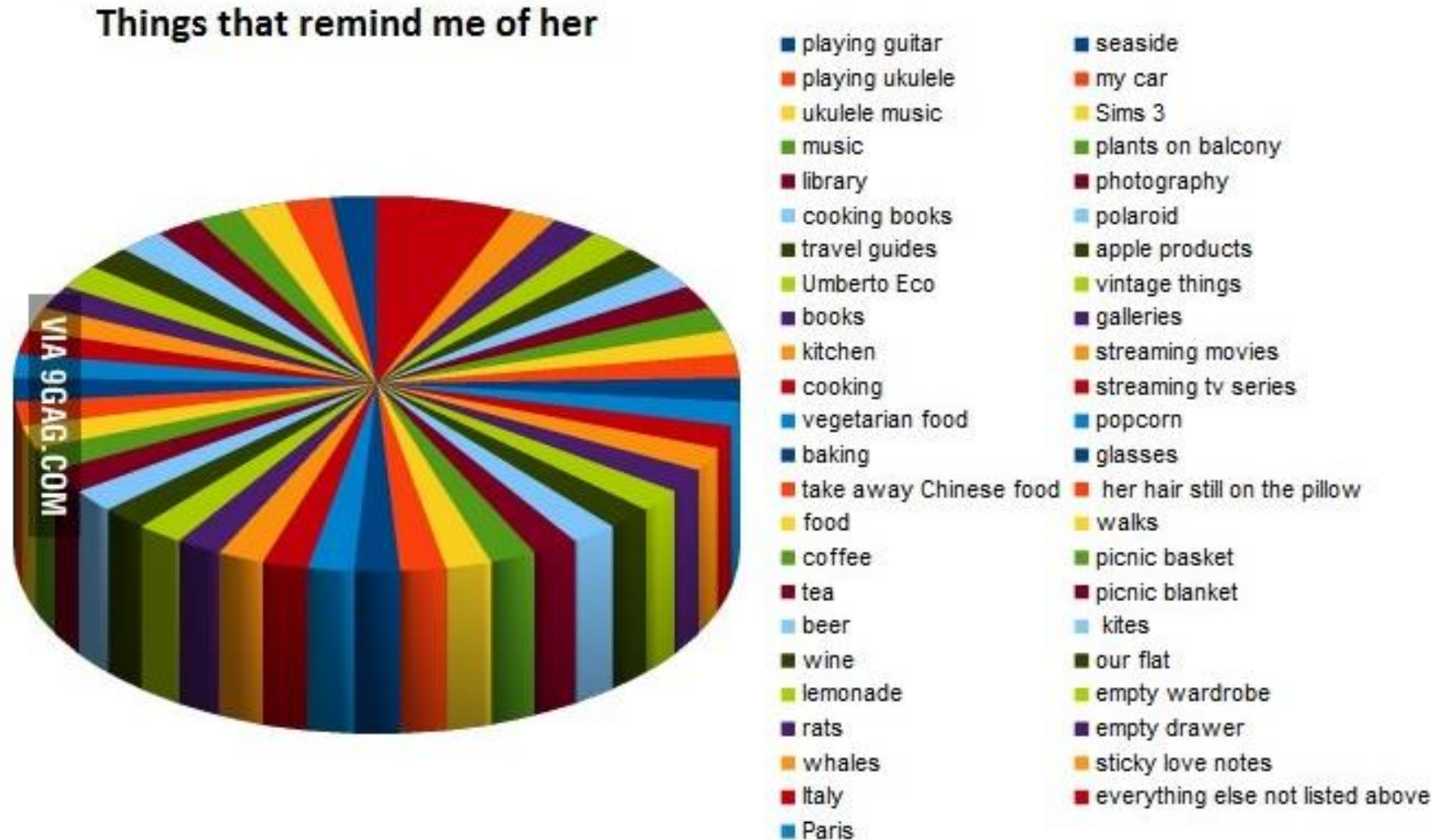


Don't: Use 3D pie charts

Perception of 3D pie charts



Don't: Use 3D pie charts with >5 categories



Don't: Use patterned 3D winding staircase lying on its side pie diagrams with >5 categories!

Anatomy of a Winning TED Talk

● 1%

Sophisticated Visual Aids

We're not sure who puts the D in TED—most of the best presentations favor tepid PowerPoint slide shows (sorry, Brené Brown), Pictionary-quality drawings (really, Simon Sinek?), or no props at all.

● 5%

Opening Joke

Remember the one about the shoe salesman who went to Africa in the 1900s? That's how Benjamin Zander opened his talk—which turned out to be about classical music.

● 5%

Spontaneous Moment

Don't overprepare. Tease the guy in the front row ("You could light up a village with this guy's eyes"). Commend the stagehand who handles the human brain you brought.

● 5%

Statement of Utter Certainty

People come for answers—give 'em what they want, as Shawn Achor did: "By training your brain ... we can reverse the formula for happiness and success."

● 12%

Snappy Refrain

The TED equivalent of "I have a dream." Example: "People don't buy what you do; they buy why you do it." Repeat 7x.

● 23%

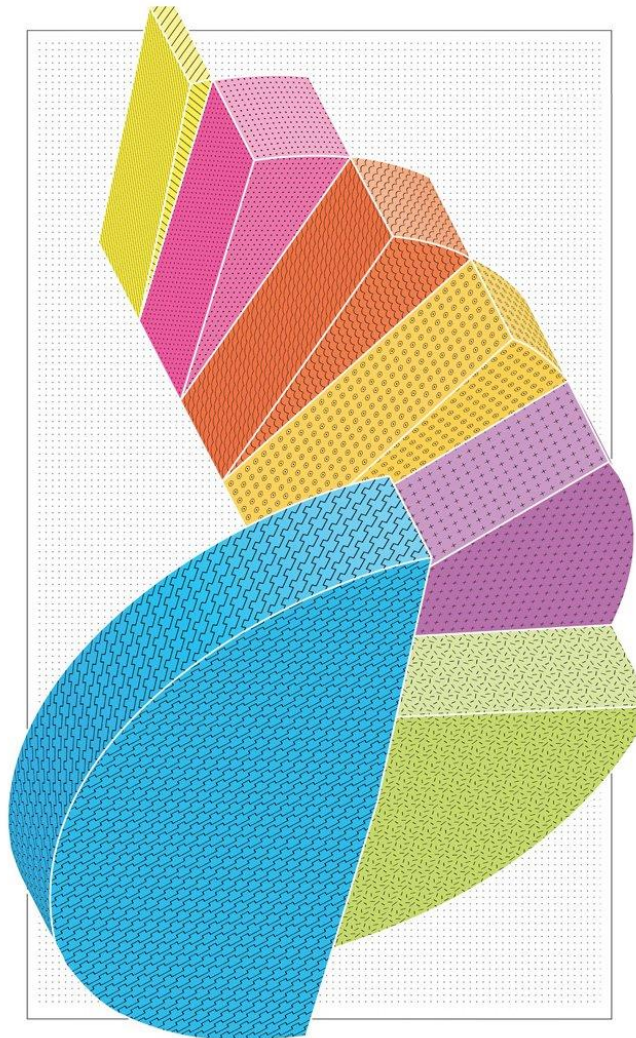
Personal Failure

Be relatable. We want to know about that nervous breakdown. Or at least the time you didn't fit in at summer camp.

● 49%

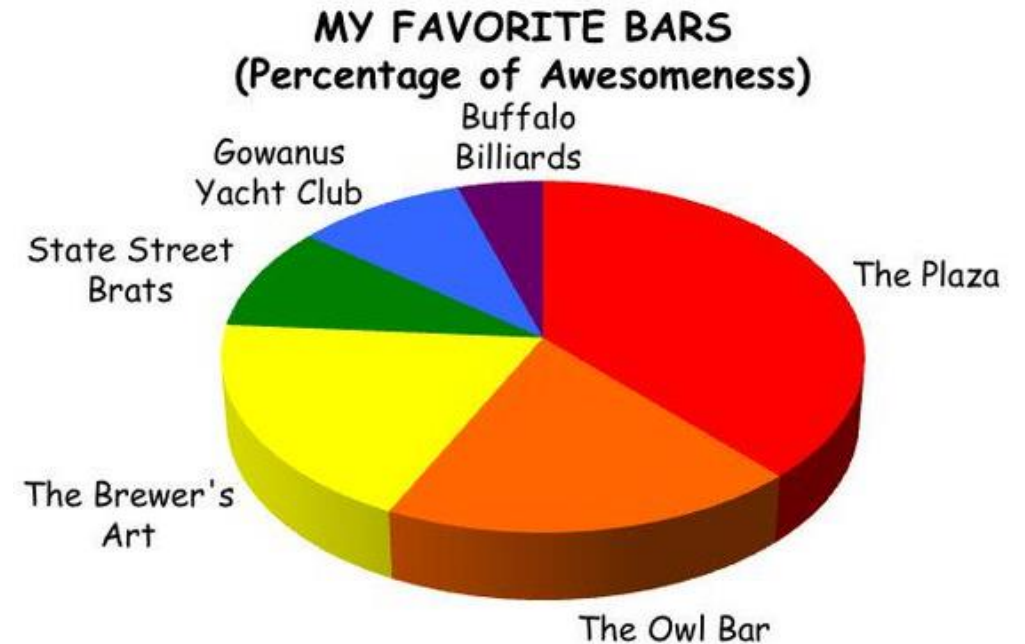
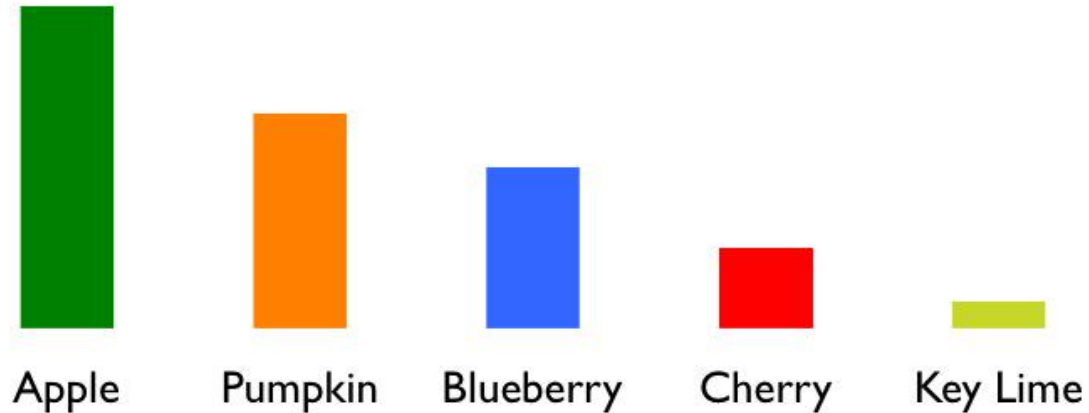
Contrarian Thesis

Wait a sec—we should be playing *more* videogames? The more choices we have, the worse off we are? TED is where conventional wisdom goes to die.



Do: Know when to choose for *bars* or *pies*

My Favorite Pies
(Percentage of Tastiness)

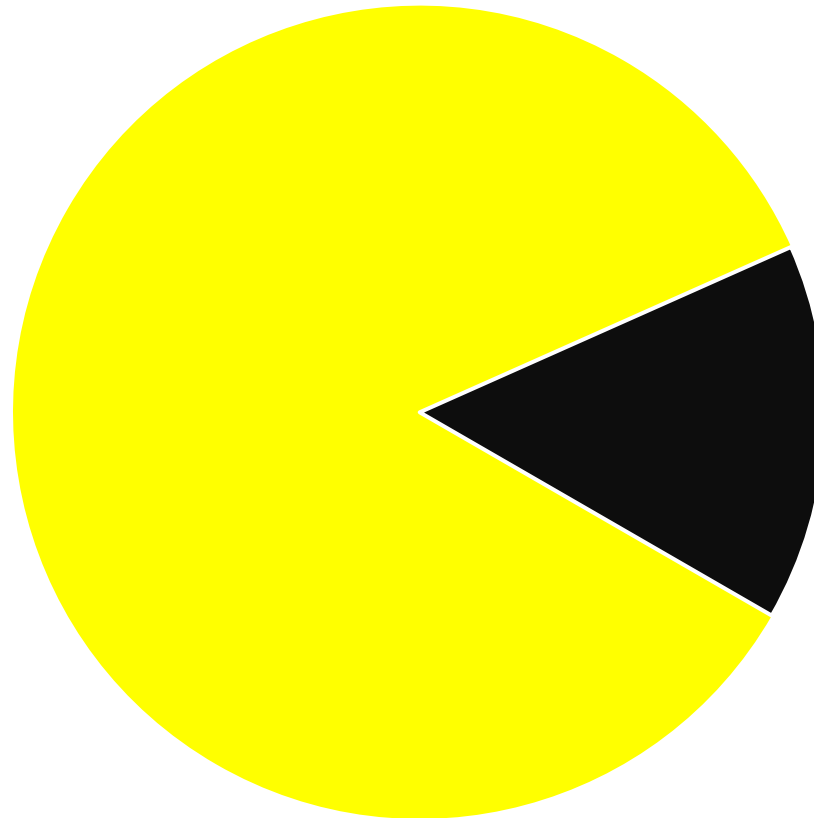


Pie charts are great to show the part-to-whole relation of 2 categories

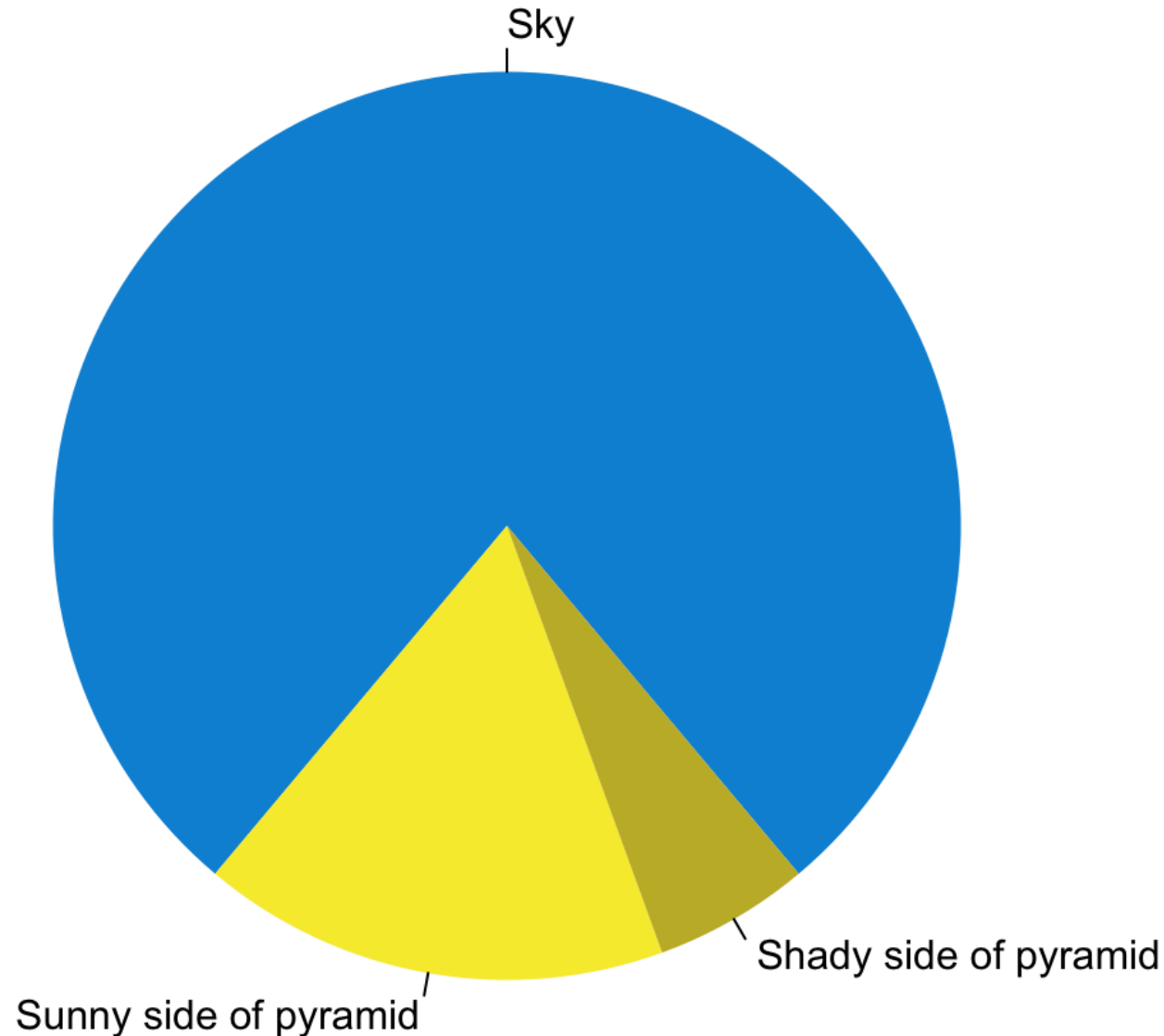
Percentage of chart that looks like Pacman

■ Pacman

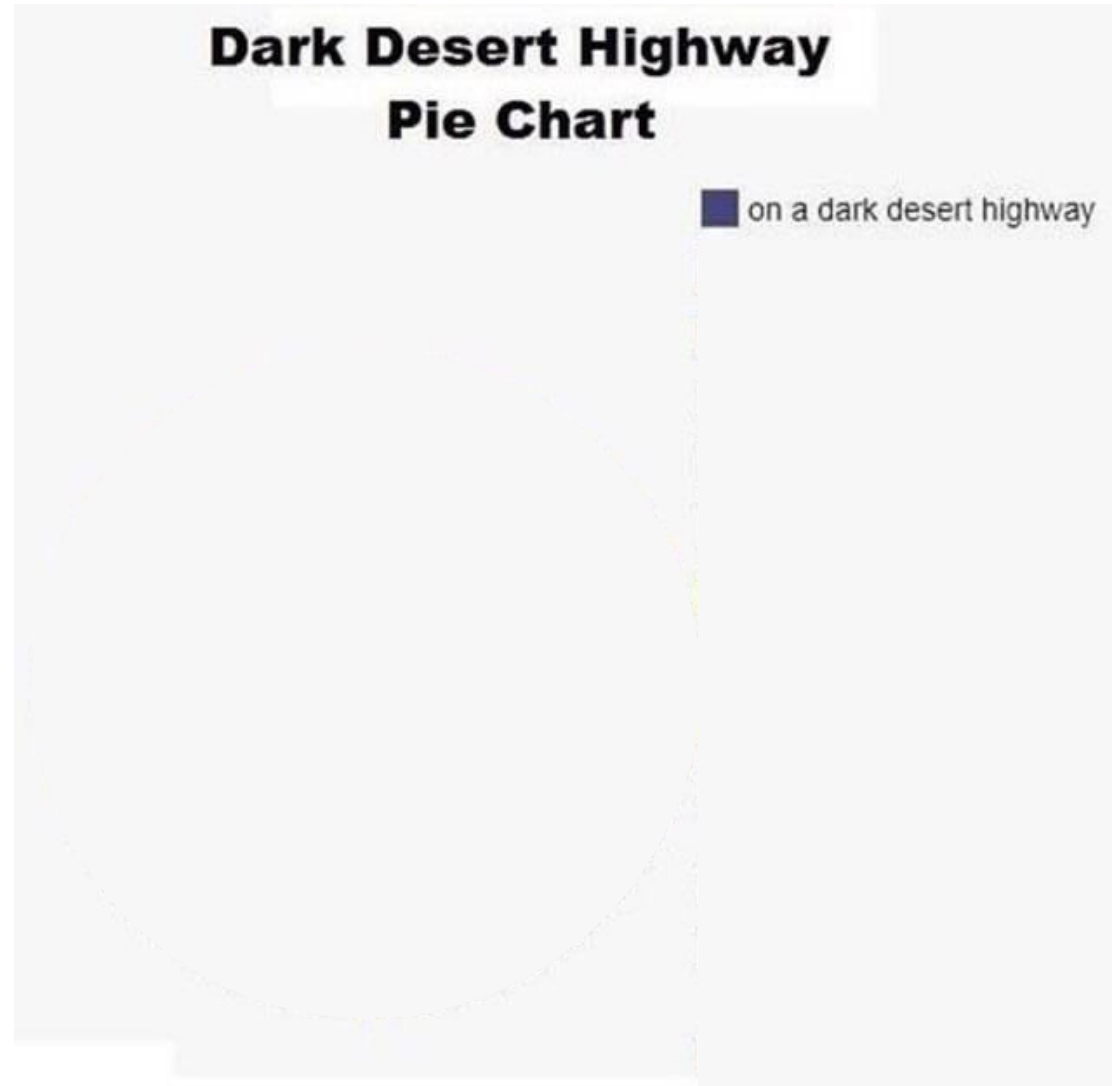
■ Not Pacman



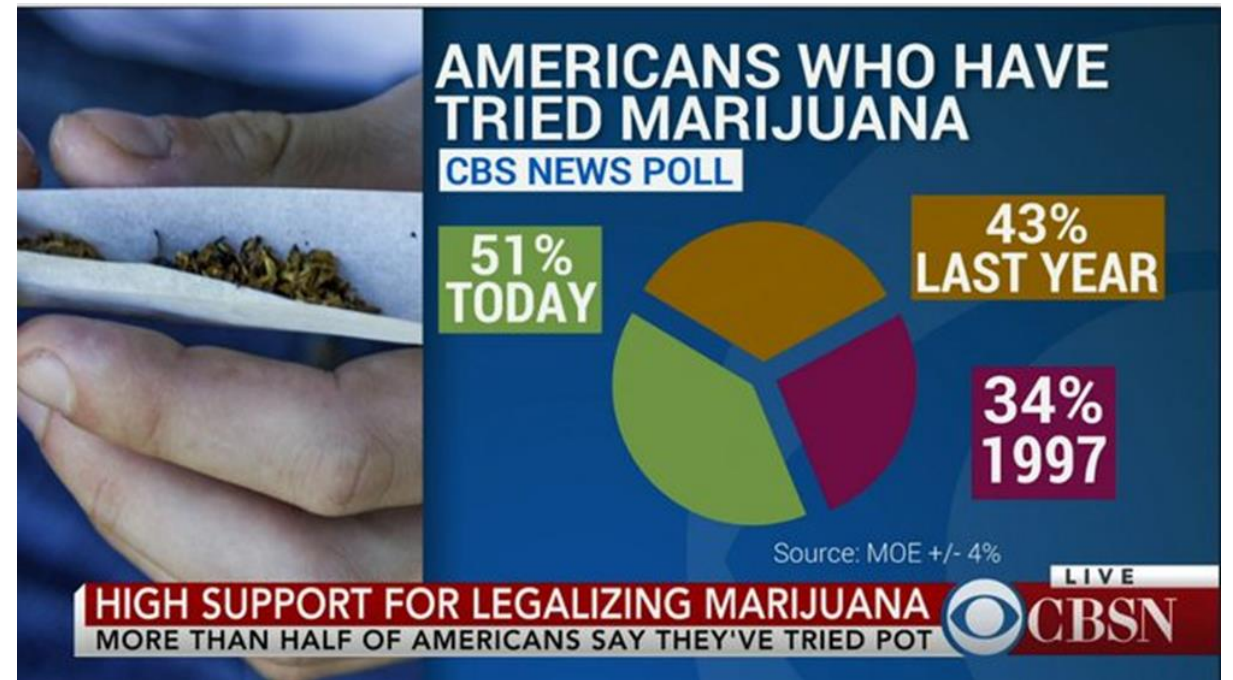
Pie charts for 3 categories are okay too



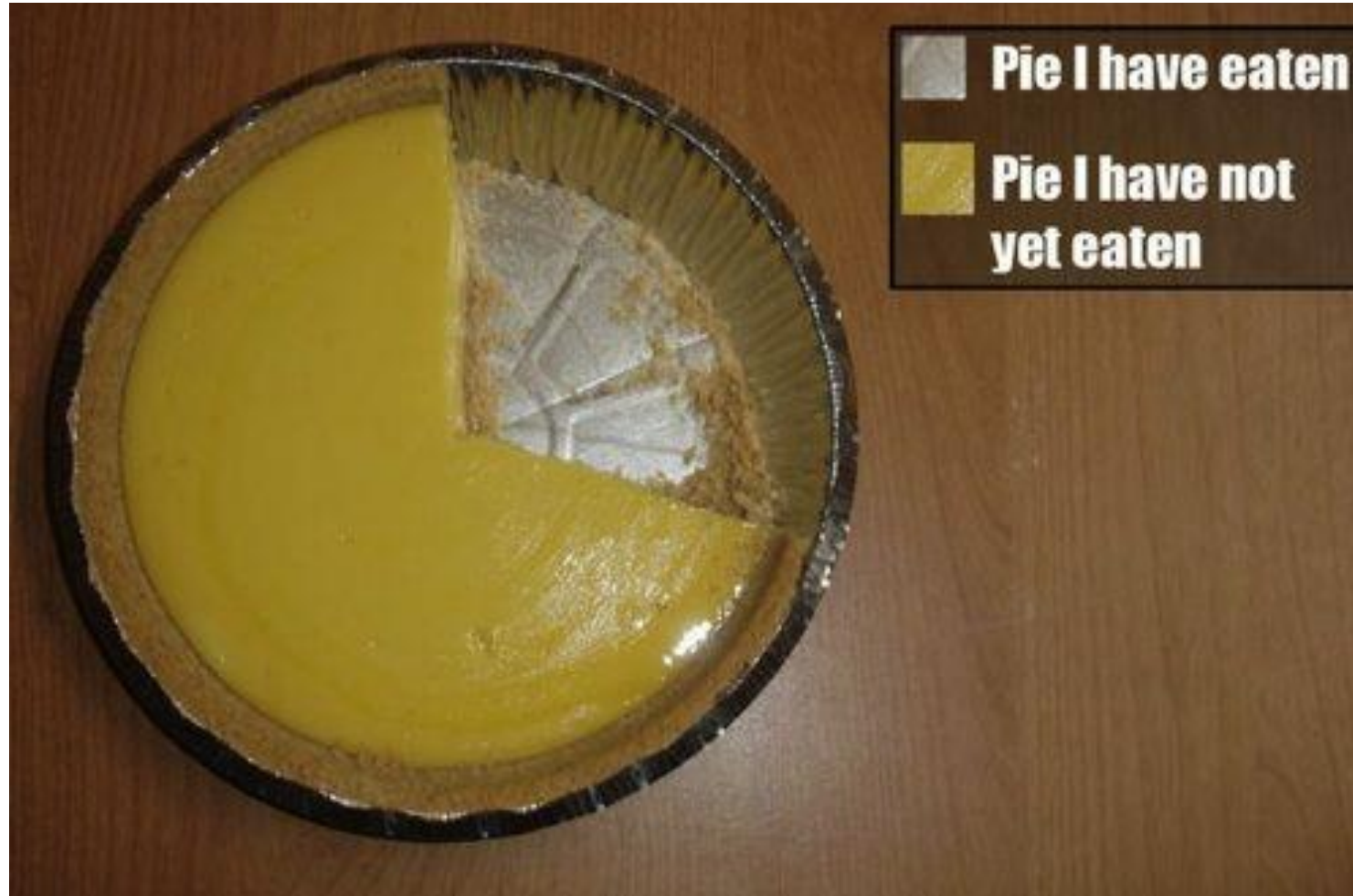
Sometimes pie charts with >5 groups work 😊



The *worst* pie chart *ever*...?



The *best* pie chart ever?



Thank you!



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Happy Viz-ing!

Links to slides and resources:

<https://github.com/IreneVDB/IreneVDB.github.io/blob/main/VLAG/>

- Presentation.pdf
- ReferenceList.pdf



<https://irenevdb.rbind.io>